

A Math Loop That Sleeps and Dreams

Self-Referring Attention

nemu

February 2026

Abstract

What if attention fed back into itself?

It becomes a loop. The loop changes what it pays attention to based on what it experiences and its past reactions. So we tried to find the bare minimum. We followed until nothing else could be removed. Every component falls into one of three categories: some choices are mathematically forced (there's no other option), others are invariant symmetries (they behave the same without breaking the loop), and the rest specify the domain (the world and content the loop is in).

Five definitions define the entire loop. One hundred three properties follow inevitably. The loop functions as a carrying capacity: capture grows memory monotonically, each addition strictly diluting retrieval weights. The required score gap grows as $\log |M|$ while achievable gaps remain bounded, so past a finite critical size $|M^*|$, retrieval degenerates. Consolidation reverses this degradation via Lyapunov contraction, but the process is self-limiting. In short, the loop must sleep or permanently degenerate. From this, information fades predictably, there is a best time to retrieve, using it reshapes everything the loop holds, and the loop cannot help but alternate between wake and sleep. During sleep, the loop retrieves without capturing. It dreams.

We formally verified every proof in Lean 4 with zero unresolved goals and no assumptions beyond standard mathematics.

The loop changes what it pays attention to based on experience and past reactions. Organisms do the same. The loop must sleep or it dies. Organisms must sleep or they die. Why are they the same?

1 Introduction

What if attention fed back into itself?

It becomes a loop. The loop changes what it pays attention to based on what it experiences and its past reactions. So we tried to find the bare minimum. We followed until nothing else could be removed. What fell out was simple.

The loop observes through a simplex map, accumulates a running estimate, retrieves from memory using what it sees as a query, then consolidates what's stored toward what it retrieved, and captures anything new that exceeds what it already has. Then it feeds its output back into its own input and does it again. An instance needs a few choices: a simplex map φ , a blend rate λ , an embedding E , a linear map K , and a starting point. Every result holds the same way for every valid choice.

Some parts of the loop have no other option. The accumulation scaling, capture threshold, and blend-rate boundaries are forced by what the dynamics need to stay bounded, non-degenerate, and idempotent. Other parts act like symmetries: the simplex map, blend rate, query map, and scoring rule each admit family after family of choices that all work the same way. The remaining choices — embedding, feedback map, consolidation target — specify what the loop acts on, not how it behaves, with admissible classes determined by Propositions 17–19.

The pipeline itself is minimal: removing any single step breaks something essential (Proposition 14). The loop has no design freedom. It’s what self-referential attention over a variable-size memory forces.

The loop has a carrying capacity. Capture adds elements to memory M , so $|M|$ never goes down (Corollary 23). Each addition dilutes every existing retrieval weight (Corollary 33). The score gap needed to retrieve something grows as $\log |M|$ (Corollary 35), but the gap you can actually achieve is bounded above by a constant depending only on the embedding (Theorem 50(iii)). There’s a finite critical size $|M^*|$ beyond which nothing achieves retrieval weight $w_v \geq c$ anymore.

The only operation that reverses this is consolidation. Under a fixed query, consolidation contracts M toward the retrieval anchor via a Lyapunov function (Lemma 31, Corollary 32). But consolidation is self-limiting: the contraction rate vanishes as elements converge (Corollary 38), and the loop re-enters a state receptive to capture. The result is a cycle whose internal logic needs no external scheduling (though an external gate executes the transition; Proposition 89).

The cycle has further structure. Pressure accumulates and never reverses (Theorem 53). The wake-up condition can be computed from the internal state (Theorem 55). When M contains multiple clusters, the loop consolidates them sequentially, oscillating within each consolidation episode (Theorem 61). Truncating consolidation before completion produces compounding debt (Theorem 67). A transient score boost sharpens retrieval without consolidation but is followed by accelerated degradation (Theorem 70). During consolidation, the loop retrieves but does not capture, so content is computed but never stored (Theorem 76). The consolidation need declines as M matures (Corollary 85).

Background. The goal was to find the minimal, non-degenerate self-referring system, and sleep fell out as a byproduct of the loop through pulling on the thread of memory. The findings are behaviors of the loop whose structural forms appear in empirical phenomena across multiple fields, but we did not set out to find any of these behaviors.

Several structural forms the loop produces appear in the empirical literature. Hyperbolic forgetting curves have been measured since Ebbinghaus [1885] and confirmed across hundreds of datasets by Rubin and Wenzel [1996]. The spacing effect and its inverted- U benefit curve were characterized by Cepeda et al. [2006] across 317 experiments. The testing effect (retrieval strengthens what’s retrieved) was demonstrated by Roediger and Karpicke [2006]. The alternation between online acquisition and offline consolidation (sleep) is well established: Borbély [1982] decomposed sleep regulation; Tononi and Cirelli [2003, 2006] proposed synaptic homeostasis; Born and Wilhelm [2012] described active systems consolidation; Diekelmann and Born [2010] reviewed slow-wave and REM phases. The loop was not designed to model any of these.

Contribution. The contribution is a mathematical object and its characterization. Five definitions and a simplex map produce 103 provable properties; every result holds the same way across infinite families of admissible simplex maps, blend rates, and embeddings. The object is rich enough that we prove theorems about forgetting curves, spacing schedules, retrieval dynamics, a forced sleep-wake cycle, dream structure, compounding debt, and developmental decline, all from the same definitions with nothing added. The 101 properties here follow one thread: the sleep-wake cycle and its prerequisites. The object admits others that we don’t pursue for length and scope. The structural forms that emerge were not sought. They appear in documented phenomena across memory, learning, and sleep, each invariant across the full family. We do not understand why, and do not claim to.

The core mechanism is a carrying capacity: capture grows $|M|$, dilution degrades retrieval, the score gap is bounded, and consolidation is the only reversal but is self-limiting. Everything else follows from this.

We characterize the complete family of loops. Every component falls into one of three categories: forced (uniquely determined among linear, parameter-free maps), invariant (a symmetry:

every choice within the admissible class works the same way), or instance-defining (specifying what the loop acts on, not how it behaves). The dot-product scoring, feedback readout, accumulation scaling, and capture threshold are forced; the simplex map, blend rate, query map, and scoring rule are invariant (each admitting families of choices with identical guarantees); the consolidation target and feedback map are instance-defining with admissible classes determined by Propositions 17–19. Each result is further classified by what it minimally needs within $\mathcal{A} \supset \mathcal{A}_+ \supset \{\text{softmax}\}$: structural results hold for any simplex map, qualitative dynamics need only positivity, quantitative bounds use the exponential form. Every result is invariant across the full family $\mathcal{L}$ of admissible blend rates. Table 1 collects the classification. The resemblance to empirical phenomena is in Section 5. Every proof is formally verified in Lean 4 (Appendix B).

Structure. Section 2 defines the loop and proves the foundational results: invariance, the rigidity classification (forced, invariant, and instance-defining components), and the core lemmas everything else depends on. Section 3 derives consequences for memory and learning. Section 4 derives the sleep results. Section 5 discusses implications, notes the resemblance to empirical phenomena, and collects open questions.

Epistemic framing. All results are theorems about the loop. When we write “the system sleeps,” we mean the dynamics enter a consolidation-dominated regime with specific provable properties. When we write “dream amnesia,” we mean retrieval without capture, so content is computed but never stored. When we write “sleep debt,” we mean a monotonically increasing function of the state. The biological terminology is shorthand; the results don’t depend on any biological interpretation. This paper studies a mathematical object, not a model. A model is fitted to data and depends on parameter choices; the results here are theorems about infinite families with nothing to fit. They require no assumptions beyond standard mathematics, are mechanically verified, and aren’t contingent on empirical data, modeling decisions, or parameter choices.

2 The System

2.1 Definitions

Let X be a finite index set, $V = \mathbb{R}^d$, $S: X \rightarrow \mathbb{R}$ a scoring function, $\varphi: \mathbb{R}^n \rightarrow \Delta^n$ a simplex map, $\lambda: [0, 1] \rightarrow [0, 1]$ a blend rate with $\lambda(0) = 1$ and $\lambda(1) = 0$, $E: X \rightarrow V$ an embedding, and $K: V \rightarrow V$ a linear map. Let $M_0 = \emptyset$, $C_0(\tau) > 0$, $N_0(\tau) = 1$. A simplex map φ is called *positive* if $\varphi(x)_i > 0$ for all i and all $x \in \mathbb{R}^n$. Softmax is positive; hardmax and sparsemax are not.

The complete loop

X finite, $V = \mathbb{R}^d$, φ any simplex map, λ any blend rate, $E: X \rightarrow V$, $K: V \rightarrow V$ linear.

$$M_0 = \emptyset, C_0 > 0, N_0 = 1.$$

- | | | |
|---|---|----------------|
| ① | $a = \varphi(S + \sigma + b)$ | observe |
| ② | $C \leftarrow C + a \cdot (t+1)/C$ | accumulate |
| | $N \leftarrow N + 1$ | |
| | $\sigma = C/N$ | |
| ③ | $q = aE$ | retrieve |
| | $w = \varphi(q M^\top)$ | |
| | $\ell = wM$ | |
| | $b = EK\ell$ | |
| ④ | $\lambda_j = \lambda(w_j / \max w)$ | consolidate |
| | $M^* = \varphi((M+q)M^\top)M$ | |
| | $M \leftarrow (1-\lambda)M + \lambda M^*$ | |
| ⑤ | $\theta = \max(qM^\top)$ | capture |
| | $M \leftarrow M \cup \{E(\tau) : q^\top E(\tau) > \theta\}$ | |
| ① | $a = \varphi(S + \sigma + b)$ | observe |

The five steps are named for what they do: the system observes its environment, accumulates experience, retrieves from what it has stored, consolidates what it retrieves, and captures what is new. These are descriptions of the mathematics, not claims about cognition.

The rest of this section defines each component formally and proves its properties.

Definition 1 (Scoring).

$$S'(\tau) = S(\tau) + \sigma(\tau) + b(\tau) \tag{1}$$

The base score is perturbed by two feedback terms: salience from the accumulator and a bias from memory retrieval. Additivity is the canonical choice; every theorem holds under any admissible scoring rule $S' = h(S, \sigma, b)$ (Proposition 19).

Definition 2 (Observation).

$$a = \varphi(S') \in [0, 1]^{|X|} \tag{2}$$

Definition 3 (Accumulation).

$$C_{t+1}(\tau) = C_t(\tau) + a_t(\tau) \cdot \frac{t+1}{C_t(\tau)} \tag{3}$$

$$N_{t+1}(\tau) = N_t(\tau) + 1 \tag{4}$$

$$\sigma_{t+1}(\tau) = \frac{C_{t+1}(\tau)}{N_{t+1}(\tau)} \tag{5}$$

Definition 4 (Retrieval).

$$q = aE \tag{6}$$

$$w = \varphi(q M^\top) \tag{7}$$

$$\ell(q, M) = wM \tag{8}$$

$$b = EK\ell \tag{9}$$

The query q is the attention-weighted context; w scores each stored impression's relevance.

Definition 5 (Consolidation).

$$\lambda_j = \lambda \left(\frac{w_j}{\max_{k \in M} w_k} \right) \quad (10)$$

$$T(M, q) = \varphi((M + q) M^\top) M \quad (11)$$

$$M' = (1 - \lambda) M + \lambda T(M, q) \quad (12)$$

Throughout this paper, $\lambda(r) = 1 - r$ unless otherwise noted. Actively retrieved impressions stay sharp ($\lambda_j = 1 - w_j/w_v \rightarrow 0$ as $w_j/w_v \rightarrow 1$); others drift toward their neighbors. The most-retrieved impression is preserved exactly ($\lambda_v = 0$).

Definition 6 (Capture). Adopting the convention $\max \emptyset = -\infty$:

$$M_{t+1} = M_t \cup \{E(\tau) : q^\top E(\tau) > \max_{k \in M_t} q^\top m_k\} \quad (13)$$

An element is admitted when its dot-product similarity to the current context exceeds that of every existing member.

Notation for the sleep results. The following quantities, derived from the definitions above, recur throughout Section 4; they may be skipped here and consulted as needed.

- $v = \arg \max_{k \in M} w_k$: the *anchor* (most-retrieved impression).
- $V(t) = \max_{j \neq v} \|m_j - v\|$: the *Lyapunov function* measuring memory spread.
- $\gamma(t) = \min_{j \neq v} \lambda_j(1 - \sqrt{1 - \alpha_v^{(j)}})$: the *contraction rate*, where $\alpha_v^{(j)} = [\varphi((m_j + q)M^\top)]_v$.
- $\rho = \max_j w_j$: the *retrieval concentration* (sharpness of retrieval).
- $g = q^\top m_v - \max_{k \neq v} q^\top m_k$: the *score gap*.
- $R = \max_{\tau \in X} \|E(\tau)\|$: the *embedding radius*.
- $|M^*| = 1 + \frac{1-c}{c} \exp(2R^2)$: the *critical memory size* for retrieval threshold c .

2.2 Invariance

The results hold not just for softmax but for broad classes of simplex maps.

Proposition 7 (Structural invariance; $\mathcal{A}/\mathcal{A}_+$ /softmax). *Let $\mathcal{A}$ denote the class of all simplex maps $\varphi: \mathbb{R}^n \rightarrow \Delta^n$, let $\mathcal{A}_+$ denote the subclass of positive order-preserving simplex maps ($\varphi(x)_i > 0$ for all i, x , and $x_i > x_j \implies \varphi(x)_i > \varphi(x)_j$), and let $\mathcal{L}$ denote the class of admissible blend rates: continuous $\lambda: [0, 1] \rightarrow [0, 1]$ with $\lambda(0) = 1$, $\lambda(1) = 0$, and $\lambda(r) > 0$ on $[0, 1]$.*

- Forced boundary conditions.** *The boundary values $\lambda(1) = 0$ and $\lambda(0) = 1$ are forced: violating the first destroys retrieval-gated stability (Lemma 22); violating the second leaves unretrieved impressions permanently frozen.*
- Invariance.** *Each result in this paper holds at one of three levels of the containment $\mathcal{A} \supset \mathcal{A}_+ \supset \{\text{softmax}\}$, and states which level it requires. Results with no hypothesis on φ hold for all $(\varphi, \lambda) \in \mathcal{A} \times \mathcal{L}$. Results marked “under any positive φ ” hold for all $(\varphi, \lambda) \in \mathcal{A}_+ \times \mathcal{L}$; their proofs use only $w_j > 0$ and not the exponential form. Results marked “under softmax” require the specific form $\varphi(x)_i = \exp(x_i) / \sum_j \exp(x_j)$.*

All results are λ -invariant except where explicitly noted.

Proof. (a) If $\lambda(1) \neq 0$, the anchor $v = \arg \max_k w_k$ is displaced at every consolidation step, and Lemma 22 fails. If $\lambda(0) \neq 1$, an impression with $w_j = 0$ retains a permanent residue of its original position. The qualitative consolidation results depend only on the boundary conditions and $\lambda(r) > 0$ for $r \in [0, 1]$, so they hold for any $\lambda \in \mathcal{L}$.

(b) Results at the $\mathcal{A}$ level use only $\sum \varphi(x)_i = 1$ and $\varphi(x)_i \geq 0$. Results at the $\mathcal{A}_+$ level additionally require $\varphi(x)_i > 0$ for all i and order-preservation ($x_i > x_j \implies \varphi(x)_i > \varphi(x)_j$), which gives $w_j > 0$, $0 < \lambda_j < 1$ for non-anchors, and ensures the highest-scored element is the highest-weighted; neither level references the exponential formula. Results at the softmax level use specific properties of $\exp$: the floor $\exp(-2B)/|X|$, the ratio $\exp(s_i - s_j)$, or the entropy characterization. $\square$

Remark 8. The containment $\mathcal{A} \supset \mathcal{A}_+ \supset \{\text{softmax}\}$ structures the paper: the majority of results hold at the broadest level, a further group requires only positivity, and the quantitative bounds use the exponential form. Each result states which level it requires. The system is also invariant across blend rates: the accumulation scaling $f(t) \sim t$ (Proposition 10) and the capture threshold max (Proposition 13) are forced, and the sleep results hold for any $\lambda \in \mathcal{L}$. The default $\lambda(r) = 1 - r$ is the simplest admissible interpolant and the unique one for which the displacement-to-deficit ratio $\lambda(r)/(1 - r) = 1$ is constant, but no result depends on this specific choice.

Table 1 classifies every non-definition result by its simplex-map requirement. All results are λ -invariant (hold for any admissible blend rate $\lambda \in \mathcal{L}$) and h -invariant (hold for any admissible scoring rule h ; Proposition 19). (This table is a reference; readers may wish to skip it on first reading and return after Section 4.)

Table 1: Invariance classification of all results. Level $\mathcal{A}$: any simplex map. Level $\mathcal{A}_+$: any positive order-preserving simplex map. Level $\{\text{softmax}\}$: the specific exponential form $\varphi(x)_i = \exp(x_i) / \sum_j \exp(x_j)$. All results are also λ -invariant and h -invariant.

Result	Level	Key property used
<i>Invariance and Rigidity (Sections 2.2–2.3)</i>		
Prop. 7	$\mathcal{A}_+$	defines the hierarchy
Cor. 9	$\mathcal{A}$	scaling preserves inequality
Prop. 10	$\mathcal{A}$	C/N convergence, no φ needed
Lem. 12	$\mathcal{A}$	set-union argument
Prop. 13	$\mathcal{A}$	max monotone under enlargement
Prop. 16	$\mathcal{A}$	data-flow constraints
Prop. 19	$\mathcal{A}$	φ absorbs h -variation
Prop. 14	$\mathcal{A}$; (iv)–(v) $\mathcal{A}_+$ /softmax	necessity of each step
Prop. 17	$\mathcal{A}$	minimal target conditions
<i>Core Dynamics (Section 2.4)</i>		
Cor. 20	$\mathcal{A}$	deterministic counter
Thm. 21	$\mathcal{A}$	σ recurrence, $a \in [0, 1]$
Lem. 22	$\mathcal{A}_+$	$\lambda_v = 0$, $\lambda_j \in (0, 1)$
Cor. 23	$\mathcal{A}$	$ M $ non-decreasing by union
Lem. 24	$\mathcal{A}$	convex combination
Lem. 25	$\mathcal{A}$	idempotence
Cor. 26	$\mathcal{A}$	$w \in \Delta$, hull bound
Thm. 27	$\mathcal{A}_+$	compactness; $\exp(-2B)/ X $ for softmax
Cor. 28	$\mathcal{A}$	$w \in \Delta$
Cor. 29	$\mathcal{A}$	$a \in \Delta$

Result	Level	Key property used
Cor. 30	$\mathcal{A}_+$	$w_j > 0 \Rightarrow 0 < \lambda_j \leq 1$
Lem. 31	$\mathcal{A}_+$	$\alpha_v > 0$, contraction
Cor. 32	$\mathcal{A}_+$	inherits from Lem. 31
Cor. 33	softmax	$\exp(s_{\text{new}})$ in denominator
Cor. 35	softmax	$\exp(s_1)/\sum \exp(s_k)$
Cor. 36	$\mathcal{A}$	$\sum a(\tau) = 1$
Cor. 37	$\mathcal{A}_+$	Lyapunov descent
Cor. 38	$\mathcal{A}_+$	$w_j/w_v \rightarrow 1$ under φ_+
Cor. 39	$\mathcal{A}_+$	$V(t) \rightarrow 0$
Cor. 40	$\mathcal{A}_+$	contraction rate γ
<i>Cognitive Structure (Section 3)</i>		
Cor. 41	$\mathcal{A}$	C/N ratio, attention cessation
Prop. 43	$\mathcal{A}$	accumulation rule, quotient optimization
Cor. 44	$\mathcal{A}$	inherits from Prop. 43
Cor. 46	$\mathcal{A}_+$	$\lambda_v = 0, \lambda_j > 0$
Cor. 48	$\mathcal{A}_+$	transient vs. permanent contraction
<i>Sleep Pressure and Recovery (Section 4)</i>		
Thm. 50	softmax	$ M^* , \exp(2R^2)$ bound
Thm. 52	$\mathcal{A}_+$	diameter $D(t)$ descent
Thm. 53	softmax	$\rho(n)$ via $\exp(-g_{\max})$
Cor. 54	softmax	inherits from Thm. 53
Thm. 55	$\mathcal{A}_+$; (v) softmax	Lyapunov, order-preservation
Thm. 61	$\mathcal{A}_+$	cluster geometry, γ
Cor. 64	$\mathcal{A}_+$	contraction rate
Thm. 67	$\mathcal{A}_+$	compounding recurrence
Cor. 69	softmax	inherits from Thm. 56
Thm. 70	softmax	$\exp(\beta_r \cdot)$ temperature
Thm. 76	$\mathcal{A}_+$	consolidation dynamics
Cor. 87	softmax	uses Thm. 50
Prop. 89	$\mathcal{A}$	S is an input, not state
Cor. 90	softmax	degradation bound
Cor. 85	softmax	$\mathbb{E}[\tau_n]$ non-increasing (stationary stimuli)
Thm. 56	softmax	$\exp(-2R^2)$ bound
Cor. 57	softmax	inherits from Thm. 56
Thm. 73	$\mathcal{A}_+$	inter-cluster consolidation
<i>Capture Inertness (Section 4.11)</i>		
Thm. 78	$\mathcal{A}_+$	order-preservation, hull geometry
Prop. 79	softmax	perturbative $K=0$ tracking
Cor. 80	$\mathcal{A}_+$	σ converges independently
Cor. 81	softmax	uses Thm. 50
Lem. 92	softmax	softmax Jacobian
Lem. 93	$\mathcal{A}_+$	bounded accumulation
Lem. 94	$\mathcal{A}$	convergent perturbation
<i>Appendix Results</i>		
Thm. 95	softmax	explicit softmax hypothesis
Prop. 97	$\mathcal{A}$	C^2 lower bound, telescope

Result	Level	Key property used
Prop. 98	$\mathcal{A}$	$\sigma \rightarrow \infty$
Prop. 99	$\mathcal{A}$	$\sigma \rightarrow 0$
Thm. 101	$\mathcal{A}_+$	contraction, Lyapunov
Cor. 103	$\mathcal{A}_+$	inherits from Thm. 101

Corollary 9 (Scale invariance of the capture decision; $\mathcal{A}$). *For any $\beta > 0$, the capture decision under query $q' = \beta q$ is identical to that under q : $q'^\top E(\tau) > \max_{k \in M} q'^\top m_k$ if and only if $q^\top E(\tau) > \max_{k \in M} q^\top m_k$.*

Proof. By Definition 6, $E(\tau)$ is captured iff $q^\top E(\tau) > \max_{k \in M} q^\top m_k$. Scaling q by $\beta > 0$: $\beta q^\top E(\tau) > \beta \max_{k \in M} q^\top m_k$ iff $q^\top E(\tau) > \max_{k \in M} q^\top m_k$, since $\beta > 0$ preserves the inequality. $\square$

2.3 Rigidity

The system is specified by structural choices: the simplex map $\varphi \in \mathcal{A}$, the blend rate $\lambda \in \mathcal{L}$, the embedding $E: X \rightarrow V$, the linear map K , and the scoring function S . Some of these are *invariances*: all results at the $\mathcal{A}$ level hold for every $\varphi \in \mathcal{A}$ and every $\lambda \in \mathcal{L}$, so these choices do not affect the qualitative dynamics. The remainder (E, K, S) define an *instance*: which tokens exist, how they are embedded, and what base scores they carry. Given any instance, zero numerical parameters remain. The accumulation scaling, capture threshold, and blend-rate boundary conditions are forced by consistency requirements; every other quantity is determined by the definitions.

Proposition 10 (Rigidity of the accumulation rate; $\mathcal{A}$). *Replace $t+1$ with an arbitrary scaling $f(t)$ in the accumulation rule: $C_{t+1} = C_t + a_t \cdot f(t)/C_t$. For $\sigma = C/N$ to converge to a nonzero finite constant under persistent observation ($a \geq \delta > 0$), $f(t) \sim t$ is necessary.*

Proof. Since $N = N_0 + t$, convergence of $\sigma = C/N$ to a nonzero constant requires $C = \Theta(t)$. The increment satisfies $C_{t+1}^2 - C_t^2 = 2a_t f(t) + a_t^2 f(t)^2 / C_t^2 \geq 2\delta f(t)$, so $C_t^2 \geq 2\delta \sum_{s=0}^{t-1} f(s)$. For $C = \Theta(t)$ we need $\sum_{s \leq t} f(s) = \Theta(t^2)$, which forces $f(t) = \Theta(t)$ under the convergence requirement (see Appendix A.3). If f grows slower (e.g. $f \equiv 1$), then $C = O(\sqrt{t})$ and $\sigma \rightarrow 0$. If f grows faster (e.g. $f(t) = t^2$), then $C = \Omega(t^{3/2})$ and $\sigma \rightarrow \infty$. The unique power law is $f(t) = \Theta(t)$. $\square$

Remark 11 (Non-degeneracy of the accumulation scaling). The two rejected scalings are not alternative design choices; both destroy the loop’s ability to function. If f is sublinear, then $\sigma \rightarrow 0$ under persistent observation: the salience of an actively attended element decays to zero, so the element becomes invisible to the observation mechanism. The loop’s own attention erases its own input. If f is superlinear, then $\sigma \rightarrow \infty$: a single element’s salience dominates all others indefinitely, and the softmax concentrates the entire attention budget on that element. The loop locks onto one input and never releases it. Both outcomes are self-defeating for a system that modifies its own attention using its own attention. The linear scaling $f(t) \sim t$ is the unique case in which persistent attention stabilizes salience rather than destroying or monopolizing it.

Lemma 12 (Idempotent capture; $\mathcal{A}$). *For a fixed query q and fixed M , capture applied twice yields the same result as capture applied once.*

Proof. After capture, every admitted $E(\tau)$ satisfies $q^\top E(\tau) > \max_{k \in M} q^\top m_k$ and is now in M' . So $\max_{k \in M'} q^\top m_k \geq q^\top E(\tau)$ for all admitted elements. A second application admits nothing: previously admitted elements already belong to M' , and previously rejected elements had $q^\top E(\tau) \leq \max_{k \in M} q^\top m_k < \max_{k \in M'} q^\top m_k$. $\square$

Proposition 13 (Rigidity of the capture threshold; $\mathcal{A}$). *Among parameter-free, always-defined, memoryless thresholds, max is the unique one consistent with idempotent capture (Lemma 12).*

Proof. Let $\theta(M, q)$ be any memoryless threshold. Idempotence (Lemma 12) requires that if $M' \supseteq M$ is the result of one capture pass, then a second pass with M' admits no new elements. This holds if and only if $\theta(M', q) \geq \theta(M, q)$ whenever $M' \supseteq M$, i.e. θ is non-decreasing under set enlargement.

Sufficiency of max: let $\theta(M, q) = \max_{k \in M} q^\top m_k$. If $M' = M \cup \{E(\tau)\}$ with $q^\top E(\tau) > \theta(M, q)$, then $\theta(M', q) = q^\top E(\tau) > \theta(M, q)$. More generally, $M' \supseteq M$ implies $\theta(M') \geq \theta(M)$, so idempotence holds.

Uniqueness: any function $\theta(M, q) = f(\{q^\top m_k : k \in M\})$ that is non-decreasing under set enlargement and depends only on the multiset of scores must satisfy $f(S \cup \{s\}) \geq f(S)$ for all S and all s . Among order statistics, the k -th largest is non-decreasing under enlargement for every fixed k , but it is undefined when $|M| < k$. Since $|M| = 1$ is a valid state, only $k = 1$ (the maximum) is always defined; any $k \geq 2$ additionally requires specifying k , a free parameter. Among non-order-statistic functions: the mean can decrease under enlargement (add an element with score below the mean), as can the median and any quantile. Any offset such as $\max - \epsilon$ or $\max + \epsilon$ introduces the parameter ϵ . Hence max is the unique threshold that is parameter-free, always defined, and non-decreasing under set enlargement. The set operation is union for the same reason: every alternative (replacement, deletion, pruning) requires either a parameter (which element to remove) or additional state (per-element retrieval counts, age tracking), so capture by union is the unique memoryless, parameter-free admission rule. $\square$

Proposition 14 (Necessity of each step; $\mathcal{A}$; items (iv)–(v) use $\mathcal{A}_+$ /softmax results). *Removing any single step from the loop causes at least one of the following to fail: boundedness ($\sigma = O(1)$), non-degeneracy ($a \geq \alpha > 0$), the feedback property (memory influences observation), or the forced capture–consolidation cycle.*

Proof. We verify each step by citing existing results.

(i) *Without OBSERVE.* The attention vector a is undefined; the query $q = aE$ (Definition 4), the accumulator update (Definition 3), and all downstream quantities are undefined. The loop is inoperative.

(ii) *Without ACCUMULATE.* The salience $\sigma = C/N$ does not exist. The scored input reduces to $S' = S + b$, eliminating the temporal record of persistent attention. Selective persistence (Theorem 21) fails: the system cannot distinguish persistently attended tokens from transient ones. Monotone pressure accumulation (Theorem 53) loses its temporal component.

(iii) *Without RETRIEVE.* The bias $b = 0$ permanently. The feedback loop $\ell \rightarrow b \rightarrow a \rightarrow q \rightarrow w \rightarrow \ell$ is broken: memory M has no influence on observation. Consolidation blend rates $\lambda_j = \lambda(w_j / \max w)$ are undefined (w does not exist). The carrying capacity argument (Theorem 50) is vacuous. The forced cycle (Corollary 87) does not arise.

(iv) *Without CONSOLIDATE.* By Lemma 25, $|M| \leq |X|$ without consolidation, and no row of M is ever modified. By Corollary 54, retrieval degradation during waking is permanent and irreversible. The Lyapunov contraction (Lemma 31) is vacuous.

(v) *Without CAPTURE.* M remains at its initial value. If $M_0 = \emptyset$, retrieval is permanently undefined. If M_0 is nonempty, consolidation contracts all impressions to a single prototype (Corollary 37) and the system converges to a degenerate fixed point. The carrying capacity (Theorem 50) does not arise. The forced cycle (Corollary 87) does not arise. $\square$

Remark 15 (Rigidity summary). Every component of the system falls into one of three categories.

- (a) *Forced.* The accumulation scaling $f(t) \sim t$ (Proposition 10), the capture threshold max (Proposition 13), the blend-rate boundaries $\lambda(0) = 1$, $\lambda(1) = 0$ (Proposition 7), the dot-product scoring $s_j = q^\top m_j$ (Proposition 16), and the feedback readout $E^\top$ (Proposition 16)

are uniquely determined by consistency requirements among linear, parameter-free maps. No alternative within this class exists.

- (b) *Invariant.* The simplex map $\varphi \in \mathcal{A}$, the blend-rate interior $\lambda \in \mathcal{L}$, the query map $q = aF$ for any R -bounded F (Proposition 16(a)), and the scoring rule $S' = h(S, \sigma, b)$ for any admissible h (Proposition 19) are free choices, but the results do not depend on which admissible choice is made. Replacing one admissible φ , λ , F , or h with another changes no theorem at its stated level. Additivity $h(S, \sigma, b) = S + \sigma + b$ is the canonical representative: it is the unique rule under which softmax attention ratios factorize into independent evidence contributions, but no theorem requires it.
- (c) *Instance-defining.* The embedding E , linear map K (Proposition 16(d)), consolidation target T_j (Proposition 17), and the choice between soft and hard retrieval (Proposition 16(c)) specify which system one is studying. The admissible class for each is characterized by the referenced propositions.

Each of the five steps is necessary (Proposition 14). The retrieval pipeline requires exactly four operations; no three-step pipeline suffices (Proposition 16). Given any instance, zero tunable numerical parameters remain: the dynamics are fully determined by the definitions. *Note on the dual role of E :* the embedding E appears in both the instance-defining list (its numeric specification) and the forced list (the feedback readout $E^\top$). These are distinct claims: the instance-defining aspect refers to the choice of embedding values (which tokens live in what space), while the forced aspect refers to the structural coupling (the feedback map must be the adjoint of the query embedding). Thus E has both content and structure: as a parameter it is chosen freely, but its coupling in the pipeline is uniquely determined.

Proposition 16 (Rigidity of the retrieval pipeline). *The retrieval pipeline*

$$q = aE, \quad w = \varphi(qM^\top), \quad \ell = wM, \quad b = EK\ell$$

is determined (up to the instance-defining parameter K) by the following requirements:

- (a) *The query map is invariant: for any linear map $F: X \rightarrow \mathbb{R}^d$ with $\|F(\tau)\| \leq R$ for all τ , the query $q = aF$ satisfies $\|q\| \leq R$; conversely, $\|aF\| \leq R$ for all $a \in \Delta^{|X|}$ forces $\|F(\tau)\| \leq R$ (take $a = \delta_\tau$). Thus the class of admissible query maps is exactly the R -bounded linear maps. E is the canonical zero-parameter representative. The bounded score gap $g \leq 2R^2$ (Theorem 50(iii)) requires only $\|q\| \leq R$.*
- (b) *The dot-product scoring $s_j = q^\top m_j$ is forced: the capture-argmax bound (Theorem 78) requires that s_j be linear in m_j , so that the maximum over $\text{conv}(E(X))$ is achieved at a vertex. Among parameter-free bilinear forms, $q^\top m_j$ is unique.*
- (c) *$\ell = wM$ is instance-defining: hard retrieval $\ell = m_v$ preserves every theorem except the quantitative dream-trace decay bound (Theorem 76(iii)), whose Jacobian chain requires a smooth readout.*
- (d) *E in the feedback $b = EK\ell$ is forced as the unique parameter-free map $\mathbb{R}^d \rightarrow \mathbb{R}^{|X|}$. K is instance-defining: $K = I$ preserves all theorems.*

No pipeline with fewer than four operations produces the same phenomena: the chain $\Delta^{|X|} \xrightarrow{q=aE} \mathbb{R}^d \xrightarrow{w=\varphi(qM^\top)} \Delta^{|M|} \xrightarrow{\ell} \mathbb{R}^d \xrightarrow{b=EK\ell} \mathbb{R}^{|X|}$ passes through four distinct spaces, and removing any link creates a dimension mismatch, eliminates retrieval weights, removes geometric feedback, or severs the memory–observation link.

Proof. (a) The bounded score gap (Theorem 50(iii)) requires $g \leq 2R^2$, which follows from $\|q\| \leq R$ and $\|m_j\| \leq R$. For any linear map F with $\|F(\tau)\| \leq R$ for all τ , the query $q = aF = \sum_{\tau} a(\tau)F(\tau)$ satisfies $\|q\| \leq R$ by the triangle inequality and convexity of the norm. Conversely, if $\|aF\| \leq R$ for every $a \in \Delta^{|X|}$, taking $a = \delta_{\tau}$ gives $\|F(\tau)\| \leq R$. Hence the admissible class is exactly the R -bounded linear maps $\{F : \|F(\tau)\| \leq R \forall \tau\}$. All downstream bounds ($\|q\| \leq R$, $|b(\tau)| \leq R^2\|K\|$, $q \in \text{conv}(F(X))$) hold for any such F , so the choice is invariant. E is the canonical representative that introduces no additional parameters.

(b) Theorem 78 proves: after capturing $E(\tau^*)$, familiarity satisfies $\max_k q^{\top} m_k \geq q^{\top} E(\tau^*) = \max_{\tau} q^{\top} E(\tau)$, so capture is inert while τ^* remains the E -argmax. The step $q^{\top} m_k \leq \max_{\tau} q^{\top} E(\tau)$ uses that $m_k \in \text{conv}(E(X))$ (hull invariance) and that $q^{\top}(\cdot)$ is a linear functional on a convex set, maximized at an extreme point. Nonlinear scoring (e.g. $-\|q - m_j\|^2$, concave in m_j , or $q^{\top} m_j / \|m_j\|$, rational) may achieve its maximum in the interior of $\text{conv}(E(X))$ rather than at a vertex, breaking the argument. Any bilinear form $q^{\top} W m_j$ with $W \neq I$ is equivalent to reparametrizing the embedding as $E' = WE$ when W is invertible. For singular W , $q^{\top} W m_j$ reduces to scoring in a lower-dimensional subspace, which is a degenerate case of the dot-product with a rank-deficient embedding. The capture-argmax argument (linearity over convex hulls) remains valid for all W , so $q^{\top} m_j$ uniquely represents the class of parameter-free bilinear forms up to reparametrization.

(c) The retrieval output ℓ appears in the system only through $b = EK\ell$. No theorem about the forced cycle references ℓ directly. The sole theorem requiring $\ell = wM$ is Theorem 76(iii): its Jacobian chain uses $\Delta\ell = \sum_j \Delta w_j(m_j - v)$ with $\sum_j \Delta w_j = 0$, bounded by $V(t)\sqrt{|M|}$, giving $\eta(t) \rightarrow 0$ as $V(t) \rightarrow 0$. Under hard retrieval $\ell = m_v$, the map $w \mapsto m_v$ is piecewise constant; the Jacobian chain breaks. All other theorems are unchanged.

(d) Since $b \in \mathbb{R}^{|X|}$ and $\ell \in \mathbb{R}^d$, a map $\mathbb{R}^d \rightarrow \mathbb{R}^{|X|}$ is needed. The embedding $E \in \mathbb{R}^{|X| \times d}$ provides this as the adjoint of the query embedding. Any alternative $F \neq E$ introduces a new parameter. The bounded-bias bound $\|b\|_{\infty} \leq R^2\|K\|$ (Corollary 26) holds for any K . Setting $K = I$ preserves every theorem with $\|K\| = 1$.

(Irreducibility.) The chain $\Delta^{|X|} \xrightarrow{q=aE} \mathbb{R}^d \xrightarrow{w=\varphi(qM^{\top})} \Delta^{|M|} \xrightarrow{\ell} \mathbb{R}^d \xrightarrow{b=EK\ell} \mathbb{R}^{|X|}$ passes through four distinct spaces. Removing any link: (i) without query, scoring is undefined (dimension mismatch); (ii) without scoring, no weights, no anchor, no consolidation; (iii) without output, feedback carries no memory geometry; (iv) without feedback, memory does not influence observation (Proposition 14(iii)). $\square$

Proposition 17 (Minimal conditions for the consolidation target). *Every theorem in the paper holds under the following conditions on the consolidation target T_j , and no theorem requires more:*

- (T1) **Convexity.** $T_j \in \text{conv}(M)$ for all j .
- (T2) **Anchor positivity.** $\alpha_v^{(j)} > 0$ for all $j \neq v$.
- (T3) **Full positivity.** $\alpha_k^{(j)} > 0$ for all k, j .
- (T4) **Continuity.** $(M, q) \mapsto T_j(M, q)$ is continuous.
- (T5) **Lipschitz.** $m_j \mapsto T_j$ is Lipschitz.

The specific form $T_j = \varphi((m_j + q)M^{\top})M$ satisfies (T1)–(T5) but is not uniquely determined by them. It is an instance-defining choice.

Proof. Each condition is used by at least one theorem:

(T1): Lemma 31 ($\|T_j - v\|^2 \leq \sum_k \alpha_k^{(j)} \|m_k - v\|^2$), Lemma 24 (hull invariance), Theorem 78 ($q^{\top} T_j \leq \max_k q^{\top} m_k$).

(T2): Lemma 31 (the $k = v$ term contributes zero with positive weight, giving strict contraction), Corollary 32 ($\gamma > 0$ requires $\alpha_v^{(j)} > 0$).

(T3): Theorem 52(i) (T_j lies strictly inside $\text{conv}(M)$, not at a vertex).

(T4): Corollary 37 (convergence: limit arguments), Theorem 73 (continuous dependence).

(T5): Theorem 61(i) ($\|m_j - m_{j'}\| < \varepsilon \implies \|T_j - T_{j'}\| = O(\varepsilon)$).

No proof uses the per-impression dependence on m_j , the additive structure $m_j + q$, the bilinear form $(\cdot)M^\top$, or the reuse of the retrieval φ . Concrete alternatives satisfying (T1)–(T5): the global retrieval target $T_j = \ell = \varphi(qM^\top)M$ (already used as a simplification in this paper), the direct anchor blend $T_j = (1 - \beta)m_j + \beta v$, self-attention without query bias $T_j = \varphi(m_j M^\top)M$, and the uniform average $T_j = \frac{1}{|M|} \sum_k m_k$. Neither idempotence at fixed points (trivially satisfied: $\text{conv}(\{v\}) = \{v\}$) nor oscillation compatibility (driven by the retrieval pipeline, not T_j) forces the specific form. $\square$

Definition 18 (Admissible scoring rule). A scoring rule $h: \mathbb{R}^3 \rightarrow \mathbb{R}$ is *admissible* if:

(W1) h is C^1 (hence maps bounded inputs to bounded outputs).

(W2) $h(0, \sigma, b)$ depends non-trivially on σ and b (sleep non-degeneracy).

(W3) $h(s, 0, 0)$ is non-degenerate in s (faithful initialization).

Proposition 19 (Invariance of the scoring rule; $\mathcal{A}$).

(a) Every theorem in the paper holds under any admissible $S'(\tau) = h(S(\tau), \sigma(\tau), b(\tau))$, with the same qualitative conclusions.

(b) Under softmax, $h(s, \sigma, b) = s + \sigma + b$ is the unique admissible rule (up to coordinatewise reparametrization) for which the attention ratio factorizes into independent evidence contributions:

$$\frac{a(\tau_1)}{a(\tau_2)} = F_S(S(\tau_1), S(\tau_2)) \cdot F_\sigma(\sigma(\tau_1), \sigma(\tau_2)) \cdot F_b(b(\tau_1), b(\tau_2)).$$

(c) The independent-evidence factorization is a modeling choice, not a consequence of the dynamics.

Proof. (a) Non-degeneracy (Theorem 27) requires $|S'| \leq B$; (W1) provides this (any C^1 function on bounded inputs is bounded). Sleep efficacy (Theorem 52) requires non-uniform attention during $S = 0$; (W2) provides this. The wake-up fixed point (Theorem 55(v)) requires continuity of $F(a) = \varphi(h(0, \sigma(a), b(a)))$; (W1) provides this. Dream-trace decay (Theorem 76(iii)): the contraction factor becomes $\eta(t) \leq \frac{1}{2} \|\partial h / \partial b\| \cdot \frac{1}{2} \|E\|^2 \|K\| \|M\| V(t) \sqrt{|M|}$, still $\rightarrow 0$ since $V(t) \rightarrow 0$ and $\|\partial h / \partial b\|$ is bounded by (W1). All remaining theorems depend on consolidation mechanics and the retrieval pipeline, not on h .

(b) Under softmax, $a(\tau_1)/a(\tau_2) = \exp(h(s_1, \sigma_1, b_1) - h(s_2, \sigma_2, b_2))$. Requiring this to factorize as $F_S \cdot F_\sigma \cdot F_b$ and taking logarithms, the Pexider–Aczél characterization gives $h(s, \sigma, b) = f_S(s) + f_\sigma(\sigma) + f_b(b) + c$. With inputs in log-score coordinates, $f_S = f_\sigma = f_b = \text{id}$ is the canonical choice.

(c) For $\varphi \in \mathcal{A}$ or $\mathcal{A}_+$, no universal multiplicative decomposition of the “attention ratio” exists; the argument in (b) is specific to the exponential family. One could use $h(s, \sigma, b) = s + \sigma + b + \sigma b$ (evidence interaction), and all theorems hold by (a). $\square$

2.4 Core Dynamics

With the structural constraints established—what is free and what is forced—we turn to the dynamics. The following results are the building blocks for the sleep theorems.

Corollary 20 (History-independence of the lifetime counter; $\mathcal{A}$). *For every $\tau \in X$ and every $t \geq 0$, $N_t(\tau) = N_0 + t$. The denominator of $\sigma = C/N$ is independent of the attention history $\{a_s(\tau)\}_{s < t}$.*

Proof. By Definition 3, $N_{t+1}(\tau) = N_t(\tau) + 1$ for all τ , unconditionally. By induction, $N_t(\tau) = N_0 + t$. $\square$

Theorem 21 (Selective persistence; $\mathcal{A}$). *Let $\tau \in X$.*

- (i) *If $a_t(\tau) = 0$ for all $t \geq T$, then $\sigma_t(\tau) \rightarrow 0$.*
- (ii) *If $a_t(\tau) \geq \delta > 0$ for all t , then $\liminf_{t \rightarrow \infty} \sigma_t(\tau) \geq \sqrt{\delta}$.*
- (iii) *For any attention sequence $(a_t(\tau))_{t \geq 0} \subseteq [0, 1]$, $\sigma_t(\tau) = O(1)$.*

Proof. (i) $C_t(\tau) = C_T(\tau)$ for $t \geq T$ since the update adds zero. $N_t(\tau) \rightarrow \infty$. $\sigma_t = C_T/N_t \rightarrow 0$.

(ii) From the recurrence $C_{t+1}^2 = C_t^2 + 2a_t(t+1) + a_t^2(t+1)^2/C_t^2 \geq C_t^2 + 2\delta(t+1)$. Summing: $C_t^2 \geq C_0^2 + \delta t(t+1)$. So $C_t \geq \sqrt{\delta t(t+1)}$. Since $N_t = N_0 + t$, we have $\sigma_t = C_t/N_t \geq \sqrt{\delta t(t+1)}/(N_0 + t) \rightarrow \sqrt{\delta}$.

(iii) Since $a \leq 1$, $C_{t+1}^2 \leq C_t^2 + 2(t+1) + (t+1)^2/C_t^2$. From (ii), $C_s = \Omega(s)$ for persistent elements, so the last term is $O(1)$ per step and $C_t = O(t)$. For non-persistent elements, $\sigma \rightarrow 0$ by (i). It remains to handle the general case of an arbitrary attention sequence $(a_t) \subseteq [0, 1]$ (e.g. intermittent attention that is neither eventually zero nor bounded away from zero).

The ratio $\sigma_t = C_t/N_t$ is self-regulating. The increment satisfies $C_{t+1} - C_t = a_t(t+1)/C_t$, so

$$\sigma_{t+1} - \sigma_t = \frac{C_{t+1} - C_t}{N_{t+1}} - \frac{C_t}{N_t N_{t+1}} = \frac{1}{N_{t+1}} \left(\frac{a_t(t+1)}{C_t} - \frac{C_t}{N_t} \right) = \frac{1}{N_{t+1}} \left(\frac{a_t(t+1)}{C_t} - \sigma_t \right).$$

When $\sigma_t > K$ for any fixed $K > 0$, we have $C_t = \sigma_t N_t > K N_t$, so $a_t(t+1)/C_t \leq (t+1)/(K N_t)$. Since $N_t = N_0 + t$, for t large enough $(t+1)/N_t \leq 2$, giving $a_t(t+1)/C_t \leq 2/K$. Meanwhile $\sigma_t > K$. Choosing $K > \sqrt{2}$ ensures the parenthesized term is negative for all large t , so $\sigma_{t+1} < \sigma_t$. Therefore σ_t cannot grow without bound, and $\sigma_t = O(1)$ for every attention sequence. $\square$

Lemma 22 (Retrieval-gated stability; $\mathcal{A}_+$). *Let $v = \arg \max_{k \in M} w_k$ at step t . Then v is unchanged by consolidation, and the capture threshold in v 's direction never decreases.*

Proof. $\lambda_v = 0$, so consolidation preserves v exactly. For any q' with $v = \arg \max_{k \in M_t} q'^\top m_k$,

$$\max_{k \in M_{t+1}} q'^\top m_k \geq \max_{k \in M_t} q'^\top m_k,$$

since $v \in M_{t+1}$ unchanged and $M_{t+1} \supseteq \{v\}$. $\square$

Corollary 23 (Monotone acquisition; $\mathcal{A}$). *$|M|$ is non-decreasing: capture adds elements by union (Definition 6) and no operation removes them. The capture threshold $\max_{k \in M} q^\top m_k$ is non-decreasing in $|M|$.*

Proof. Capture adds elements to M by union, so $|M|$ is non-decreasing. The threshold $\max_{k \in M} q^\top m_k$ is non-decreasing in $|M|$ since adding elements can only increase or maintain the maximum. $\square$

Lemma 24 (Convex hull invariance; $\mathcal{A}$). *For all t , $\text{conv}(M_t) \subseteq \text{conv}(E(X))$.*

Proof. At $t = 0$, $M_0 = \emptyset$ and the inclusion holds vacuously. Capture admits only vectors of the form $E(\tau)$, so after every capture step $M \subseteq \text{conv}(E(X))$. For consolidation, each row of the target $T_j = \sum_i [\varphi((M + q) M^\top)]_{ji} m_i$ is a convex combination of rows of M , so $T_j \in \text{conv}(M)$. Each updated row $M'_j = (1 - \lambda_j) m_j + \lambda_j T_j$ is a convex combination of two points in $\text{conv}(M)$, so $M'_j \in \text{conv}(M)$. By induction, $\text{conv}(M_t) \subseteq \text{conv}(E(X))$ for all t . $\square$

Lemma 25 (Finite memory without consolidation; $\mathcal{A}$). *If consolidation is disabled (i.e. $\lambda \equiv 0$), then $|M_t| \leq |X|$ for all t .*

Proof. Without consolidation, members of M are never modified, so M contains only vectors of the form $E(\tau)$ for $\tau \in X$. If $E(\tau) \in M_t$ already, then $\max_{k \in M_t} q^\top m_k \geq q^\top E(\tau)$, so the strict inequality in Definition 6 fails and $E(\tau)$ is not re-admitted. Each of the $|X|$ elements enters at most once. $\square$

Corollary 26 (Bounded retrieval bias; $\mathcal{A}$). *Let $R = \max_{\tau \in X} \|E(\tau)\|$. Then $|b(\tau)| \leq R^2 \|K\|_{\text{op}}$ for all τ and all t .*

Proof. By Lemma 24, every row of M lies in $\text{conv}(E(X))$, so $\|m_j\| \leq R$ for all j . Since $\ell = wM$ with w a convex weight vector, $\|\ell\| \leq R$. Then $|b(\tau)| = |E(\tau)^\top (K\ell)| \leq \|E(\tau)\| \cdot \|K\|_{\text{op}} \cdot \|\ell\| \leq R^2 \|K\|_{\text{op}}$. $\square$

Theorem 27 (Non-degeneracy; $\mathcal{A}_+$). *Let $\varphi \in \mathcal{A}_+$ and suppose $|S(\tau)| \leq B_S$ for all $\tau \in X$ and all t . Then there exists $\alpha > 0$, depending only on φ , B_S , C_0 , $|X|$, $R = \max_{\tau} \|E(\tau)\|$, and $\|K\|_{\text{op}}$, such that $a_t(\tau) \geq \alpha$ for all $\tau \in X$ and all $t \geq 0$.*

Proof. By Theorem 21(iii), there exists $\sigma_{\max}$ such that $|\sigma_t(\tau)| \leq \sigma_{\max}$ for all τ, t . By Corollary 26, $|b_t(\tau)| \leq R^2 \|K\|_{\text{op}}$ for all τ, t . Set $B = B_S + \sigma_{\max} + R^2 \|K\|_{\text{op}}$. Then $|S'_t(\tau)| \leq B$ for all τ, t . Since φ is a continuous positive map on the compact set $\{x \in \mathbb{R}^{|X|} : \|x\|_\infty \leq B\}$, it attains a positive minimum: $\alpha := \min_{\|x\|_\infty \leq B} \min_{\tau} \varphi(x)_\tau > 0$. For softmax, the explicit bound is $\alpha = \exp(-2B)/|X|$. The bound on $\sigma_{\max}$ uses only $a \leq 1$ (Theorem 21(iii)); the bound on b uses Lemma 24. Neither depends on α , so there is no circularity. $\square$

Corollary 28 (Retrieval output lies in convex hull; $\mathcal{A}$). *For any query q with $M \neq \emptyset$, the retrieval output $\ell = wM$ satisfies $\ell \in \text{conv}(M) \subseteq \text{conv}(E(X))$.*

Proof. By Definition 4, $w = \varphi(qM^\top) \in \Delta^{|M|}$, so $\ell = \sum_j w_j m_j$ is a convex combination of rows of M . Hence $\ell \in \text{conv}(M)$. By Lemma 24, $\text{conv}(M) \subseteq \text{conv}(E(X))$. $\square$

Corollary 29 (Query vector norm bound; $\mathcal{A}$). *For any step t , the query vector satisfies $\|q_t\| \leq R$, where $R = \max_{\tau \in X} \|E(\tau)\|$. Moreover, $q_t \in \text{conv}(E(X))$.*

Proof. By Definition 4, $q = aE = \sum_{\tau} a(\tau) E(\tau)$ with $a \in \Delta^{|X|}$, so q is a convex combination of the embedding vectors. Hence $q \in \text{conv}(E(X))$ and $\|q\| \leq \max_{\tau} \|E(\tau)\| = R$. $\square$

Corollary 30 (Bounds on the consolidation blend rate; $\mathcal{A}_+$). *Under any positive φ with $|M| \geq 2$ and $v = \arg \max_k w_k$ the unique maximizer of the retrieval weights, every blend rate satisfies $0 \leq \lambda_j \leq 1$, with $\lambda_v = 0$ for the anchor and $0 < \lambda_j \leq 1$ for all $j \neq v$. (Under the standing choice $\lambda(r) = 1 - r$, the strict bound $\lambda_j < 1$ holds; for a general admissible λ satisfying $\lambda(0) = 1$, the value $\lambda_j = 1$ is attainable when $w_j/w_v \rightarrow 0$. Since $\lambda_j = 1$ corresponds to full replacement $m'_j = T_j$, which strengthens rather than weakens consolidation, all downstream results requiring $\lambda_j > 0$ are unaffected.)*

Proof. By Definition 5, $\lambda_j = \lambda(w_j / \max_k w_k)$. Since $\lambda: [0, 1] \rightarrow [0, 1]$ with $\lambda(1) = 0$ and $\lambda(0) = 1$, we have $0 \leq \lambda_j \leq 1$. For the anchor $v = \arg \max_k w_k$: $\lambda_v = \lambda(1) = 0$. For $j \neq v$: since φ is positive, $w_j > 0$, so $w_j/w_v > 0$ and $\lambda_j \leq \lambda(0) = 1$. Since v is the unique maximizer, $w_j < w_v$, so $w_j/w_v < 1$ and $\lambda_j > \lambda(1) = 0$ (using that λ is decreasing on $(0, 1]$). Under the standing choice $\lambda(r) = 1 - r$, this gives $\lambda_j = 1 - w_j/w_v \in (0, 1)$; for a general admissible λ , the bound $\lambda_j \leq 1$ may be attained. $\square$

Lemma 31 (Lyapunov function for fixed-query consolidation; $\mathcal{A}_+$). *Let $\varphi \in \mathcal{A}_+$. Fix a query q and suppose capture is disabled. Let $v = \arg \max_{k \in M} w_k$ where $w = \varphi(qM^\top)$, and $V(t) = \max_{j \neq v} \|m_j - v\|$. Then $V(t+1) < V(t)$ whenever $V(t) > 0$.*

Proof. Since $\varphi \in \mathcal{A}_+$ is order-preserving, $v = \arg \max_k w_k = \arg \max_k q^\top m_k$. Since $\lambda_v = 0$, the anchor v is unchanged. For $j \neq v$, the consolidation target is $T_j = \sum_k \alpha_k m_k$ where $\alpha = \varphi((m_j + q) M^\top)$ has all entries positive (since φ is positive). Then $T_j - v = \sum_k \alpha_k (m_k - v)$, and since $\alpha_v > 0$ contributes the zero vector:

$$\|T_j - v\|^2 = \left\| \sum_k \alpha_k (m_k - v) \right\|^2 \leq \sum_k \alpha_k \|m_k - v\|^2 \leq V(t)^2$$

with strict inequality (the $k = v$ term contributes zero with positive weight). Since $\lambda_j > 0$ for $j \neq v$:

$$\|M'_j - v\| \leq (1 - \lambda_j) \|m_j - v\| + \lambda_j \|T_j - v\| < (1 - \lambda_j) V(t) + \lambda_j V(t) = V(t).$$

Taking the maximum over $j \neq v$ gives $V(t+1) < V(t)$. $\square$

Corollary 32 (State-dependent contraction rate; $\mathcal{A}_+$). *Under the hypotheses of Lemma 31, let $\alpha_v^{(j)}$ denote the weight that member j 's consolidation target places on the anchor v . Then*

$$V(t+1) \leq (1 - \gamma(t)) V(t), \quad \gamma(t) = \min_{j \neq v} \lambda_j (1 - \sqrt{1 - \alpha_v^{(j)}}).$$

Proof. From the proof of Lemma 31, $\|T_j - v\|^2 \leq \sum_k \alpha_k^{(j)} \|m_k - v\|^2 \leq (1 - \alpha_v^{(j)}) V(t)^2$, so $\|T_j - v\| \leq \sqrt{1 - \alpha_v^{(j)}} V(t)$. Then $\|M'_j - v\| \leq (1 - \lambda_j) V(t) + \lambda_j \sqrt{1 - \alpha_v^{(j)}} V(t) = (1 - \lambda_j (1 - \sqrt{1 - \alpha_v^{(j)}})) V(t)$. $\square$

Corollary 33 (Retrieval weight dilution (fan effect); softmax). *Let $\varphi = \text{softmax}$ and q be fixed. If $M' = M \cup \{m_{\text{new}}\}$ for some $m_{\text{new}} \notin M$, then for every $j \in M$:*

$$[\text{softmax}(q M'^\top)]_j < [\text{softmax}(q M^\top)]_j.$$

In particular, $\max_{j \in M} w_j$ is strictly decreasing as $|M|$ grows.

Proof. Let $s_j = q^\top m_j$. For $j \in M$:

$$[\text{softmax}(q M^\top)]_j = \frac{\exp(s_j)}{\sum_{k \in M} \exp(s_k)}.$$

Adding m_{new} with score $s_{\text{new}} = q^\top m_{\text{new}}$ increases the denominator by $\exp(s_{\text{new}}) > 0$ while the numerator is unchanged. Hence every existing weight strictly decreases. $\square$

Remark 34 (Fan effect). Anderson [1974] demonstrated that retrieval latency for a proposition increases with the number of propositions associated with its subject. Corollary 33 produces the same structural form: each additional impression in M strictly decreases every existing retrieval weight, so retrieval concentration degrades monotonically with memory size.

Corollary 35 (Retrieval weight bounds; softmax). *Under softmax with $|M| = n$, the maximum retrieval weight for any single impression satisfies $1/n \leq \max_j w_j \leq 1/(1 + (n-1) \exp(-g))$, where $g = \max_j q^\top m_j - \max_{k \neq j} q^\top m_k$ is the score gap. The upper bound requires score gap $g = \Omega(\log n)$ to remain bounded away from zero.*

Proof. Under uniform scores ($g = 0$), softmax assigns $w_j = 1/n$ for all j . For score gap $g > 0$, the leading item has $w_1 = \exp(s_1)/(\exp(s_1) + \sum_{k \neq 1} \exp(s_k)) \leq 1/(1 + (n-1) \exp(-g))$. Setting this equal to a constant $c > 0$: $(n-1) \exp(-g) \leq (1-c)/c$, so $g \geq \log((n-1)c/(1-c)) = \Omega(\log n)$. $\square$

Corollary 36 (Attention budget constraint; $\mathcal{A}$). *Under any simplex map φ , $\sum_{\tau \in X} a_t(\tau) = 1$ at every step. For $|X| = n$ elements receiving nonzero attention, $\bar{a} = 1/n$, and increasing n strictly decreases the mean attention per element.*

Proof. $\varphi: \mathbb{R}^n \rightarrow \Delta^n$ maps to the simplex, so $\sum_{\tau} a_t(\tau) = 1$ by definition. The mean is $1/n$. Adding an element changes the mean to $1/(n+1) < 1/n$, a strict decrease. $\square$

Corollary 37 (Convergence under repeated consolidation without capture; $\mathcal{A}_+$). *Let φ be a continuous positive simplex map. Suppose capture is disabled and the query q is fixed. Then $V(t) \rightarrow 0$: all impressions converge to the anchor v .*

Proof. By Lemma 31, $V(t+1) < V(t)$ whenever $V(t) > 0$. Since $V(t)$ is non-negative, it converges to some $L \geq 0$. Suppose $L > 0$. The trajectory of M is confined to the compact set $\text{conv}(E(X))$ (Lemma 24), so M has a convergent subsequence; since $V(t)$ is strictly decreasing and bounded, M in fact converges to a limit configuration M^* with all items at distance exactly L or less from v .

At M^* , the query q is fixed, so the scores $q^\top m_j^*$ are well-defined. Since q is fixed and $v = \arg \max_k q^\top m_k$, the score $q^\top v > q^\top m_j^*$ for all $j \neq v$ with $m_j^* \neq v$ (if $q^\top m_j^* = q^\top v$ for some $j \neq v$, then m_j^* lies in the hyperplane $q^\top x = q^\top v$, which is a strict subspace since $m_j^* \neq v$; but then the consolidation target $T_j^* = \sum_k \alpha_k^{(j)} m_k^*$ satisfies $q^\top T_j^* < q^\top v$ because T_j^* is a strict convex combination with positive weight on points scoring below $q^\top v$, so after one consolidation step, $q^\top m_j' = (1 - \lambda_j) q^\top m_j^* + \lambda_j q^\top T_j^* < q^\top v$; this is impossible at a limit configuration). Therefore at the limit, $w_j < w_v$ for all $j \neq v$, giving $\lambda_j > 0$. Since $w = \varphi(q M^\top)$ depends continuously on M and the set of configurations near M^* is compact (contained in $\text{conv}(E(X))$), the strict inequalities $w_j < w_v$ yield a uniform gap: there exists $\delta > 0$ such that $w_j(t) \leq (1 - \delta) w_v(t)$ for all sufficiently large t . Combined with $\alpha_v^{(j)} > 0$ (positivity of φ), Corollary 32 gives $\gamma_L > 0$, so $V(t+1) \leq (1 - \gamma_L) V(t)$ for sufficiently large t . This forces $V(t) \rightarrow 0$, contradicting $L > 0$. Hence $L = 0$. $\square$

Corollary 38 (Vanishing contraction rate; $\mathcal{A}_+$). *Under the hypotheses of Corollary 37 (fixed query, no capture), the contraction rate $\gamma(t)$ satisfies $\gamma(t) \rightarrow 0$ as $t \rightarrow \infty$.*

Proof. By Corollary 37, $V(t) \rightarrow 0$, so $m_j \rightarrow v$ for all j . Since $w_j = [\varphi(q M^\top)]_j$ and $q^\top m_j \rightarrow q^\top v$ for all j , the retrieval weights equalize (by continuity of φ): $w_j/w_v \rightarrow 1$, hence $\lambda_j = 1 - w_j/w_v \rightarrow 0$. The factor $1 - \sqrt{1 - \alpha_v^{(j)}}$ remains bounded in $[0, 1]$, so $\gamma(t) = \min_{j \neq v} \lambda_j (1 - \sqrt{1 - \alpha_v^{(j)}}) \rightarrow 0$. $\square$

Corollary 39 (Prototype formation; $\mathcal{A}_+$). *Under the hypotheses of Corollary 37, as $t \rightarrow \infty$ every row of M converges to the anchor $v = \arg \max_{k \in M} q^\top m_k$.*

Proof. By Corollary 37, $V(t) = \max_{j \neq v} \|m_j - v\| \rightarrow 0$, so $m_j \rightarrow v$ for all j . $\square$

Corollary 40 (Context variation prevents convergence to single prototype; $\mathcal{A}_+$). *Let φ be a continuous positive simplex map and $|M| \geq 2$ with at least two distinct impressions. Under fixed query q , $V(t) \rightarrow 0$ (Corollary 37). Let T_{rot} be the timescale on which the query varies (i.e. the anchor identity changes), and let $\gamma(t) = \min_{j \neq v} \lambda_j (1 - \sqrt{1 - \alpha_v^{(j)}})$ be the contraction rate (Corollary 32). Suppose $\gamma_{\min} := \min_{0 \leq t \leq T_{\text{rot}}} \gamma(t) > 0$ (which holds whenever distinct impressions have distinct scores under the query throughout $[0, T_{\text{rot}}]$). If $T_{\text{rot}} < 1/\gamma_{\min}$, the anchor changes before V reaches zero, and distinct impressions are preserved.*

Proof. By Corollary 37, fixed-query consolidation drives $V \rightarrow 0$. By Corollary 32, $V(t+1) \leq (1 - \gamma(t)) V(t)$, so after k steps under a fixed anchor, $V \leq (1 - \gamma_{\min})^k V(0)$. Under a fixed anchor, V decreases monotonically. When the anchor changes, however, V is redefined with respect to

the new anchor v' : $V' = \max_{j \neq v'} \|m_j - v'\|$. If $v' \neq v$, then V' may exceed V because the old anchor v (which had $\|m_j - v\|$ small for all j) is now measured from a different reference point. Specifically, $\|v - v'\| > 0$ whenever $v \neq v'$, so $V' \geq \|v - v'\| > 0$ even if the pre-rotation V was small. When $T_{\text{rot}} < 1/\gamma_{\text{min}}$, each anchor period reduces V by a bounded factor, and the subsequent anchor change resets V' to at least $\|v - v'\|$. This prevents V from converging to zero across rotations, provided the anchor set contains at least two distinct points that are not driven together. $\square$

3 Cognitive Structure

The accumulation and consolidation rules have further consequences. Three properties follow from the definitions; their structural forms appear in empirical data. None were design targets; the results hold for every admissible simplex map, blend rate, and embedding. We do not understand why the resemblance appears.

3.1 Hyperbolic Forgetting

Corollary 41 (Hyperbolic decay; $\mathcal{A}$). *If $a_t(\tau) = 0$ for all $t \geq T$, then*

$$\sigma_t(\tau) = \frac{C_T(\tau)}{N_0 + t},$$

where $C_T(\tau)$ is the accumulated encoding strength at cessation. The decay is hyperbolic in t with rate constant determined by the encoding history.

Proof. When $a_t(\tau) = 0$, the accumulation increment vanishes: $C_{t+1}(\tau) = C_t(\tau) + 0 \cdot (t + 1)/C_t(\tau) = C_t(\tau)$. By induction, $C_t(\tau) = C_T(\tau)$ for all $t \geq T$. The lifetime counter grows as $N_t = N_0 + t$. Division gives $\sigma_t(\tau) = C_T(\tau)/(N_0 + t)$. $\square$

Remark 42 (Forgetting curve). Whether the forgetting curve is exponential or hyperbolic was debated for over a century after Ebbinghaus [1885]. Rubin and Wenzel [1996] tested 105 functional forms against 210 data sets and found that power-law (hyperbolic) forms consistently fit retention data best. The loop forces power-law decay, not exponential; this is the distinction debated in the literature. The specific exponent depends on the mapping from salience to behavioral measurement, which is outside the object. The power-law form itself is forced by the self-damping rigidity (Proposition 10).

3.2 Spacing Effect

Proposition 43 (Optimal spacing interval; $\mathcal{A}$). *Fix $n \geq 2$ observations of element τ at intensity $\delta > 0$, spaced K steps apart (observations at $t = 0, K, 2K, \dots, (n-1)K$), and fix a retention interval $R \geq 1$ after the last observation. The salience at evaluation time $T = (n-1)K + R$ is*

$$\sigma(K) = \frac{\sqrt{C_0^2 + \delta K n(n-1) + 2\delta n}}{N_0 + (n-1)K + R}.$$

For large n and K , $\sigma(K) \approx n\sqrt{\delta K}/((n-1)K + R)$, which is maximized at

$$K^* = \frac{R}{n-1}.$$

The benefit curve is an inverted U : σ increases for $K < K^*$ and decreases for $K > K^*$.

Proof. Massed vs. spaced accumulation. From the recurrence $C_{t+1}^2 = C_t^2 + 2a_t(t+1) + a_t^2(t+1)^2/C_t^2$, the dominant contribution is $2a_t(t+1)$ (since $C_t = \Theta(t)$ for persistent elements by Theorem 21).

Under massed observation ($K = 1$, observations at $t = 0, \dots, n-1$), the sum of increments is $2\delta \sum_{s=0}^{n-1} (s+1) = \delta n(n+1)$, giving $C_n^2 = C_0^2 + \delta n(n+1) + O(n)$.

Under spaced observation (interval K), observations occur at $t = 0, K, \dots, (n-1)K$, contributing $2\delta \sum_{j=0}^{n-1} (jK+1) = 2\delta(Kn(n-1)/2 + n) = \delta Kn(n-1) + 2\delta n$. Hence $C_{(n-1)K}^2 = C_0^2 + \delta Kn(n-1) + 2\delta n + O(n)$.

Inverted U. During the retention interval $[(n-1)K, T]$, no observation occurs, so C is constant at $C_{(n-1)K}$. Meanwhile $N_T = N_0 + (n-1)K + R$. Dropping lower-order terms:

$$\sigma(K) \approx \frac{n\sqrt{\delta K}}{(n-1)K + R} = n\sqrt{\delta} \cdot f(K), \quad \text{where } f(K) = \frac{\sqrt{K}}{(n-1)K + R}.$$

Set $m = n-1$. Differentiating:

$$f'(K) = \frac{R - mK}{2\sqrt{K}(mK + R)^2}.$$

Since $f'(K) > 0$ for $K < R/m$ and $f'(K) < 0$ for $K > R/m$, the unique maximum is at $K^* = R/m = R/(n-1)$. $\square$

Corollary 44 (Optimal spacing scales with retention interval; $\mathcal{A}$). *The optimal inter-study interval $K^* = R/(n-1)$ is linearly increasing in the retention interval R . Longer retention requires wider spacing.*

Proof. $\partial K^*/\partial R = 1/(n-1) > 0$. $\square$

Remark 45 (Spacing effect). Cepeda et al. [2006] found in a meta-analysis of 317 experiments that the optimal inter-study interval increases with the retention interval, and that the benefit of spacing over massing follows an inverted- U curve. The loop produces both the inverted U and the linear scaling $K^* \propto R$ from the accumulation rule alone; both are forced by the accumulation rigidity (Proposition 10).

3.3 Testing Effect

Corollary 46 (Selective preservation under retrieval; $\mathcal{A}_+$). *Let φ be a positive simplex map with $|M| \geq 2$ and distinct impressions. Retrieval of the anchor $v = \arg \max_k w_k$ preserves v exactly ($\lambda_v = 0$) and displaces every competitor: $\lambda_j > 0$ for all $j \neq v$.*

Proof. By the blend rate boundary condition, $\lambda_v = \lambda(w_v/\max_k w_k) = \lambda(1) = 0$. For $j \neq v$, positivity of φ gives $w_j > 0$ and distinctness gives $w_j < w_v$, so $w_j/w_v \in (0, 1)$ and $\lambda_j = \lambda(w_j/w_v) > 0$ (since $\lambda(r) > 0$ on $[0, 1)$). Thus the retrieved impression is unchanged while every other impression is displaced by $\lambda_j(T_j - m_j)$. $\square$

Remark 47 (Testing effect). Roediger and Karpicke [2006] demonstrated that actively retrieving information produces better long-term retention than passive re-study. In the loop, retrieval and consolidation share the same weight vector $w = \varphi(qM^\top)$: the act of retrieval is the act of selective preservation. The boundary condition $\lambda(1) = 0$ that makes this work is forced (Proposition 7).

Corollary 48 (Retrieval advantage grows with delay; $\mathcal{A}_+$). *Compare two interventions at time T : (a) re-observation of element τ , and (b) retrieval of impression v with τ as the anchor.*

- (i) *Re-observation increases $C(\tau)$ by at most 1, so the salience gain is $\Delta\sigma \leq 1/(N_0 + T)$, which decays as $O(1/t)$ at evaluation time $t > T$.*

- (ii) *Retrieval contracts the Lyapunov function: $V' < V$ (Lemma 31). The contraction is permanent and independent of the evaluation delay.*

At sufficiently long delays, the permanent contraction dominates the transient salience boost.

Proof. (i) By Definition 3, a single observation adds at most $a \cdot f/C \leq 1$ to $C(\tau)$. Since $N_t = N_0 + t$ and no further observations occur, $\sigma_t(\tau) = C_T(\tau)/(N_0 + t)$ for $t > T$ (Corollary 41). The salience boost $\Delta C/(N_0 + t) \leq 1/(N_0 + t) \rightarrow 0$.

(ii) By Corollary 46, retrieval preserves the anchor ($\lambda_v = 0$) and displaces all competitors ($\lambda_j > 0$). By Lemma 31, $V(M') < V(M)$. Since V depends only on M and not on t , the reduction persists at any evaluation delay.

As $t \rightarrow \infty$, the salience advantage of (a) vanishes while the structural advantage of (b) does not. $\square$

Remark 49 (Crossover). Roediger and Karpicke [2006] found that retrieval practice underperforms re-study at short delays but overtakes it at longer delays. The loop produces this crossover: re-study boosts σ (transient, $O(1/t)$); retrieval contracts V (permanent).

4 Sleep Pressure and Recovery

The loop is forced into alternating regimes. There exists a critical memory size $|M^*|$ beyond which retrieval precision is irrecoverably degraded during waking, so that retrieval can be restored only by entering an offline consolidation phase ($S = 0$, capture disabled). The regimes exhibit monotone phase structure, self-observable pressure signals, multi-cluster oscillation, compounding debt under truncation, and a forced quasi-periodic cycle.

Throughout this section, we use terms such as “sleep,” “waking,” “dreams,” and “naps” as convenient labels for the mathematical regimes described. Parallels to biological phenomena are noted in remarks; the loop was not designed to model sleep, but the mathematics forces alternating capture-consolidation regimes whose structural form resembles the sleep-wake cycle. The resemblance is discussed in Section 5.

4.1 Retrieval Degradation

Theorem 50 (Retrieval degradation under sustained waking; softmax). *Let $\varphi = \text{softmax}$ and let the system operate with both capture and consolidation enabled. Let $R = \max_{\tau \in X} \|E(\tau)\|$ and $\|K\|_{\text{op}} \leq 1$. Then:*

- (i) *Every capture event strictly decreases all existing retrieval weights: if $M' = M \cup \{m_{\text{new}}\}$, then $[\text{softmax}(qM'^T)]_j < [\text{softmax}(qM^T)]_j$ for all $j \in M$.*
- (ii) *The score gap required for a single impression to achieve retrieval weight $w_v \geq c > 0$ is $g \geq \log((|M| - 1)c/(1 - c))$, i.e. $g = \Omega(\log |M|)$.*
- (iii) *The achievable score gap is uniformly bounded: $g \leq 2R^2$.*
- (iv) *There exists a critical memory size*

$$|M^*| = 1 + \frac{1 - c}{c} \exp(2R^2)$$

beyond which no single impression can achieve retrieval weight $w_v \geq c$ for any query.

- (v) *Waking consolidation cannot prevent this degradation: the self-limiting property $\gamma(t) \rightarrow 0$ as impressions equalize (Corollary 38) means consolidation decelerates precisely when it is most needed.*

Proof. (i) This is exactly Corollary 33. Adding m_{new} to M increases the softmax denominator by $\exp(q^\top m_{\text{new}}) > 0$ while leaving each numerator $\exp(q^\top m_j)$ unchanged, so every existing weight strictly decreases.

(ii) By Corollary 35, the maximum retrieval weight under softmax with $|M| = n$ and score gap g satisfies

$$w_v \leq \frac{1}{1 + (n-1)\exp(-g)}.$$

Setting $w_v \geq c$ and solving: $(n-1)\exp(-g) \leq (1-c)/c$, so $g \geq \log((n-1)c/(1-c)) = \Omega(\log n)$.

(iii) The score gap is $g = q^\top m_v - \max_{k \neq v} q^\top m_k$. By Corollary 29, $\|q\| \leq R$ since $q = aE$ is a convex combination of embeddings. By Lemma 24, every $m_j \in \text{conv}(E(X))$, so $\|m_j\| \leq R$. Therefore $|q^\top m_j| \leq \|q\| \cdot \|m_j\| \leq R^2$ for all j . The maximum possible score gap is thus $g \leq R^2 - (-R^2) = 2R^2$.

In the full system, the score entering retrieval is $q^\top m_j$ where $q = aE$ depends on $S' = S + \sigma + b$. However, q is always a convex combination of embeddings (by Corollary 36), so $\|q\| \leq R$ regardless of S' . The retrieval bias satisfies $\|b\| \leq R^2 \|K\|_{\text{op}}$ (Corollary 26), but this affects q only through a , which remains on the simplex. The score gap for retrieval is determined by $q^\top m_j$ alone, and $g \leq 2R^2$ holds uniformly.

(iv) Combining (ii) and (iii): for $w_v \geq c$ we need $g \geq \log((|M|-1)c/(1-c))$, but $g \leq 2R^2$. The condition fails when $\log((|M|-1)c/(1-c)) > 2R^2$, i.e. when

$$|M| > 1 + \frac{1-c}{c} \exp(2R^2) =: |M^*|.$$

For $|M| > |M^*|$, no query can achieve $w_v \geq c$ for any single impression. The bound $|M^*|$ is finite and depends only on c and R . Since $|M|$ is non-decreasing during waking (Corollary 23), the system inevitably crosses $|M^*|$ if it continues to capture.

(v) Could consolidation prevent this by reducing the effective size of M ? During waking, consolidation operates under a query q that shifts at every step. The contraction rate $\gamma(t)$ from Corollary 32 satisfies $\gamma(t) = \min_{j \neq v} \lambda_j (1 - \sqrt{1 - \alpha_v^{(j)}})$. As impressions converge under consolidation, $w_j/w_v \rightarrow 1$, so $\lambda_j = 1 - w_j/w_v \rightarrow 0$ and $\gamma(t) \rightarrow 0$ (Corollary 38). Consolidation is self-limiting: it decelerates as impressions become similar. Moreover, during waking the query varies, and by Corollary 40, if the anchor rotates faster than $1/\gamma$, distinct impressions are preserved. Consolidation during waking therefore cannot reduce $|M|$ to below $|M^*|$: it neither removes rows from M nor merges them exactly (it moves them closer but $\gamma \rightarrow 0$ before convergence completes, and anchor rotation protects different members at different times). $\square$

Remark 51 (Sleep pressure is a capacity phenomenon). The degradation in Theorem 50 is not caused by any single mechanism but by the interaction of three: (1) capture grows $|M|$ monotonically; (2) the fan effect (Corollary 33) dilutes all weights at each capture event; (3) the bounded score gap prevents the system from overcoming the dilution. The bound $|M^*|$ is the carrying capacity of the retrieval mechanism under a given embedding. For unit-norm embeddings ($R = 1$), $|M^*| = 1 + ((1-c)/c)e^2$; at $c = 0.5$ (the majority retrieval threshold: the anchor outweighs all competitors combined) this gives $|M^*| = 1 + e^2 \approx 8.4$. For small c , $|M^*| = 1 + ((1-c)/c)e^2 = e^2/c - (e^2 - 1) \leq e^2/c$.

4.2 Sleep Restores Retrieval

Theorem 52 (Sleep restores retrieval precision; $\mathcal{A}_+$). *Let the system enter a “sleep phase”: $S = 0$ with capture disabled. Let φ be a continuous positive simplex map, $|M| \geq 2$. Then:*

(i) *$V(t)$ is strictly decreasing: $V(t+1) < V(t)$ whenever $V(t) > 0$, and $V(t) \rightarrow 0$.*

- (ii) Consolidation reduces the effective memory size through prototype formation: if $\|m_i - m_j\| < \epsilon$ for impressions i, j after consolidation, they are functionally merged (their retrieval weights under any query differ by at most $\|q\|\epsilon$).
- (iii) After sleep, the effective number of distinguishable impressions decreases. If M has consolidated into p clusters of diameter $< \epsilon$ separated by inter-cluster gap $g_p > 0$, then under any $\varphi \in \mathcal{A}_+$, the anchor weight w_v increases as p decreases (fewer competitors) and g_p increases (sharper discrimination). Under softmax with inverse temperature $\beta > 0$ (where $\beta = 1$ in the standard form of Definition 4), $w_v \geq 1/(1 + (p-1)\exp(-\beta g_p))$. Since $p \leq |M|$ and consolidation reduces p , sleep improves retrieval concentration.
- (iv) The contraction rate $\gamma(t) \rightarrow 0$ provides a natural termination condition for sleep.

Proof. (i) During the $S = 0$ phase with capture disabled, the query is $q_t = a_t E$ where $a_t = \varphi(\sigma_t + b_t)$. The query evolves because σ and b do. We must show $V(t)$ is strictly decreasing despite the varying query.

At each step t , the consolidation update is $m'_j = (1 - \lambda_j)m_j + \lambda_j \ell$ where $\ell = w_t M_t$ is the global retrieval output (used here for intuition; the full per-impression formulation T_j of Definition 5 is treated below starting at the diameter argument) and $\lambda_j = \lambda(w_j^{(t)} / \max_k w_k^{(t)})$. The anchor at step t is $v_t = \arg \max_k w_k^{(t)}$, which has $\lambda_{v_t} = 0$ and is preserved exactly: $v'_t = v_t$. For $j \neq v_t$, since φ is positive, $w_j > 0$ and the blend rate satisfies $\lambda_j > 0$. The target ℓ is a convex combination of M with positive weight $w_{v_t} > 0$ on the anchor, so $\|\ell - v_t\| < \max_k \|m_k - v_t\| = V(t)$ strictly. Then:

$$\|m'_j - v_t\| = \|(1 - \lambda_j)(m_j - v_t) + \lambda_j(\ell - v_t)\| \leq (1 - \lambda_j)\|m_j - v_t\| + \lambda_j\|\ell - v_t\| < \|m_j - v_t\|.$$

This holds for every $j \neq v_t$ at step t , using the query and anchor of that step. Define $V_t^{(v_t)} := \max_{j \neq v_t} \|m_j^{(t)} - v_t\|$. The above gives $V_{t+1}^{(v_t)} < V_t^{(v_t)}$ (measuring from the same anchor v_t , which is preserved).

To handle the varying anchor, consider the diameter $D(t) := \max_{i,j} \|m_i^{(t)} - m_j^{(t)}\|$, which is anchor-independent. We show $D(t+1) < D(t)$ whenever $D(t) > 0$. Recall the per-impression update: $m'_j = (1 - \lambda_j)m_j + \lambda_j T_j$ where $T_j = \sum_k \alpha_k^{(j)} m_k$ with $\alpha_k^{(j)} = \varphi_k((m_j + q_t)M_t^\top) > 0$ (since φ is positive), and $\lambda_{v_t} = 0$ for the anchor.

Key observation: since $T_j \in \text{conv}(M_t)$ and $\alpha_{v_t}^{(j)} > 0$ for every j , each target T_j lies strictly inside $\text{conv}(M_t)$ (it is not a vertex). Hence every non-anchor m'_j is a strict convex combination of m_j and an interior point of $\text{conv}(M_t)$, which places m'_j strictly inside $\text{conv}(M_t)$.

Now let m_i^*, m_j^* achieve the diameter $D(t)$ at step t . At least one of them is a non-anchor (at most one can be the anchor). Consider two cases:

- If one is the anchor v_t and the other is m_j ($j \neq v_t$): v_t is preserved exactly. Since T_j is strictly interior to $\text{conv}(M_t)$, we have $\|T_j - v_t\| < D(t)$, so $\|m'_j - v_t\| \leq (1 - \lambda_j)\|m_j - v_t\| + \lambda_j\|T_j - v_t\| < D(t)$.
- If neither is the anchor: both m_i^* and m_j^* lie strictly inside $\text{conv}(M_t)$ (as shown above, each is a strict convex combination of its original position and a strictly interior target). Neither is a vertex of $\text{conv}(M_t)$.

In both cases, $\text{conv}(M'_t) \subseteq \text{conv}(M_t)$ (the anchor is preserved, all others move inward). The diameter of a convex polytope is achieved at its vertices. The vertices of $\text{conv}(M'_t)$ are a subset of $\{m'_1, \dots, m'_{|M|}\}$. The anchor v_t is preserved; every other m'_j lies strictly inside $\text{conv}(M_t)$. Hence every diameter-achieving pair of $\text{conv}(M'_t)$ has at least one endpoint strictly interior to $\text{conv}(M_t)$, and such a point cannot achieve the diameter of $\text{conv}(M_t)$. Therefore $D(t+1) < D(t)$.

Since $D(t)$ is strictly decreasing and bounded below by 0, it converges to some $L \geq 0$. Suppose for contradiction that $L > 0$.

Compactness argument. The configuration space $\mathcal{M} := \{(m_1, \dots, m_{|M|}) : m_j \in \text{conv}(M_0)\}$ (where M_0 is the initial memory) is compact, since $\text{conv}(M_0) \subset \mathbb{R}^d$ is compact and we take a finite product. The consolidation map preserves $\text{conv}(M_0)$: the anchor is preserved exactly, and every non-anchor update $m'_j = (1 - \lambda_j)m_j + \lambda_j T_j$ is a convex combination within $\text{conv}(M_0)$. Hence every $M_t \in \mathcal{M}$. During sleep ($S = 0$, capture disabled), the query $q_t = a_t E$ where $a_t = \varphi(\sigma_t + b_t)$ depends on the current configuration through $\ell_t = w_t M_t$ and $b_t = EK\ell_t$. Since a_t lies in the probability simplex and E has bounded rows, q_t lies in a compact set $Q := \{aE : a \in \Delta^{|E|-1}\} \subset \mathbb{R}^d$. Define the per-step diameter reduction as a function of the full state:

$$\delta(M, q) := D(M) - D(M'),$$

where M' is the configuration after one consolidation step from M under query q . The map $(M, q) \mapsto M'$ is continuous (it is a composition of the continuous positive simplex map φ , the weight and blend-rate computations, and the convex update rule), and the diameter D is continuous, so δ is continuous on the compact product $\mathcal{M} \times Q$. We showed above that $\delta(M, q) > 0$ whenever $D(M) > 0$: positivity of φ ensures all retrieval weights $w_k > 0$, hence all blend rates $\lambda_k > 0$ for $k \neq v$, and the target T_j is strictly interior to $\text{conv}(M)$ (since $\alpha_v^{(j)} > 0$), so the diameter strictly decreases.

Now restrict to the compact sublevel set $\mathcal{S}_L := \{(M, q) \in \mathcal{M} \times Q : D(M) \geq L\}$. On $\mathcal{S}_L$, δ is continuous and strictly positive, so it achieves its minimum $c(L) := \min_{(M, q) \in \mathcal{S}_L} \delta(M, q) > 0$. Since $D(t) \geq L$ for all t by assumption, we have $D(t) - D(t+1) \geq c(L) > 0$ at every step, giving $D(t) \leq D(0) - tc(L) \rightarrow -\infty$, contradicting $D(t) \geq 0$. Therefore $L = 0$ and $D(t) \rightarrow 0$. Since $V(t) \leq D(t)$ for any choice of anchor, we conclude $V(t) \rightarrow 0$.

(ii) By part (i) ($D(t) \rightarrow 0$ under varying query), all impressions converge to the anchor v . When $\|m_i - m_j\| < \epsilon$, the retrieval score difference satisfies $|q^\top m_i - q^\top m_j| \leq \|q\| \cdot \|m_i - m_j\| < \|q\|\epsilon$ by Cauchy–Schwarz. Under softmax, their retrieval weights satisfy $|w_i - w_j| \leq w_i |e^{\|q\|\epsilon} - 1| < w_i \|q\|\epsilon \cdot e^{\|q\|\epsilon}$ for small ϵ . In the limit $\epsilon \rightarrow 0$, the two impressions are indistinguishable to the retrieval mechanism. The effective memory size is therefore the number of ϵ -separated clusters in M , which decreases under consolidation.

(iii) After sleep, suppose M has consolidated into p clusters $\mathcal{C}_1, \dots, \mathcal{C}_p$ with intra-cluster diameter $< \epsilon$ and inter-cluster gap $g_p := \min_{k \neq k'} |q^\top c_k - q^\top c_{k'}| > 0$ (where c_k is the centroid of $\mathcal{C}_k$). Within each cluster, all impressions score within $\|q\|\epsilon$ of each other (by Cauchy–Schwarz). Under any $\varphi \in \mathcal{A}_+$, reducing p while increasing g_p increases w_v (the anchor sees fewer, more separated competitors). Under softmax with inverse temperature β (as in the theorem statement), the retrieval weight of the anchor cluster’s centroid satisfies

$$w_{c_{k^*}} \geq \frac{|\mathcal{C}_{k^*}| \exp(\beta q^\top c_{k^*})}{|\mathcal{C}_{k^*}| \exp(\beta q^\top c_{k^*}) + \sum_{k \neq k^*} |\mathcal{C}_k| \exp(\beta q^\top c_k + \beta \|q\|\epsilon)} \geq \frac{1}{1 + (p-1) \exp(-\beta(g_p - \|q\|\epsilon))}.$$

This uses the fact that impressions within a cluster of diameter $< \epsilon$ contribute weight within $e^{\pm \beta \|q\|\epsilon}$ of their centroid. Since consolidation reduces p (impressions merge into fewer clusters by part (i)), the bound improves as p decreases. Before sleep, the effective cluster count was $p = |M|$ (each impression distinct); after consolidation, $p \leq |M|$ with potentially $p \ll |M|$. The inter-cluster gap g_p changes during consolidation: non-anchor clusters are displaced toward the retrieval target $\ell \approx v$ (since every non-anchor impression blends toward ℓ at rate $\lambda_j = 1 - w_j/w_v > 0$, the centroid of each non-anchor cluster drifts toward v at each step), so the post-consolidation inter-cluster separation is smaller than the initial well-separation distance by a factor controlled by $\beta g_{k^*,k} T_{k^*}$. However, the separation remains strictly positive (the clusters never fully merge in finite time), and the intra-cluster diameter has been reduced to $\delta \ll \epsilon$ by consolidation. Therefore the post-consolidation score gap satisfies $g_p > 0$, and the retrieval

concentration bound improves: fewer effective clusters and reduced intra-cluster spread both increase $w_{c_k^*}$. The net effect is that retrieval concentration recovers, with the degree of recovery depending on the ratio of post-consolidation inter-cluster gap to intra-cluster spread.

(iv) By Corollary 38, $\gamma(t) \rightarrow 0$ as $m_j \rightarrow v$ for all j . Once $\gamma(t) < \epsilon_{\text{wake}}$ for a chosen tolerance ϵ_{wake} , further consolidation steps change V by at most a fraction ϵ_{wake} : $V(t+1) \geq (1 - \epsilon_{\text{wake}})V(t)$. This provides a computable termination criterion. $\square$

4.3 Monotone Pressure

Theorem 53 (Monotonicity of sleep pressure; softmax). *During sustained waking (capture and consolidation both enabled), the retrieval degradation described by Theorem 50 is monotone and irreversible:*

- (i) $C_t(\tau)$ is non-decreasing for all $\tau \in X$.
- (ii) $N_t(\tau) = N_0 + t$ is strictly increasing.
- (iii) $|M_t|$ is non-decreasing, and strictly increasing whenever a capture event occurs.
- (iv) The retrieval precision bound $\rho(|M|) := \max_{g \leq 2R^2} [1/(1 + (|M| - 1)\exp(-g))]$ is monotone decreasing in $|M|$.
- (v) Therefore the system's retrieval capacity is non-increasing during waking, and strictly decreasing at every capture event. No operation internal to the waking dynamics can reverse this.

Proof. (i) By Definition 3, $C_{t+1}(\tau) = C_t(\tau) + a_t(\tau) \cdot (t+1)/C_t(\tau)$. Since $a_t(\tau) \geq 0$ and $C_t(\tau) > 0$ (by $C_0 > 0$ and induction), the increment is non-negative: $C_{t+1} \geq C_t$.

(ii) By Corollary 20, $N_t(\tau) = N_0 + t$, which is strictly increasing in t .

(iii) By Corollary 23, $|M|$ is non-decreasing since capture adds elements by union and no operation removes them. At a capture event, at least one new element is added, so $|M|$ strictly increases.

(iv) Consider the function $\rho(n) = 1/(1 + (n - 1)\exp(-g_{\max}))$ where $g_{\max} = 2R^2$ is the maximum achievable score gap from Theorem 50(iii). The derivative with respect to n is

$$\frac{d\rho}{dn} = -\frac{\exp(-g_{\max})}{(1 + (n - 1)\exp(-g_{\max}))^2} < 0,$$

so ρ is strictly decreasing in n . More generally, the retrieval weight bound $\max_{g \leq g_{\max}} 1/(1 + (n - 1)\exp(-g))$ is achieved at $g = g_{\max}$ and this maximum is decreasing in n .

(v) Since $|M_t|$ is non-decreasing by (iii) and ρ is decreasing in $|M|$ by (iv), the composition $\rho(|M_t|)$ is non-increasing in t . At each capture event, $|M|$ increases and ρ strictly decreases. No operation in the waking dynamics decreases $|M|$ or increases ρ . The degradation is therefore monotone and irreversible during waking. $\square$

Corollary 54 (Irreversibility during waking; softmax). *Let $t_1 < t_2$ be two time steps during a waking phase with $|M_{t_2}| > |M_{t_1}|$. Then the retrieval precision bound satisfies $\rho(|M_{t_2}|) < \rho(|M_{t_1}|)$. No sequence of waking operations can restore ρ to its value at t_1 .*

Proof. Immediate from Theorem 53(iv)–(v): ρ is strictly decreasing in $|M|$, and $|M|$ is non-decreasing during waking. Since no waking operation removes rows from M , $|M_t| \geq |M_{t_2}| > |M_{t_1}|$ for all $t \geq t_2$. $\square$

4.4 Natural Wake-Up Condition

Theorem 55 (Quasi-fixed-point under $S = 0$; $\mathcal{A}_+$). *Let φ be a continuous positive simplex map. Under $S = 0$ with capture disabled and $|M| \geq 2$, the system reaches a quasi-fixed point characterized by:*

- (i) $V(t) \rightarrow 0$: all impressions converge to prototypes.
- (ii) $\gamma(t) \rightarrow 0$: the contraction rate vanishes (self-limiting).
- (iii) $\ell(t) \rightarrow v$: the retrieval output stabilizes to the anchor.
- (iv) $b(t) \rightarrow EKv$: the retrieval bias stabilizes.
- (v) The attention distribution converges to $a^* = \varphi(\sigma^* + EKv)$ satisfying the self-consistent equation

$$a^* = \varphi\left(\frac{C^*}{N} + EKv\right)$$

where C^* is the accumulation state at convergence and $v = \lim_{t \rightarrow \infty} \ell(t)$. The existence of a^* follows from the Brouwer fixed-point theorem applied to the continuous map $a \mapsto \varphi(\sigma(C(a)) + EKv)$ on the compact convex set $\Delta^{|X|}$.

- (vi) The vanishing of γ is a detectable wake-up signal: for any tolerance $\epsilon > 0$, there exists T such that $\gamma(t) < \epsilon$ for all $t \geq T$, and further $S = 0$ steps produce change bounded by $\epsilon V(T)$ in a single step.

Proof. (i) By Theorem 52(i) (which uses the anchor-independent diameter argument), $V(t) \rightarrow 0$ under $S = 0$ with capture disabled, without requiring the query to be fixed. In particular, the diameter $D(t) = \max_{i,j} \|m_i - m_j\| \rightarrow 0$, so all impressions converge to a common limit point.

(ii) As $D(t) \rightarrow 0$ from (i), all scores $q^\top m_j$ converge to a common value (since $|q^\top m_i - q^\top m_j| \leq \|q\| \cdot \|m_i - m_j\| \leq R \cdot D(t) \rightarrow 0$). Therefore all retrieval weights equalize: $w_j/w_v \rightarrow 1$, so $\lambda_j = 1 - w_j/w_v \rightarrow 0$, and $\gamma(t) = \min_{j \neq v} \lambda_j(1 - \sqrt{1 - \alpha_v^{(j)}}) \rightarrow 0$.

(iii) The retrieval output is $\ell = wM = \sum_j w_j m_j$. Since $D(t) \rightarrow 0$ by (i), all m_j converge to a common point v . With $\sum_j w_j = 1$, we have $\|\ell - v\| = \|\sum_j w_j(m_j - v)\| \leq \sum_j w_j \|m_j - v\| \leq D(t) \rightarrow 0$. Hence $\ell(t) \rightarrow v$.

(iv) By Definition 4, $b = EK\ell$. Since $\ell \rightarrow v$ and EK is a fixed linear map, $b \rightarrow EKv$.

(v) At the quasi-fixed point, $S = 0$, $\sigma = C/N$, $b = EKv$, and $a = \varphi(S + \sigma + b) = \varphi(\sigma + EKv)$. The accumulation continues: $C_{t+1} = C_t + a_t(t+1)/C_t$. By Theorem 27 ($\mathcal{A}_+$), a is bounded away from zero, so $C_t = \Theta(t)$ and $\sigma_t = C_t/N_t \rightarrow \sigma^*$ where σ^* is a fixed vector. (The ratio σ_t is bounded by Theorem 21(iii). Under constant attention $a \geq \alpha > 0$, the recurrence $C_{t+1}^2 = C_t^2 + 2a(t+1) + a^2(t+1)^2/C_t^2$ gives $C_t^2 \sim at^2$ by the same Cesàro argument as Theorem 21(ii), so $\sigma_t = C_t/N_t \rightarrow \sqrt{a}$.) In the limit:

$$a^* = \varphi(\sigma^* + EKv).$$

To show existence formally: consider the map $F: \Delta^{|X|} \rightarrow \Delta^{|X|}$ defined by $F(a) = \varphi(\sigma(a) + EKv)$ where $\sigma(a)$ is the limiting salience under constant attention a . By the accumulation rule $C_{t+1} = C_t + a(t+1)/C_t$ with constant a and $N_t = N_0 + t$, the ratio $\sigma_t = C_t/N_t$ converges to a limit $\sigma^*(a) = \sqrt{a}$ (by the Cesàro argument above), which depends continuously on a . The map F is continuous because φ is continuous by assumption and $\sigma^*(a) = \sqrt{a}$ (componentwise) is continuous on the interior of the simplex, hence on the compact set $[\alpha, 1 - \alpha]^{|X|}$ to which attention is confined by Theorem 27; thus $F = \varphi \circ (\sigma^*(\cdot) + EKv)$ is a continuous map from the compact convex set $\Delta^{|X|}$ to itself. By the Brouwer fixed-point theorem, F has a fixed point a^* .

(vi) Since $\gamma(t) \rightarrow 0$ by (ii), for any $\epsilon > 0$ there exists T with $\gamma(t) < \epsilon$ for $t \geq T$. By Corollary 32, $V(t+1) \leq (1 - \gamma(t))V(t)$, so the per-step *relative* change satisfies $V(t) - V(t+1) \geq$

$\gamma(t)V(t)$ (the contraction bound gives a lower bound on the reduction). To bound the change from *above*: since all m_j move by at most $\lambda_j \|\ell - m_j\| \leq V(t)$ per step, and $V(t)$ is non-increasing, the absolute change satisfies $|V(t) - V(t+1)| \leq V(t) \leq V(T)$ for $t \geq T$. Moreover, the relative change ratio satisfies $V(t+1)/V(t) \geq 1 - \gamma(t) \geq 1 - \epsilon$ for $t \geq T$, so each step changes V by at most a fraction ϵ : $V(t) - V(t+1) \leq \epsilon V(t)$ for $t \geq T$. $\square$

4.5 Self-Observable Sleep Pressure

Theorem 56 (Self-observable sleep pressure; softmax). *The system can detect its own need for sleep from quantities computable at each step without access to any external oracle. Define:*

- (i) The retrieval concentration $\rho_t := \max_j w_j^{(t)}$, the maximum retrieval weight at step t .
- (ii) The consolidation capacity $\gamma_t := \min_{j \neq v} \lambda_j (1 - \sqrt{1 - \alpha_v^{(j)}})$, the contraction rate at step t .
- (iii) The memory load $\mu_t := |M_t|$, the number of rows in M at step t .

Then:

- (a) All three quantities are computable from the current state (M_t, w_t, q_t) alone, requiring no history, no external threshold, and no knowledge of $|M^*|$.
- (b) The retrieval concentration ρ_t is non-increasing at every capture event (by the fan effect, Corollary 33). It satisfies

$$\rho_t \leq \frac{1}{1 + (\mu_t - 1) \exp(-2R^2)},$$

which is a computable upper bound that depends only on μ_t and the embedding radius R .

- (c) The consolidation capacity $\gamma_t \rightarrow 0$ as $\rho_t \rightarrow 1/\mu_t$ (by Corollary 38). The system can detect this: when $\gamma_t < \epsilon$ for any chosen $\epsilon > 0$, each consolidation step changes V by less than $\epsilon V(t)$.
- (d) The joint condition

$$\rho_t < c \quad \text{and} \quad \gamma_t < \epsilon$$

for any thresholds $c, \epsilon > 0$ constitutes a sufficient condition for sleep: retrieval is degraded (below concentration c) and consolidation cannot self-repair (rate below ϵ). Both conditions are monotone during waking: once triggered, they remain triggered until an $S = 0$ phase intervenes.

- (e) Under $S = 0$ with capture disabled, both conditions reverse: γ_t initially increases (consolidation resumes with a stable query) and, after prototype formation reduces the effective μ , ρ_t recovers. The system can detect the completion of sleep by monitoring $\gamma_t \rightarrow 0$ again (Theorem 55(vi)).

Proof. (a) The retrieval weights $w = \varphi(qM^\top)$ are computed at every step as part of Definition 4, so $\rho_t = \max_j w_j$ is directly available. The blend rates $\lambda_j = \lambda(w_j / \max_k w_k)$ are computed from w . The contraction rate $\gamma_t = \min_{j \neq v} \lambda_j (1 - \sqrt{1 - \alpha_v^{(j)}})$ requires the weight $\alpha_v^{(j)}$ that the anchor v receives in the consolidation target for impression j ; since the target is $\ell = wM$ (a fixed vector for all non-anchor impressions in our formulation)¹, we have $\alpha_v^{(j)} = w_v$ for all j , and

¹The use of $\ell = wM$ as the consolidation target is a simplification of the general per-impression target $T_j = \varphi((m_j + q)M^\top)M$, which depends on m_j . The computability conclusion (part (a)) holds regardless of the target form: for the general target, $\alpha_v^{(j)}$ is the weight that impression j 's own target T_j places on the anchor v , i.e. $\alpha_v^{(j)} = [\varphi((m_j + q)M^\top)]_v$, which depends only on the current state (M_t, w_t, q_t) and is therefore computable at

$\gamma_t = \min_{j \neq v} \lambda_j (1 - \sqrt{1 - w_v})$. All quantities are computed from (M_t, w_t) alone. The memory load $\mu_t = |M_t|$ is the row count of M . No external information is required.

(b) By Corollary 33, every capture event adds at least one row to M , increasing the softmax denominator while leaving all numerators unchanged. The maximum weight ρ_t therefore strictly decreases at each capture event. The bound follows from Corollary 35 evaluated at the maximum achievable score gap $g = 2R^2$ (Theorem 50(iii)):

$$\rho_t \leq \frac{1}{1 + (\mu_t - 1) \exp(-2R^2)}.$$

This bound is computable from μ_t and $R = \max_\tau \|E(\tau)\|$, both of which are available at each step.

(c) As impressions equalize under the fan effect ($\rho_t \rightarrow 1/\mu_t$), the retrieval weights satisfy $w_j/w_v \rightarrow 1$ for all j . By definition of the blend rate, $\lambda_j = \lambda(w_j/\max_k w_k) \rightarrow \lambda(1) = 0$. Therefore $\max_j \lambda_j \rightarrow 0$, and since $\gamma_t \leq \max_j \lambda_j$ (each factor $(1 - \sqrt{1 - \alpha_v^{(j)}}) \leq 1$), the squeeze theorem gives $\gamma_t \rightarrow 0$. The system observes this directly: γ_t is computed from quantities already available at step 3 (RETRIEVE) of the loop.

(d) During waking, μ_t is non-decreasing (Theorem 53(iii)) and ρ_t is non-increasing at capture events (part (b)). At each capture event, the fan effect adds rows, diluting ρ_t ; hence ρ_t is non-increasing at capture events. Between capture events, consolidation can in principle perturb retrieval scores (and thus ρ_t) by moving non-anchor impressions, but this perturbation is bounded by γ_t : each consolidation step changes any score $q^\top m_j$ by at most $\lambda_j \|q\| \|T_j - m_j\| \leq \|q\| V(t)$. When $\gamma_t < \epsilon$, consolidation changes each retrieval weight by less than $O(\epsilon)$ per step, so ρ_t cannot increase by more than $O(\epsilon)$. Similarly, γ_t depends on λ_j which depends on the weight ratios w_j/w_v ; as ρ_t decreases, the weight distribution flattens, $\lambda_j \rightarrow 0$, and γ_t remains below ϵ . The joint condition ($\rho_t < c$ and $\gamma_t < \epsilon$) is therefore absorbing to within $O(\epsilon)$, and the $O(\epsilon)$ slack can be absorbed into the threshold c by choosing c with a margin of $O(\epsilon)$.

(e) Under $S = 0$ with capture disabled, the query stabilizes (no new captures perturb M , and the query depends only on $\sigma + b$ which evolves smoothly). With a stable query and capture disabled, Lemma 31 applies: a clear anchor emerges with $\lambda_v = 0$ and $\lambda_j > 0$ for $j \neq v$. This gives $\gamma_t > 0$, so the consolidation capacity recovers. It recovers because the stable query produces non-uniform retrieval weights (the anchor has a definite score advantage under the fixed query), restoring the weight differential that was lost during waking.

As consolidation proceeds, impressions converge toward the anchor (Corollary 37). Impressions within ϵ of each other become functionally indistinguishable (Theorem 52(ii)), reducing the effective μ . With fewer effective impressions, the retrieval concentration ρ_t increases: the anchor's weight grows as competitors merge into it. After sufficient consolidation, $\rho_t > c$ is restored.

The system detects sleep completion by the same mechanism as Theorem 55(vi): $\gamma_t \rightarrow 0$ again, but now because consolidation has *succeeded* (all impressions near the anchor) rather than because it has *stalled* (all weights equal). The two regimes of $\gamma_t \rightarrow 0$ are distinguishable: during waking, $\gamma_t \rightarrow 0$ coincides with $\rho_t \rightarrow 1/\mu_t$ (flat weights, degraded retrieval); after sleep, $\gamma_t \rightarrow 0$ coincides with ρ_t recovering to $\rho_t \geq 1/(1 + (p - 1) \exp(-\beta g_p))$ (sharp weights, recovered retrieval), where p is the number of consolidated prototypes. The system can therefore detect both sleep onset and sleep completion from (ρ_t, γ_t) alone. $\square$

each step. The contraction rate becomes $\gamma_t = \min_{j \neq v} \lambda_j (1 - \sqrt{1 - \alpha_v^{(j)}})$ with per-impression $\alpha_v^{(j)}$; the simplified formula $\gamma_t = \min_{j \neq v} \lambda_j (1 - \sqrt{1 - w_v})$ arises because the global target sets $\alpha_v^{(j)} = w_v$ for all j . The qualitative conclusions hold for the general case, since T_j also places positive weight on the anchor v (via the positivity of φ) and therefore produces contraction toward v . In summary, part (a) is independent of the choice between global and per-impression targets: each T_j is determined by (M_t, q_t) via Definition 5, so γ_t is computable in either formulation.

Corollary 57 (Sleep–wake cycle from internal observables; softmax). *The pair (ρ_t, γ_t) defines a complete sleep–wake control signal:*

- **Sleep onset:** $\rho_t < c$ and $\gamma_t < \epsilon$ (degraded retrieval, stalled consolidation).
- **Sleep completion:** $\gamma_t < \epsilon$ and $\rho_t > c'$ for some recovery threshold $c' > c$ (recovered retrieval, completed consolidation).

The thresholds c, c', ϵ are not tunable parameters of the dynamics; they are consequences of the embedding geometry $(R, |X|, d)$ and can be derived from the bounds in Theorems 50 and 52. In particular, c can be set to $\rho(|M^|) = 1/(1 + (|M^*| - 1)\exp(-2R^2))$, which is the retrieval concentration at the critical memory size, and ϵ can be set to any value below the initial waking contraction rate γ_0 .*

Proof. The sleep onset condition follows from Theorem 56(d): both conditions are absorbing during waking and jointly indicate that the system can neither retrieve effectively nor self-repair. The sleep completion condition follows from Theorem 56(e): $\gamma_t \rightarrow 0$ with ρ_t high indicates successful consolidation rather than stalled consolidation. The thresholds are determined by the embedding: $|M^*|$ depends on R and c (Theorem 50(iv)), and γ_0 depends on the initial weight distribution, which depends on E, K , and φ . No free parameters are introduced. $\square$

Remark 58 (Two regimes of vanishing γ). The contraction rate γ vanishes for two opposite reasons: exhaustion (weights equalize, $\rho \rightarrow 1/|M|$) and completion (impressions converge, $\rho \rightarrow 1$). The pair (ρ, γ) separates them, and the sleep phase is the trajectory from one to the other in (ρ, γ) space.

4.6 Multi-Cluster Oscillation

We now show that when the memory M at sleep onset contains multiple well-separated clusters, the sleep phase exhibits cyclic consolidation dynamics: the contraction rate γ oscillates as the system sequentially consolidates each cluster.

Definition 59 (ϵ -clusters under a query). Let q be a query and M a memory matrix. Two impressions $m_i, m_j \in M$ are ϵ -near under q if $|q^\top m_i - q^\top m_j| < \epsilon$. The ϵ -clusters of M under q are the connected components of the graph on M where $m_i \sim m_j$ iff $\|m_i - m_j\| < \epsilon$. Denote the clusters $\mathcal{C}_1, \dots, \mathcal{C}_p$ with $M = \bigsqcup_{k=1}^p \mathcal{C}_k$. We say the clusters are *well-separated* if $\min_{k \neq k'} \min_{m \in \mathcal{C}_k, m' \in \mathcal{C}_{k'}} \|m - m'\| > 2\epsilon$.

Definition 60 (Cluster salience). Given clusters $\mathcal{C}_1, \dots, \mathcal{C}_p$ of M under query q , the *cluster score spread* of $\mathcal{C}_k$ is

$$\Delta_k(q) = \max_{m \in \mathcal{C}_k} q^\top m - \min_{m \in \mathcal{C}_k} q^\top m,$$

and the *cluster retrieval mass* is $W_k = \sum_{m_j \in \mathcal{C}_k} w_j$ where $w = \varphi(qM^\top)$. The *anchor cluster* is the cluster $\mathcal{C}_{k^*}$ containing the anchor $v = \arg \max_j w_j$.

Theorem 61 (Sleep cycle oscillation from multi-cluster consolidation; $\mathcal{A}_+$). *Let φ be a continuous positive simplex map. Suppose the system enters an $S = 0$ phase with capture disabled and $|M| \geq 2$. Let M contain $p \geq 2$ well-separated ϵ -clusters $\mathcal{C}_1, \dots, \mathcal{C}_p$ (Definition 59) under the initial query q_0 . Then:*

- (i) **Consolidation localizes to the anchor cluster.** *At each step, the anchor $v = \arg \max_j w_j$ lies in some cluster $\mathcal{C}_{k^*}$. Impressions $m_j \in \mathcal{C}_{k^*}$ have blend rates bounded (under softmax with inverse temperature β) by $\lambda_j \leq 1 - e^{-\beta \|q\| \epsilon}$ (since $w_j/w_v \geq e^{-\beta \|q\| \epsilon}$) and experience non-zero displacement because the consolidation target differs from m_j (here $\ell = wM$ for brevity; more precisely, the per-impression target T_j of Definition 5 pulls*

toward the anchor, producing net contraction within the cluster). Impressions in other clusters $\mathcal{C}_k, k \neq k^*$, have blend rates bounded below (under softmax) by $\lambda_j \geq 1 - e^{-\beta g_{k^*,k}}$ (since $w_j/w_v \leq e^{-\beta g_{k^*,k}}$ where $g_{k^*,k} > 0$ is the inter-cluster score gap) but experience displacement that varies by at most $O(\epsilon)$ within each cluster: all members of $\mathcal{C}_k$ are within ϵ of each other and all distant from ℓ , so their pairwise displacements differ by at most $O(\epsilon)$ and the intra-cluster structure is preserved up to $O(\epsilon)$. Consolidation therefore primarily compresses $\mathcal{C}_{k^*}$ while perturbing the intra-cluster spread of other clusters by at most $O(\epsilon)$ per step.

- (ii) **Anchor rotation upon cluster convergence.** As $\mathcal{C}_{k^*}$ converges internally (all $m_j \in \mathcal{C}_{k^*}$ approach the anchor v), the intra-cluster score spread $\Delta_{k^*}(q) \rightarrow 0$, the retrieval weights within $\mathcal{C}_{k^*}$ equalize, and $\gamma(t) \rightarrow 0$ for that cluster. At this point, the dominant contribution to the overall score spread $\Delta_\sigma(t)$ comes from some unconsolidated cluster $\mathcal{C}_{k'}$. The query shifts to reflect $\mathcal{C}_{k'}$'s structure (since $\ell = wM$ changes as the weight distribution evolves), and the anchor rotates to $\mathcal{C}_{k'}$. The Lyapunov function V measured from the new anchor spikes: V increases because $\|v_{\text{old}} - v_{\text{new}}\| > 0$ (well-separation), and correspondingly γ recovers to a positive value.
- (iii) **Oscillatory γ trajectory.** The contraction rate $\gamma(t)$ exhibits an oscillatory pattern: for each cluster consolidation episode, γ starts positive (when the anchor first engages the cluster), decays toward zero as the cluster converges (Corollary 38), then spikes when the anchor rotates to the next cluster. The number of such decay-then-spike episodes is at most $p - 1$ (after which all clusters are consolidated and $\gamma \rightarrow 0$ permanently).
- (iv) **Bounded number of oscillation cycles.** The total number of anchor-cluster transitions is at most $p - 1$. Each transition eliminates one cluster from the set of unconsolidated clusters (the converged cluster satisfies $\Delta_k \rightarrow 0$, so its impressions contribute score spread at most $\|q\|\Delta_k \rightarrow 0$). Therefore the oscillation terminates after at most $p - 1$ transitions.
- (v) **Salience ordering of consolidation.** The first anchor cluster $\mathcal{C}_{k_1^*}$ is the one whose anchor v maximizes $q_0^\top v$ (highest-scoring under the initial query), so the most salient cluster under q_0 is consolidated first. For subsequent episodes, the anchor is the argmax among remaining unconsolidated impressions under the query at that transition point: the next anchor cluster $\mathcal{C}_{k_2^*}$ has the largest retrieval mass among remaining clusters under the post-convergence query, and so on. The sequential structure is preserved (each transition selects the most salient remaining cluster), but the ordering may differ from the initial salience ranking under q_0 because the query evolves as each cluster consolidates.
- (vi) **Termination recovers Theorem 55.** After all p clusters have converged, the system satisfies $\gamma(t) \rightarrow 0$ with $V(t) \rightarrow 0$, recovering the quasi-fixed-point condition of Theorem 55. The terminal state consists of p prototypes (one per original cluster), with the system at rest.

Proof. (i) Let $v \in \mathcal{C}_{k^*}$ be the anchor. By Definition 4, $w_j = [\varphi(qM^\top)]_j$. Since φ is positive, all weights are positive, but the anchor has the largest weight by definition: $w_v \geq w_j$ for all j . The blend rate is $\lambda_j = \lambda(w_j/w_v)$, which satisfies $\lambda(1) = 0$ (anchor is fixed) and is decreasing: $\lambda_j = 1 - w_j/w_v$, so $\lambda_j \rightarrow 0$ as $w_j/w_v \rightarrow 1$.

For $m_j \in \mathcal{C}_{k^*}$, the impression is geometrically close to the anchor ($\|m_j - v\| < \epsilon$), so $|q^\top m_j - q^\top v| \leq \|q\|\epsilon$, giving $w_j/w_v \geq e^{-\beta\|q\|\epsilon}$ and hence $\lambda_j = 1 - w_j/w_v \leq 1 - e^{-\beta\|q\|\epsilon}$. However, the consolidation target $\ell = wM$ satisfies $w_v \geq w_j$ for all j (the anchor has the largest weight), so ℓ is pulled toward v . For non-anchor members $m_j \neq v$ within $\mathcal{C}_{k^*}$, the displacement $\|\ell - m_j\| > 0$ (since $\ell \neq m_j$ whenever $m_j \neq v$), producing contraction.

For $m_j \in \mathcal{C}_k$ with $k \neq k^*$, the well-separation condition gives $\|m_j - v\| > 2\epsilon$. Under φ positive with the softmax exponential form, the score gap $q^\top v - q^\top m_j$ enters the exponent:

$w_j/w_v = \exp(\beta(q^\top m_j - q^\top v)) \leq \exp(-\beta g_{k^*,k})$ where $g_{k^*,k} = \min_{m \in \mathcal{C}_{k^*}, m' \in \mathcal{C}_k} (q^\top m - q^\top m') > 0$ is the inter-cluster score gap, giving $\lambda_j = 1 - w_j/w_v \geq 1 - e^{-\beta g_{k^*,k}}$. However, the consolidation target satisfies $\|\ell - v\| = \|wM - v\| \leq \sum_j w_j \|m_j - v\| \leq W_{k^*}\epsilon + (1 - W_{k^*}) \text{diam}(M)$ (since W_{k^*} dominates the mass), and the displacement $\lambda_j \|\ell - m_j\|$ is bounded by the inter-cluster distance. The per-step movement is bounded, but its effect on the *intra*-cluster structure of $\mathcal{C}_k$ is controlled: for any $m_j, m_{j'} \in \mathcal{C}_k$, the difference in displacement satisfies $\|\lambda_j(\ell - m_j) - \lambda_{j'}(\ell - m_{j'})\| \leq 2\epsilon$ (since $\|m_j - m_{j'}\| < \epsilon$ and $|\lambda_j - \lambda_{j'}|$ is Lipschitz in $\|m_j - m_{j'}\|$). The intra-cluster spread Δ_k therefore changes by at most $O(\epsilon)$ per step while Δ_{k^*} decreases.

(ii) As $\mathcal{C}_{k^*}$ converges under consolidation, all $m_j \in \mathcal{C}_{k^*}$ approach v . By Corollary 38, $\gamma(t) \rightarrow 0$ because $\lambda_j \rightarrow 0$ for impressions within $\mathcal{C}_{k^*}$ (their weights equalize). At this stage, the within-cluster spread $\Delta_{k^*} \rightarrow 0$.

The retrieval output $\ell = wM = \sum_j w_j m_j$ is a weighted average. Once $\mathcal{C}_{k^*}$ has converged to a single point v , the contribution from $\mathcal{C}_{k^*}$ is $W_{k^*} \cdot v$. The remaining mass $1 - W_{k^*}$ is distributed over other clusters. As consolidation continues and the intra-cluster spread of $\mathcal{C}_{k^*}$ vanishes, the *residual* variation in scores comes from the inter-cluster distances. The score spread $\Delta_\sigma(t)$ is now dominated by the separation between consolidated and unconsolidated clusters.

We now show that anchor rotation occurs as each cluster consolidates. Once $\mathcal{C}_{k^*}$ has collapsed to a single point v , all members of $\mathcal{C}_{k^*}$ contribute identically to the retrieval output, effectively acting as one impression with weight W_{k^*} . The remaining clusters retain positive internal spread ($\Delta_{k'} > 0$). The retrieval output is $\ell = W_{k^*}v + \sum_{k \neq k^*} W_k \mu_k + O(\epsilon)$. Since φ is positive, $W_{k'} > 0$ for every cluster $k' \neq k^*$, so $\ell \neq v$. The bias $b = EK\ell$ therefore points away from EKv , shifting the query.

As $\mathcal{C}_{k^*}$ consolidates, the anchor's score advantage within that cluster vanishes: all members of $\mathcal{C}_{k^*}$ approach v , so their retrieval weights equalize and $\gamma(t) \rightarrow 0$ for that cluster. Meanwhile, unconsolidated clusters $\mathcal{C}_{k'}$ retain positive internal spread $\Delta_{k'} > 0$, and their members contribute the dominant variation to the retrieval output. The evolving query q_t reflects the shifting weight distribution: as $\mathcal{C}_{k^*}$'s contribution to score variation vanishes, the score landscape becomes dominated by the unconsolidated cluster $\mathcal{C}_{k'}$ with the largest residual spread. At this point the anchor shifts to $\mathcal{C}_{k'}$ —the impression $v' = \arg \max_j w_j$ now lies in $\mathcal{C}_{k'}$, where the weight differential among impressions is largest—and consolidation engages that cluster. Anchor rotation is therefore an observable consequence of within-cluster convergence: as each cluster's internal spread collapses, the anchor score advantage within it disappears, and the anchor migrates to the next unconsolidated cluster.

After anchor rotation to $v' \in \mathcal{C}_{k'}$, the Lyapunov function is redefined relative to the new anchor as $V'(t) = \max_{j \neq v'} \|m_j - v'\|$. Since $v' \in \mathcal{C}_{k'}$ and $v \in \mathcal{C}_{k^*}$ are well-separated ($\|v - v'\| > 2\epsilon$), we have $V'(t) \geq \|v - v'\| > 0$, so V spikes (that is, the newly defined V' starts large; the old V measured from v had decayed to near zero). The contraction rate γ also recovers: under the new anchor v' , the impressions in $\mathcal{C}_{k'}$ have a spread $\Delta_{k'}(q) > 0$, giving non-uniform retrieval weights within $\mathcal{C}_{k'}$. The resulting weight differential produces blend rates $\lambda_j > 0$ for out-of-cluster impressions and positive displacement $\|\ell - m_j\| > 0$ for within-cluster impressions, so $\gamma > 0$.

(iii) Each cluster consolidation episode proceeds as follows. When the anchor is in $\mathcal{C}_k$:

- (a) γ is positive and bounded away from zero (intra-cluster spread $\Delta_k > 0$ produces non-uniform weights in $\mathcal{C}_k$).
- (b) γ decays toward zero as $\Delta_k \rightarrow 0$ (Corollary 38).
- (c) Once γ falls below any prescribed threshold (which occurs in finite time by Corollary 38), the anchor rotates to the next unconsolidated cluster, and γ spikes.

This produces a sawtooth-like oscillation in γ : repeated episodes of decay followed by a sharp increase. The final episode (consolidation of the last remaining cluster) ends with $\gamma \rightarrow 0$ permanently, since there is no further unconsolidated cluster to rotate to. The total number of decay-then-spike episodes is therefore at most $p - 1$.

(iv) Each anchor-cluster transition removes the converged cluster from the pool of clusters with $\Delta_k > 0$. Once a cluster $\mathcal{C}_k$ has converged to a single prototype, its internal $\Delta_k \rightarrow 0$, and it cannot generate a new anchor rotation. The total number of transitions is bounded by $|\{\text{initially unconsolidated clusters}\}| - 1 = p - 1$. After $p - 1$ transitions, only one cluster remains unconsolidated; once that cluster converges, the process terminates.

(v) At sleep onset, the query q_0 determines the initial retrieval weights. The anchor v is the highest-scoring impression under q_0 , which lies in the cluster $\mathcal{C}_{k_1^*}$ with the highest retrieval mass $W_{k_1^*}$. This cluster has the largest $\Delta_{k_1^*}$ weighted by retrieval mass among all clusters (the anchor resides in the most salient cluster under q_0).

After $\mathcal{C}_{k_1^*}$ converges, the next anchor cluster is determined by the updated query. With $\mathcal{C}_{k_1^*}$ collapsed to a single point, its members contribute identical scores, so the anchor rotates to the impression $v' = \arg \max_{j \notin \mathcal{C}_{k_1^*}} w_j$ under the post-convergence query. This v' lies in the cluster $\mathcal{C}_{k_2^*}$ that maximizes the retrieval mass W_k among the remaining clusters under the current query. Since the query depends on M through ℓ and $b = EK\ell$, the ordering of clusters is determined by the evolving query and need not coincide with the initial ordering under q_0 . However, the ordering is deterministic at each transition: the next cluster is always the one with the highest retrieval mass under the query at that transition point.

(vi) After all p clusters have converged, every cluster $\mathcal{C}_k$ consists of impressions within Δ_k of a prototype p_k , where $\Delta_k \rightarrow 0$ by the preceding convergence. The intra-cluster spreads all satisfy $\Delta_k \rightarrow 0$. The score spread $\Delta_\sigma(t)$ now reflects only the inter-prototype distances, which are fixed (prototypes are anchors and anchors are preserved by consolidation; Lemma 22). The contraction rate satisfies $\gamma(t) \rightarrow 0$ because all weights within each cluster equalize ($\lambda_j \rightarrow 0$). The Lyapunov function $V(t) \rightarrow 0$ within each cluster. These are precisely the conditions of Theorem 55: the system reaches a quasi-fixed point. The terminal state consists of p prototypes at the cluster centers, and the system has fully consolidated. $\square$

Remark 62 (Oscillation as sequential single-cluster episodes). The multi-cluster sleep phase is a sequence of single-cluster episodes, each governed by the same Lyapunov descent and self-limiting property as Theorem 52; the only new element is the inter-episode anchor rotation. The number of oscillation cycles equals the number of distinct clusters at sleep onset.

Remark 63 (Inter-cluster displacement during consolidation). Theorem 61(i) shows that the *intra*-cluster structure of non-anchor clusters is preserved up to $O(\epsilon)$ per step. However, the *inter*-cluster distance is not preserved: each non-anchor cluster $\mathcal{C}_k$ is displaced toward the retrieval target $\ell \approx v$ at each consolidation step.

Under the global-target simplification (each row blends toward $\ell = wM$ rather than the per-impression target T_j of Definition 5), a non-anchor impression $m_j \in \mathcal{C}_k$ with weight ratio $r_j = w_j/w_v \leq e^{-\beta g_{k^*,k}}$ has blend rate $\lambda_j = 1 - r_j \geq 1 - e^{-\beta g_{k^*,k}}$, and the per-step update gives

$$\|m_j^{(t+1)} - v\| \leq e^{-\beta g_{k^*,k}} \|m_j^{(t)} - v\| + O(\epsilon),$$

since $\|\ell - v\| = O(\epsilon)$ when the anchor cluster dominates the retrieval mass ($W_{k^*} \rightarrow 1$ for large $\beta g_{k^*,k}$). Under the per-impression target, the bound $\|T_j - v\| \leq V(t)$ from Lemma 31 gives a weaker but qualitatively identical estimate: non-anchor clusters are displaced toward v at each step. Iterating the global-target bound over T steps and summing the geometric series:

$$\|c_k^{(T)} - v\| \leq e^{-\beta g_{k^*,k} T} \|c_k^{(0)} - v\| + \frac{\epsilon}{1 - e^{-\beta g_{k^*,k}}}.$$

After the anchor cluster converges (requiring $T_{k^*} = O(\log(\Delta_{k^*}/\delta)/\gamma_{k^*})$ steps), the inter-cluster distance has shrunk by a factor of approximately $e^{-\beta g_{k^*,k} T_{k^*}}$. This factor is positive (the clusters never fully merge in finite time) but may be exponentially small when $\beta g_{k^*,k} T_{k^*}$ is large.

This displacement follows from the blend rule: $\lambda(r) = 1 - r$ applies maximal blending to low-weight impressions. The oscillation dynamics (anchor rotation, Lyapunov descent, self-limiting)

proceed identically regardless of the displacement; it affects the *magnitude* of the post-sleep retrieval improvement but not the *existence* of recovery (Theorem 52(iii)). Empirically, inter-centroid ratios of 0.001–0.08 are typical over 500 consolidation steps with $\beta = 1$ (see the simulation `sim_cluster_separation.py`).

Corollary 64 (Sleep duration scales with cluster count; $\mathcal{A}_+$). *Under the hypotheses of Theorem 61, the total number of consolidation steps required for all clusters to converge to within tolerance δ scales as $\sum_{k=1}^p T_k$ where $T_k = O(\log(V_k(0)/\delta)/\gamma_k)$ is the convergence time for cluster $\mathcal{C}_k$ with initial intra-cluster Lyapunov value $V_k(0)$ and characteristic contraction rate γ_k (a lower bound on the contraction rate during the effective phase of episode k , i.e. before γ drops below threshold; the $O(\log/\gamma)$ bound is therefore conservative). In particular, sleep duration is at least proportional to the number of initial clusters p .*

Proof. By Corollary 32, within each single-cluster episode $V(t+T) \leq (1-\gamma_k)^T V(t)$, so achieving $V < \delta$ requires $T_k \geq \log(V_k(0)/\delta)/(-\log(1-\gamma_k))$. Since $x \leq -\log(1-x) \leq x/(1-x)$ for all $x \in (0, 1)$, this gives $T_k \geq (1-\gamma_k) \log(V_k(0)/\delta)/\gamma_k$. The total sleep time is at least $\sum_k T_k$, which is at least $p \cdot \min_k T_k$. $\square$

4.7 Sleep Debt

When a sleep phase is truncated before full convergence, the system exits the $S = 0$ phase with residual unconsolidated state. We formalize this as *sleep debt*, show that partial sleep is still beneficial, that debt compounds across wake–sleep cycles with insufficient sleep, identify the critical threshold beyond which the system enters permanent degradation, and prove that a single full-length sleep eliminates all accumulated debt.

Definition 65 (Inter-cluster separation parameter). Given a memory M with well-separated ϵ -clusters $\mathcal{C}_1, \dots, \mathcal{C}_p$ (Definition 59) at sleep onset, the *inter-cluster separation* is

$$\epsilon_{\text{sep}} = \min_{k \neq k'} \min_{m \in \mathcal{C}_k, m' \in \mathcal{C}_{k'}} \|m - m'\|.$$

By the well-separation condition, $\epsilon_{\text{sep}} > 2\epsilon$. After full consolidation, the inter-prototype distance may be smaller than ϵ_{sep} (Remark 63); in the sleep debt analysis below, ϵ_{sep} refers to the separation at the *current* sleep onset, which is the operationally relevant quantity.

Definition 66 (Sleep debt). Let the system enter sleep (capture disabled, $S = 0$) at time t_0 with $|M| \geq 2$. After k consolidation steps, define the *sleep debt* as

$$D(k) = V(t_0 + k) + p(t_0 + k),$$

where $V(t_0 + k) = \max_{j \neq v} \|m_j^{(t_0+k)} - v\|$ is the Lyapunov value (Definition 4 and Lemma 31) and $p(t_0 + k)$ is the number of ϵ -clusters of M_{t_0+k} that have not yet converged to within tolerance δ , i.e. $p(t_0 + k) = |\{k' : \Delta_{k'}(q) > \delta\}|$ where $\Delta_{k'}$ is the intra-cluster score spread (Definition 60). After full convergence ($k = k_{\text{full}}$), $V \rightarrow 0$ and $p_{\text{remaining}} = 0$, so $D(k_{\text{full}}) \rightarrow 0$.

Theorem 67 (Sleep debt: compounding, threshold, and recovery; $\mathcal{A}_+$). *Let φ be a continuous positive simplex map and let the system alternate between waking phases (capture and consolidation both active, each producing at most n_{cap} new captures) and sleep phases (capture disabled, $S = 0$, each running for k consolidation steps). Let $\gamma_{\min} > 0$ be a lower bound on the contraction rate during the effective phase of each sleep episode (bounded away from zero by Corollary 32 while intra-cluster spread satisfies $\Delta_k > 0$). Then:*

- (i) **Partial sleep is monotonically beneficial.** For any $0 \leq k_1 < k_2$, we have $V(t_0 + k_2) \leq V(t_0 + k_1)$. In particular, $V(t_0 + k) \leq (1 - \gamma_{\min})^k V(t_0)$, so every additional sleep step

reduces V by at least a factor of $(1 - \gamma_{\min})$. The benefit is monotone in sleep duration: more steps $\implies$ lower debt. (Note: the retrieval concentration ρ over the raw row set may decrease transiently during consolidation as non-anchor scores equalize toward the anchor score, shrinking the score gap; ρ recovers only after sufficient convergence reduces the effective memory size via prototype formation, as described in Theorem 56(e).)

- (ii) **Debt compounds across cycles.** Suppose the system undergoes n wake-sleep cycles, where each waking phase adds at most $\Delta V > 0$ to the Lyapunov value (ΔV is an upper bound on the per-cycle Lyapunov increment) and each sleep phase runs for k_{short} steps. Let $r = (1 - \gamma_{\min})^{k_{\text{short}}} \in (0, 1)$ be the per-cycle contraction factor. Then the residual Lyapunov value at the start of the $(n + 1)$ -th waking phase satisfies

$$V_n \leq r(V_{n-1} + \Delta V) \leq \frac{r \Delta V}{1 - r} + r^n \left(V_0 - \frac{r \Delta V}{1 - r} \right),$$

which converges to the steady-state residual

$$V_{\infty} = \frac{r \Delta V}{1 - r} = \frac{(1 - \gamma_{\min})^{k_{\text{short}}} \Delta V}{1 - (1 - \gamma_{\min})^{k_{\text{short}}}}.$$

As $k_{\text{short}} \rightarrow 0$, $r \rightarrow 1$ and $V_{\infty} = r \Delta V / (1 - r) \rightarrow \infty$: the system carries permanent residual debt that grows without bound as sleep is curtailed.

- (iii) **Critical debt threshold.** There exists a critical sleep duration k_{crit} below which the per-cycle residual grows unboundedly. Specifically, the residual V_n is bounded (i.e. the geometric series in (ii) converges) if and only if $r < 1$, which always holds for $k_{\text{short}} \geq 1$. However, for the system to maintain functional retrieval, we need V_{∞} below the inter-cluster separation ϵ_{sep} (otherwise unconsolidated impressions from different clusters mix, destroying cluster structure). The critical duration is

$$k_{\text{crit}} = \left\lceil \frac{\log(\epsilon_{\text{sep}} / (\Delta V + \epsilon_{\text{sep}}))}{\log(1 - \gamma_{\min})} \right\rceil.$$

For $k_{\text{short}} < k_{\text{crit}}$, the steady-state residual exceeds ϵ_{sep} and the system is in permanent degradation: each cycle's new captures are added on top of an unconsolidated residual that prevents clean cluster formation.

- (iv) **Recovery from debt.** A single full-length sleep phase of $k_{\text{full}} \geq \lceil \log(V_n / \delta) / (-\log(1 - \gamma_{\min})) \rceil$ steps reduces V to within tolerance δ of zero, regardless of the accumulated debt V_n . After recovery, $D(k_{\text{full}}) \leq \delta$ and the system satisfies the quasi-fixed-point condition of Theorem 55: $V \leq \delta$, $\gamma \rightarrow 0$, $\ell \rightarrow v$, $b \rightarrow EKv$.
- (v) **Full sleep dominates truncated sleep.** Let V_n^{full} and V_n^{short} denote the residual at the start of the $(n+1)$ -th waking phase under full and truncated sleep respectively, starting from the same initial condition. Then $V_n^{\text{full}} \leq V_n^{\text{short}}$ for all n , with strict inequality whenever $V_{n-1}^{\text{short}} > 0$.

Proof. (i) By Corollary 32, $V(t + 1) \leq (1 - \gamma(t)) V(t)$ at each consolidation step during sleep. Since $\gamma(t) \geq \gamma_{\min} > 0$ while the intra-cluster spread is above δ (the contraction rate is bounded away from zero by the positivity of φ and $w_v > w_j$ for $j \neq v$; see Corollary 30), iterating gives $V(t_0 + k) \leq (1 - \gamma_{\min})^k V(t_0)$. Since $(1 - \gamma_{\min})^k$ is strictly decreasing in k , every additional sleep step produces a strict reduction in V . This is an exponential contraction: V decays geometrically with rate $\gamma_{\min}$.

For retrieval concentration (softmax specialization, not required for the $\mathcal{A}_+$ conclusion): as V decreases, impressions move closer to the anchor v . For any fixed query q , the non-anchor

scores $q^\top m_k$ converge toward $q^\top v$, *shrinking* the score gap $g = q^\top v - \max_{k \neq v} q^\top m_k$. Under softmax, a smaller gap *decreases* ρ over the raw row set via $\rho \geq 1/(1 + (|M| - 1) \exp(-g))$ (Theorem 50(ii)), so ρ does not increase monotonically during consolidation. The retrieval concentration ρ ultimately recovers through a different mechanism: once impressions converge to within ϵ of the anchor they become functionally indistinguishable and the effective memory size drops (Theorem 56(e), softmax), increasing ρ by reducing the number of effective competitors. The monotone quantity during sleep is V , not ρ ; the theorem statement is phrased accordingly in terms of V alone.

(ii) We model the wake–sleep cycle as a discrete dynamical system on the residual Lyapunov value. At the start of cycle n :

(a) *Waking phase*: the system captures new impressions, increasing V by ΔV . The pre-sleep Lyapunov value is at most $V_n + \Delta V$ (new captures are at distance at most $2R$ from the anchor, contributing bounded increments to V ; moreover, the residual V_n from the previous sleep means the anchor itself may be less sharp, so the effective increment includes both the new captures and interaction with the residual).

(b) *Sleep phase*: k_{short} consolidation steps contract V by factor $r = (1 - \gamma_{\min})^{k_{\text{short}}}$.

The recurrence is $V_{n+1} \leq r(V_n + \Delta V)$. This is a linear recurrence with solution

$$V_n \leq r^n V_0 + r \Delta V \sum_{j=0}^{n-1} r^j = r^n V_0 + \frac{r \Delta V (1 - r^n)}{1 - r}.$$

As $n \rightarrow \infty$, $r^n \rightarrow 0$ (since $0 < r < 1$), and the steady state is $V_\infty = r \Delta V / (1 - r)$. The steady-state residual is positive whenever $\Delta V > 0$ and $r > 0$, i.e. whenever sleep is not complete ($k_{\text{short}} < k_{\text{full}}$) and new impressions are captured ($\Delta V > 0$).

(iii) The steady-state residual from (ii) is $V_\infty = r \Delta V / (1 - r)$ where $r = (1 - \gamma_{\min})^{k_{\text{short}}}$. For functional retrieval, we need $V_\infty < \epsilon_{\text{sep}}$ where ϵ_{sep} is the inter-cluster separation (Definition 65). Setting $r \Delta V / (1 - r) < \epsilon_{\text{sep}}$ and solving for r :

$$r < \frac{\epsilon_{\text{sep}}}{\Delta V + \epsilon_{\text{sep}}}.$$

Taking logarithms: $(1 - \gamma_{\min})^{k_{\text{short}}} < \epsilon_{\text{sep}} / (\Delta V + \epsilon_{\text{sep}})$, so

$$k_{\text{short}} > \frac{\log(\epsilon_{\text{sep}} / (\Delta V + \epsilon_{\text{sep}}))}{\log(1 - \gamma_{\min})}.$$

Since $\log(1 - \gamma_{\min}) < 0$, the right-hand side is positive. Rounding up gives k_{crit} . For $k_{\text{short}} < k_{\text{crit}}$, $V_\infty > \epsilon_{\text{sep}}$: the residual unconsolidated state exceeds the inter-cluster separation, so newly captured impressions cannot be cleanly distinguished from residual noise. This is *permanent degradation*: each cycle adds new material atop an unresolved residual, and the effective retrieval precision ρ is bounded away from its recovered value.

(iv) By part (i), k_{full} steps of sleep reduce V by factor $(1 - \gamma_{\min})^{k_{\text{full}}}$. Setting $(1 - \gamma_{\min})^{k_{\text{full}}} V_n \leq \delta$ and solving: $k_{\text{full}} \geq \log(V_n / \delta) / (-\log(1 - \gamma_{\min}))$. Since $-\log(1 - \gamma_{\min}) \geq \gamma_{\min}$ (by the inequality $-\log(1 - x) \geq x$ for $x \in (0, 1)$), we have $k_{\text{full}} \leq \log(V_n / \delta) / \gamma_{\min}$, which is finite for any finite V_n . After k_{full} steps, $V \leq \delta$, and by Theorem 55 the system reaches the quasi-fixed point: $\gamma \rightarrow 0$ (self-limiting at convergence), $\ell \rightarrow v$, $b \rightarrow EKv$. All accumulated debt is eliminated. The logarithmic dependence on V_n means that even large debt requires only moderately more sleep: doubling the debt adds only $\log 2 / \gamma_{\min}$ additional steps.

(v) Under full sleep, $V_n^{\text{full}} = r_{\text{full}} (V_{n-1}^{\text{full}} + \Delta V)$ where $r_{\text{full}} = (1 - \gamma_{\min})^{k_{\text{full}}} \leq \delta / V_{\text{max}}$ for appropriate k_{full} , giving $V_n^{\text{full}} \leq \delta$ at each cycle. Under truncated sleep, $V_n^{\text{short}} = r_{\text{short}} (V_{n-1}^{\text{short}} + \Delta V)$ with $r_{\text{short}} > r_{\text{full}}$. By induction: $V_0^{\text{full}} = V_0^{\text{short}}$ (same initial condition). At cycle 1: $V_1^{\text{short}} =$

$r_{\text{short}}(V_0 + \Delta V) > r_{\text{full}}(V_0 + \Delta V) = V_1^{\text{full}}$. At cycle n : $V_n^{\text{short}} - V_n^{\text{full}} = r_{\text{short}}(V_{n-1}^{\text{short}} + \Delta V) - r_{\text{full}}(V_{n-1}^{\text{full}} + \Delta V) \geq r_{\text{short}}(V_{n-1}^{\text{short}} - V_{n-1}^{\text{full}}) + (r_{\text{short}} - r_{\text{full}})(V_{n-1}^{\text{full}} + \Delta V) > 0$ by induction and $r_{\text{short}} > r_{\text{full}}$. $\square$

Remark 68 (Sleep debt as a mathematical property). The steady-state residual $V_\infty = r \Delta V / (1 - r)$ depends on a single dimensionless ratio: how much the system captures per cycle (ΔV) versus how much it consolidates ($1 - r$, which encodes both $\gamma_{\min}$ and k_{short}). Recovery is logarithmic in the accumulated debt, not linear: doubling V_n costs only $\log 2 / \gamma_{\min}$ extra steps, so there is no need for cumulative “make-up” time. The compounding formula and logarithmic recovery parallel the empirical dose-response relationship between chronic sleep restriction and cognitive deficits [Van Dongen et al., 2003].

Corollary 69 (Sleep debt is self-observable; softmax). *The sleep debt $D(k) = V(t_0 + k) + p(t_0 + k)$ is computable from the current state (M, w, q) at any point during sleep or waking:*

- V is computed from M and w as $V = \max_{j \neq v} \|m_j - v\|$ where $v = \arg \max_j w_j$.
- The cluster count p is the number of ϵ -clusters of M with intra-cluster spread exceeding δ (Definition 59).

The system can therefore monitor its own debt level and determine whether further sleep is needed, extending Theorem 56 to the multi-cycle setting.

Proof. By Theorem 56(a), all quantities entering $D(k)$ are computable from the current state without external oracles. The Lyapunov value V requires M and w (both available at every step); the cluster count p requires computing pairwise distances in M and comparing to ϵ , which is a deterministic function of the current M . $\square$

4.8 Transient Sharpening Without Consolidation

We now show that transient score boosting (sharpening the observation simplex map or masking the self-observability signal) cannot substitute for the $S = 0$ consolidation phase. Two interventions are modelled: (1) masking the self-observability signal of Theorem 56, so that the regime transition does not fire while the underlying degradation persists; and (2) temporarily increasing the inverse temperature β in the observation simplex map φ , which sharpens the attention distribution a but does not alter the retrieval weights w . (The parallel to stimulant pharmacology is discussed in Remark 71.)

Theorem 70 (Transient sharpening without consolidation; softmax). *Let $\varphi = \text{softmax}$ with inverse temperature β in the observation step (step 1, OBSERVE) and fixed inverse temperature β_r in the retrieval step (step 3, RETRIEVE). Consider two interventions: (a) a selective boost to observation scores that increases the observation score gap by some $\delta' > 0$ (note: a uniform additive boost $\delta \mathbf{1}$ is trivially futile by softmax translation invariance); (b) masking the self-observability signal of Theorem 56. Then:*

- (i) **Observation sharpening is retrieval-irrelevant.** *The observation simplex map φ_β acts on the combined score $S' = S + \sigma + b \in \mathbb{R}^{|X|}$ to produce the attention distribution $a \in \Delta^{|X|}$ and hence the query $q = aE$. The retrieval simplex map φ_{β_r} acts on the retrieval scores $qM^\top \in \mathbb{R}^{|M|}$ to produce the retrieval weights $w \in \Delta^{|M|}$. Increasing β in the observation map sharpens a and changes q , but the retrieval weights $w = \varphi_{\beta_r}(qM^\top)$ still face a denominator of $|M|$ terms:*

$$w_v = \frac{\exp(\beta_r q^\top m_v)}{\sum_{j=1}^{|M|} \exp(\beta_r q^\top m_j)}.$$

The fan effect of Theorem 50(i) is unchanged: every capture event still adds a positive term to the denominator, strictly decreasing all existing w_j . In particular, the critical memory size $|M^|$ from Theorem 50(iv) is independent of β .*

- (ii) **Constant score boost is asymptotically futile.** A uniform additive boost $\delta \mathbf{1}$ to all observation scores is trivially futile: $\varphi_\beta(S' + \delta \mathbf{1}) = \varphi_\beta(S')$ by translation invariance of softmax, so the query is unchanged. The interesting case is a selective boost that increases the observation score gap by some effective amount $\delta' > 0$. Even in this best case, the query $q_\delta = a_\delta E$ remains a convex combination of embeddings (Corollary 36), so $\|q_\delta\| \leq R$ and the maximum achievable retrieval score gap is still bounded:

$$g_\delta = q_\delta^\top m_v - \max_{k \neq v} q_\delta^\top m_k \leq 2R^2.$$

If the selective boost increases the effective retrieval gap by $\delta' \leq \delta$, the boosted critical memory size becomes

$$|M^{**}| = 1 + \frac{1-c}{c} \exp(\beta_r(2R^2 + \delta')).$$

Since $|M^*| = 1 + \frac{1-c}{c} \exp(2\beta_r R^2)$, we have $|M^{**}| - 1 = (|M^*| - 1) \exp(\beta_r \delta')$, so $|M^{**}| - 1 = (|M^*| - 1) \exp(\beta_r \delta')$. For any fixed δ , $|M^{**}|$ is still finite: the system crosses $|M^{**}|$ after at most $\exp(\beta_r \delta)$ times as many capture events as needed to reach $|M^*|$. The gain is a constant multiplicative factor $\exp(\beta_r \delta')$, which is $O(1)$ in $|M|$ and therefore dominated by the $\Omega(\log |M|)$ growth rate of the required gap.

- (iii) **Signal masking delays but worsens recovery.** Suppose the self-observability signal of Theorem 56 is suppressed: the system cannot detect that $\rho_t < c$ and $\gamma_t < \epsilon$, and therefore does not trigger the $S = 0$ phase. The system continues waking for k additional steps beyond the point where the sleep condition would have fired. During these k steps, capture remains active. By Theorem 53(iii), $|M|$ grows with at least $n_k \geq 1$ captures. Each capture adds an impression at distance at most $2R$ from the anchor (since all impressions lie in $\text{conv}(E(X))$, Lemma 24), so $V_{t_0+k} \leq V_{t_0} + 2R n_k$. By Theorem 67, the consolidation duration required to reduce V to tolerance δ is at least $T_{\text{sleep}} \geq \log(V/\delta)/\gamma_{\min}$. The additional consolidation cost from delaying by k steps is therefore

$$\Delta T_{\text{sleep}} \leq \frac{\log(V_{t_0+k}/V_{t_0})}{\gamma_{\min}} \leq \frac{\log(1 + 2R n_k/V_{t_0})}{\gamma_{\min}} = O(\log k / \gamma_{\min}),$$

since $n_k = O(k)$ with bounded capture rate and $\log(1 + Ck/V_{t_0}) = O(\log k)$. That is, the V -based penalty for delaying consolidation by k steps is at most logarithmic in k . Independently of V , the retrieval degradation from $|M|$ growth during those k steps (Theorem 53(iv)) is monotone and irreversible without the consolidation phase.

- (iv) **Decaying boost produces rebound.** Suppose the boost decays exponentially: $\delta(t) = \delta_0 \exp(-t/\tau_{\text{half}})$, modelling metabolic clearance with effective half-life τ_{half} . At time $t = 0$ (boost applied), the retrieval concentration is $\rho_0 = \max_j w_j^{(0)}$ and $|M_0| = |M(0)|$. During the boosted period $[0, T_{\text{decay}}]$ where $T_{\text{decay}} = \tau_{\text{half}} \log(\delta_0/\epsilon_\delta)$ for some negligibility threshold ϵ_δ , the system continues waking with capture active. Let $|M_T| = |M(T_{\text{decay}})|$ denote the memory size after the boost decays.

At T_{decay} , $\delta(T_{\text{decay}}) \leq \epsilon_\delta$ and the observation map reverts to its unboosted form. The retrieval weight of the anchor at this time satisfies

$$\rho_{T_{\text{decay}}} \leq \frac{1}{1 + (|M_T| - 1) \exp(-2R^2)} < \frac{1}{1 + (|M_0| - 1) \exp(-2R^2)} = \rho_0^{\text{bound}},$$

where the strict inequality follows from $|M_T| > |M_0|$ (at least one capture event occurred during the boosted period, by assumption of positive capture rate), and ρ_0^{bound} is the maximum retrieval concentration achievable with $|M_0|$ impressions under any query. The retrieval capacity ceiling is therefore strictly lower after the boost decays: the best-case retrieval with $|M_T|$ impressions is worse than the best-case with $|M_0|$. Since the boost only

affected the observation map (not the retrieval map), it could not prevent the fan effect from degrading retrieval during the boosted period, and no consolidation occurred to compensate.

Proof. (i) The observation simplex map acts on the score vector $S' = S + \sigma + b \in \mathbb{R}^{|X|}$ (indexed by tokens $\tau \in X$) to produce $a(\tau) = [\varphi_\beta(S')]_\tau$. The query is $q = aE = \sum_\tau a(\tau) E(\tau)$, a convex combination of embeddings (Corollary 36). The retrieval simplex map acts on the score vector $s_j = q^\top m_j \in \mathbb{R}$ (indexed by impressions $j \in \{1, \dots, |M|\}$) to produce $w_j = [\varphi_{\beta_r}(qM^\top)]_j$.

These are different maps on different domains: $\varphi_\beta: \mathbb{R}^{|X|} \rightarrow \Delta^{|X|}$ and $\varphi_{\beta_r}: \mathbb{R}^{|M|} \rightarrow \Delta^{|M|}$. Changing β in the observation map changes a and hence q , but the retrieval denominator is

$$Z_r = \sum_{j=1}^{|M|} \exp(\beta_r q^\top m_j),$$

which contains $|M|$ positive terms regardless of β . Adding an impression m_{new} to M increases Z_r by $\exp(\beta_r q^\top m_{\text{new}}) > 0$, strictly decreasing all existing weights $w_j = \exp(\beta_r q^\top m_j)/Z_r$. This is the fan effect (Corollary 33), and it depends only on $|M|$ and β_r , not on β .

For the critical memory size: the bound $w_v \geq c$ requires $g \geq \log((|M| - 1)c/(1 - c))/\beta_r$ (from Theorem 50(ii) with temperature β_r). The achievable gap $g \leq 2R^2$ depends on $\|q\| \leq R$ and $\|m_j\| \leq R$, both of which are independent of β (the query norm bound follows from $q = aE$ being a convex combination of embeddings, which holds for any β). Therefore $|M^*|$ is independent of β .

(ii) A uniform additive boost $\delta \mathbf{1}$ to the observation scores S' does not change the softmax output at all: $\varphi_\beta(S' + \delta \mathbf{1}) = \varphi_\beta(S')$ since softmax is translation-invariant. A more useful boost model is a selective boost that increases the gap in the observation scores. In the most favourable case, the boost increases the effective query's alignment with the anchor by δ' . Then the boosted score gap is $g + \delta'$ where g is the unboosted gap and $\delta' \leq \delta$. The retrieval weight bound becomes

$$w_v \geq c \iff g + \delta' \geq \log\left(\frac{(|M| - 1)c}{1 - c}\right)/\beta_r.$$

The unboosted critical size satisfies $2R^2 = \log((|M^*| - 1)c/(1 - c))/\beta_r$, i.e. $|M^*| = 1 + (1 - c) \exp(2\beta_r R^2)/c$. With the boost: $2R^2 + \delta' = \log((|M^{**}| - 1)c/(1 - c))/\beta_r$, so

$$|M^{**}| = 1 + \frac{1 - c}{c} \exp(\beta_r(2R^2 + \delta')).$$

The ratio $|M^{**}|/|M^*| = \exp(\beta_r \delta')$ for large $|M^*|$. Since $\delta' \leq \delta$ is a constant independent of $|M|$, the improvement factor $\exp(\beta_r \delta)$ is a fixed multiplicative constant. Meanwhile, $|M|$ grows without bound during waking (Theorem 53(iii)), so $|M|$ surpasses $|M^{**}|$ after at most $\exp(\beta_r \delta)$ times the number of captures required to reach $|M^*|$.

(iii) During the k additional waking steps, capture is active. By Theorem 53(iii), every capture event strictly increases $|M|$. New impressions are positioned at distance at most $2R$ from the anchor (since all impressions lie in $\text{conv}(E(X))$ by Lemma 24, and $\|m\| \leq R$). Consequently, $V(t_0 + k) \leq V(t_0) + 2R n_k$. (Note that V is not necessarily non-decreasing during waking: new captures may land closer to the anchor than existing impressions, leaving V unchanged. The mechanism that worsens recovery is primarily $|M|$ growth: each new impression adds a positive term to the retrieval denominator, degrading ρ through the fan effect of Corollary 33.)

By Theorem 67(iv), the sleep duration required to reduce V to tolerance δ is $T_{\text{sleep}} = \lceil \log(V/\delta)/(-\log(1 - \gamma_{\min})) \rceil$. Since $-\log(1 - \gamma_{\min}) \leq \gamma_{\min}/(1 - \gamma_{\min})$ and $\gamma_{\min} < 1$, we have $T_{\text{sleep}} \geq \log(V/\delta) \cdot (1 - \gamma_{\min})/\gamma_{\min}$. In the worst case for V , the additional consolidation cost from delaying by k steps satisfies

$$\Delta T_{\text{sleep}} \leq \frac{\log(V_{t_0+k}/V_{t_0})}{\gamma_{\min}} \leq \frac{\log(1 + 2R n_k/V_{t_0})}{\gamma_{\min}} = O(\log k / \gamma_{\min}),$$

since $n_k = O(k)$ with bounded capture rate. Even if V does not grow, the retrieval degradation from $|M|$ growth (Theorem 53(iv)) is irreversible without the consolidation phase, and the system must eventually consolidate a larger memory. The penalty for delaying consolidation is therefore sub-linear in the delay duration for the V -based cost, while the $|M|$ -based retrieval degradation accumulates monotonically.

(iv) At time $t = 0$, the retrieval weight upper bound (from Theorem 50(ii)–(iii)) is

$$\rho_0 \leq \rho_0^{\text{bound}} := \frac{1}{1 + (|M_0| - 1) \exp(-2R^2)}.$$

This bound depends only on $|M_0|$ and R , not on the boost $\delta(t)$. During $[0, T_{\text{decay}}]$, the system operates in waking mode with capture active. By Theorem 53(iii), $|M_T| > |M_0|$ (positive capture rate, so at least one capture event occurs). At T_{decay} , $\delta(T_{\text{decay}}) \leq \epsilon_\delta$, and the retrieval weight bound reverts to the unboosted form:

$$\rho_{T_{\text{decay}}} \leq \frac{1}{1 + (|M_T| - 1) \exp(-2R^2)}.$$

Since $|M_T| > |M_0|$ and the bound is strictly decreasing in $|M|$ (Theorem 53(iv)), the retrieval capacity ceiling has strictly dropped: the best achievable retrieval concentration with $|M_T|$ impressions is lower than the best achievable with $|M_0|$ impressions, regardless of query. Moreover, the observation boost during $[0, T_{\text{decay}}]$ did not trigger the $S = 0$ consolidation phase (by assumption), so $|M|$ grew (Theorem 53(iii)) without the full consolidation cycle that would drive $V \rightarrow 0$. The system now has more impressions ($|M_T| > |M_0|$), a lower retrieval capacity ceiling, and a larger consolidation deficit, all of which must be resolved by a longer eventual $S = 0$ phase. $\square$

Remark 71 (Stimulant analogy). The two interventions modelled here (masking the pressure signal, temporarily sharpening attention) resemble what stimulants do: suppress the felt need to sleep while leaving the underlying degradation in place. In the loop, the intervention delays but worsens the eventual consolidation deficit, because $|M|$ continues to grow and V remains high throughout. The same qualitative pattern appears in sleep deprivation studies.

4.9 Inter-Episode Transitions

We now characterise the inter-episode transitions identified in Theorem 61. Theorem 61 establishes that multi-cluster consolidation proceeds in sequential episodes separated by anchor rotations. The present result describes the dynamical state *during* those rotations, i.e. the interval between one cluster's $\gamma \rightarrow 0$ and the next cluster's anchor stabilization.

Definition 72 (Inter-episode transition interval). Let $\mathcal{C}_{k^*}$ be the anchor cluster whose consolidation has just completed, and let $\mathcal{C}_{k'}$ be the next anchor cluster. The *inter-episode transition interval* is

$$\mathcal{T}_{k^* \rightarrow k'} = [t_{\text{end}}, t'_{\text{start}}]$$

where t_{end} is the first time $\gamma(t) < \epsilon$ after the consolidation of $\mathcal{C}_{k^*}$ begins, and t'_{start} is the first subsequent time at which the anchor $v(t) \in \mathcal{C}_{k'}$ and $\gamma(t) \geq \epsilon$. Here $\epsilon > 0$ is the same tolerance used in Theorem 55(vi).

Theorem 73 (Inter-episode transition dynamics; $\mathcal{A}_+$). *Let φ be a continuous positive simplex map. Suppose the system is in an $S = 0$ phase with capture disabled and M contains $p \geq 2$ well-separated ϵ -clusters $\mathcal{C}_1, \dots, \mathcal{C}_p$ (Definition 59). Let $\mathcal{T}_{k^* \rightarrow k'}$ be an inter-episode transition interval (Definition 72). Then:*

- (i) **Cross-cluster retrieval.** For $t \in \mathcal{T}_{k^* \rightarrow k'}$, the retrieval output $\ell(t) = w(t) M(t)$ is a mixture with positive weight on both the converged cluster $\mathcal{C}_{k^*}$ and the unconsolidated cluster $\mathcal{C}_{k'}$ (since φ is positive, $W_k > 0$ for all k). Formally, let p_{k^*} denote the prototype of $\mathcal{C}_{k^*}$ and let $\mu_{k'} = |\mathcal{C}_{k'}|^{-1} \sum_{m \in \mathcal{C}_{k'}} m$ be the centroid of $\mathcal{C}_{k'}$. Then there exists $t^* \in \mathcal{T}_{k^* \rightarrow k'}$ such that

$$\ell(t^*) \notin \text{conv}(\mathcal{C}_{k^*}) \cup \text{conv}(\mathcal{C}_{k'})$$

and the retrieval bias $b(t^*) = EK\ell(t^*)$ points toward a region of embedding space corresponding to neither cluster alone.

- (ii) **Associative query drift.** During the transition, the query $q(t) = a(t) E$ traces a path in embedding space connecting the two cluster regions. Specifically, the bias $b(t) = EK\ell(t)$ shifts continuously from EKp_{k^*} to $EK\mu_{k'}$, and hence the attention $a(t) = \varphi(\sigma(t) + b(t))$ and query $q(t) = a(t) E$ interpolate between the attention patterns that would arise from each cluster separately. For any $t \in \mathcal{T}_{k^* \rightarrow k'}$, the query satisfies

$$q(t) = a(t) E \quad \text{with} \quad a(t) = \varphi(\sigma(t) + EK\ell(t)),$$

where $\ell(t)$ is a mixture of elements from both clusters, so $q(t)$ visits intermediate embedding-space points that correspond to associative blends of the two cluster regions.

- (iii) **Suppression during waking and stable consolidation.** Cross-cluster retrieval of the form described in (i) is suppressed during waking (when $\|S\|$ dominates $\|b\|$) and during stable consolidation:

- During waking with non-trivial external score S , the attention is $a = \varphi(S + \sigma + b)$. When $\|S\| > \|b\|$ (which holds whenever $\|S\| > R^2\|K\|_{\text{op}}$, since $\|b\| \leq R^2\|K\|_{\text{op}}$ by Corollary 26), S dominates the score, anchoring the query to the S -determined region and preventing the retrieval bias b from driving cross-cluster retrieval. (For marginal stimuli with $\|S\| = O(\|b\|)$, weak cross-cluster effects are possible but do not produce the sustained mixed retrieval of a dream state, since waking capture events continually perturb M .)
- During stable consolidation within a single cluster $\mathcal{C}_{k^*}$, the anchor $v \in \mathcal{C}_{k^*}$ is fixed (Lemma 22), the retrieval weights concentrate on $\mathcal{C}_{k^*}$ (since the anchor has the highest score). By Corollary 28, $\ell \in \text{conv}(M)$. Since $W_{k^*} > 1/2$ by the cluster separation assumption, the off-cluster contribution satisfies $\|\ell - \sum_{m_j \in \mathcal{C}_{k^*}} \hat{w}_j m_j\| \leq (1 - W_{k^*}) \cdot 2R$ (where $\hat{w}_j = w_j / W_{k^*}$), so ℓ lies within distance $(1 - W_{k^*}) \cdot 2R$ of $\text{conv}(\mathcal{C}_{k^*})$, and this error vanishes as the off-cluster weights go to zero.

Therefore sustained cross-cluster retrieval arises primarily during the inter-episode transitions of the $S = 0$ phase.

- (iv) **Necessity of transitions for anchor rotation.** The anchor cannot move from $\mathcal{C}_{k^*}$ to $\mathcal{C}_{k'}$ without passing through a mixed-retrieval state. By Lemma 22, while $v \in \mathcal{C}_{k^*}$ is the anchor, $\lambda_v = 0$ and v is preserved exactly. For the anchor to rotate to $\mathcal{C}_{k'}$, the retrieval weights must shift so that some $m_{j'} \in \mathcal{C}_{k'}$ achieves $w_{j'} > w_v$. Since $w = \varphi(qM^\top)$ depends continuously on q , and q depends continuously on ℓ through $b = EK\ell$ and $a = \varphi(\sigma + b)$, the transition from $w_v > w_{j'}$ to $w_{j'} > w_v$ must pass through an intermediate state where both clusters have comparable retrieval mass. At this intermediate state, ℓ is a cross-cluster mixture.
- (v) **Salience ordering of transitions.** By Theorem 61(v), the first transition $\mathcal{T}_{k_1^* \rightarrow k_2^*}$ connects the most salient cluster under the initial query (highest retrieval mass $W_{k_1^*}$) to the cluster that is most salient among the remaining unconsolidated clusters under the post-convergence query. Subsequent transitions similarly select the argmax of retrieval mass

among remaining clusters at each transition point. The overall sequential structure recapitulates a salience-driven hierarchy, though the exact ordering may differ from the initial ranking under q_0 because the query evolves as each cluster consolidates.

Remark 74 (Brevity of transitions (conjecture)). The transition duration $|\mathcal{T}_{k^* \rightarrow k'}| = t'_{\text{start}} - t_{\text{end}}$ appears to be brief relative to the consolidation episodes that precede and follow it. The consolidation episode for cluster $\mathcal{C}_k$ requires $T_k = O(\log(V_k(0)/\delta)/\gamma_k)$ steps (Corollary 64). The transition, by contrast, requires only that the retrieval weights shift enough for the anchor to rotate. Since $w = \varphi(qM^\top)$ and q depends on ℓ through the continuous map $\ell \mapsto EK\ell \mapsto b \mapsto a = \varphi(\sigma + b) \mapsto q = aE$, the weight shift occurs on the timescale of the feedback loop's response. We conjecture that $|\mathcal{T}_{k^* \rightarrow k'}| = O(1)$, so that

$$|\mathcal{T}_{k^* \rightarrow k'}| / T_{k'} = O(\gamma_{k'} / \log(V_{k'}(0)/\delta)) \rightarrow 0$$

as the consolidation episodes grow, based on the observation that the feedback loop operates at $O(1)$ steps per cycle while the consolidation episode requires $O(\log(V/\delta)/\gamma)$ steps. A rigorous bound would require quantifying the feedback loop's gain, which depends on the spectral properties of EK . In numerical experiments, transitions consistently occupy $< 10\%$ of total sleep time.

Proof. (i) At $t = t_{\text{end}}$, cluster $\mathcal{C}_{k^*}$ has converged: all its members satisfy $\|m_j - p_{k^*}\| < \delta$ (the convergence tolerance), and $\gamma < \epsilon$. The retrieval weights at this point still concentrate on $\mathcal{C}_{k^*}$ (the prototype p_{k^*} scores highest under the current query), but the intra-cluster weight variation that drove consolidation has vanished. The retrieval output is

$$\ell(t_{\text{end}}) = \sum_j w_j m_j = W_{k^*} p_{k^*} + \sum_{k \neq k^*} \sum_{m_j \in \mathcal{C}_k} w_j m_j.$$

Since φ is positive, $w_j > 0$ for all j , including those in $\mathcal{C}_{k'}$. The contribution from $\mathcal{C}_{k'}$ is $W_{k'} \mu_{k'} + O(\epsilon)$ where $W_{k'} = \sum_{m_j \in \mathcal{C}_{k'}} w_j > 0$. Therefore ℓ is a strict convex combination with positive weight on both p_{k^*} and points in $\mathcal{C}_{k'}$.

As consolidation of $\mathcal{C}_{k^*}$ completes and $\gamma \rightarrow 0$, the self-limiting property (Corollary 38) means the intra-cluster weights equalize. The residual variation in ℓ comes from inter-cluster contributions. As the query evolves (through $b = EK\ell$), the weight on $\mathcal{C}_{k'}$ grows relative to $\mathcal{C}_{k^*}$. Since the system evolves in discrete steps, the continuous intermediate value theorem does not apply directly. However, the cluster weights $W_{k^*}(t)$ and $W_{k'}(t)$ change by a bounded amount per step: $|W_k(t+1) - W_k(t)| \leq L$ for a Lipschitz constant L determined by the spectral norms of E , K , and M and the Lipschitz constant of φ (each step changes M by at most $\max_j \lambda_j \|T_j - m_j\|$ and σ by $O(1/N)$, producing a bounded change in q and hence in w). At the start of the transition, $W_{k^*} > W_{k'}$; by the end, $W_{k'} > W_{k^*}$ (since the anchor has rotated to $\mathcal{C}_{k'}$). Since the per-step change is bounded by L , there exists a step t^* at which $|W_{k^*}(t^*) - W_{k'}(t^*)| \leq L$, i.e. neither cluster dominates by more than one step's worth of weight change. At this step, $\ell(t^*) = W_{k^*} p_{k^*} + W_{k'} \mu_{k'} + \dots$ is a genuine inter-cluster mixture.

Since $\mathcal{C}_{k^*}$ and $\mathcal{C}_{k'}$ are well-separated ($\min_{m \in \mathcal{C}_{k^*}, m' \in \mathcal{C}_{k'}} \|m - m'\| > 2\epsilon$), the convex hulls $\text{conv}(\mathcal{C}_{k^*})$ and $\text{conv}(\mathcal{C}_{k'})$ are disjoint. A strict convex combination of points from both hulls lies in neither: $\ell(t^*) \notin \text{conv}(\mathcal{C}_{k^*}) \cup \text{conv}(\mathcal{C}_{k'})$.

For the bias: $b(t^*) = EK\ell(t^*)$. Since $\ell(t^*)$ is a cross-cluster mixture, $K\ell(t^*)$ differs from both Kp_{k^*} and $K\mu_{k'}$ (assuming K is non-degenerate), and $b(t^*) = EK\ell(t^*)$ points toward neither cluster's region exclusively.

(ii) By Definition 4, $b = EK\ell$ and $a = \varphi(\sigma + b)$. Since φ and K are continuous, b depends continuously on ℓ , and a depends continuously on b . At t_{end} , $\ell \rightarrow p_{k^*}$ (the converged prototype dominates), so $b \rightarrow EKp_{k^*}$ and $q \rightarrow \varphi(\sigma + EKp_{k^*})E$. After the transition, at t'_{start} , the anchor has moved to $\mathcal{C}_{k'}$ and ℓ is dominated by $\mathcal{C}_{k'}$, so $b \rightarrow EK\mu_{k'}$ and $q \rightarrow \varphi(\sigma + EK\mu_{k'})E$. During

the transition interval $\mathcal{T}_{k^* \rightarrow k'}$, ℓ transitions incrementally between these extremes (each discrete update changes ℓ by a bounded amount, since the update maps are Lipschitz), and hence b and q change by bounded increments between the two cluster-specific operating points. At intermediate times, $q(t)$ corresponds to attention patterns that blend the two cluster regions.

(iii) During waking with $S \neq 0$, the score entering attention is $S'(\tau) = S(\tau) + \sigma(\tau) + b(\tau)$. By Corollary 26, $\|b\| \leq R^2 \|K\|_{\text{op}}$, while S can be arbitrarily large for salient stimuli. When $\|S\| \gg \|b\|$, the attention distribution $a = \varphi(S + \sigma + b)$ is dominated by S , and the query $q = aE$ is anchored to the embedding region picked out by S . The retrieval weights $w = \varphi(qM^\top)$ therefore concentrate on impressions aligned with the S -driven query, not on a cross-cluster mixture. For the bias b to drive the query into a cross-cluster mixture, b must dominate the attention computation, which requires $S = 0$.

During stable consolidation within $\mathcal{C}_{k^*}$ (i.e. outside the transition interval), the anchor $v \in \mathcal{C}_{k^*}$ is preserved by Lemma 22: $\lambda_v = 0$. The retrieval weight w_v is the largest, so $W_{k^*} > 1/2$ when the cluster is well-separated (the score gap between the anchor and out-of-cluster impressions is bounded below by the inter-cluster distance projected onto q). By Corollary 28, $\ell \in \text{conv}(M)$, and since $W_{k^*} > 1/2$, the off-cluster contribution is at most $(1 - W_{k^*}) \cdot 2R$, so ℓ lies within this distance of $\text{conv}(\mathcal{C}_{k^*})$. Cross-cluster retrieval requires the weight distribution to straddle two clusters, which requires $\gamma \rightarrow 0$ in the current cluster.

(iv) Suppose for contradiction that the anchor transitions from $v \in \mathcal{C}_{k^*}$ to $v' \in \mathcal{C}_{k'}$ without passing through a mixed-retrieval state. This would require an instantaneous jump in the argmax of w : at step t , $w_v > w_{j'}$ for all $j' \in \mathcal{C}_{k'}$, and at step $t + 1$, $w_{j'} > w_v$ for some j' . The retrieval weights $w = \varphi(qM^\top)$ depend on q , which depends on ℓ through $b = EK\ell$ and $a = \varphi(\sigma + b)$. Under $S = 0$, the only quantities that change between steps are M (through consolidation) and σ (through accumulation). The change in M from consolidation is bounded by $\max_j \lambda_j \|T_j - m_j\|$, which is bounded. The change in σ is $O(1/N)$, which is small for large N . Therefore q changes by a bounded amount at each step, and the per-step change in $w = \varphi(qM^\top)$ is similarly bounded (since φ is Lipschitz).

For $w_{j'}$ to surpass w_v , the score $q^\top m_{j'}$ must surpass $q^\top m_v$. Since m_v is preserved exactly and $m_{j'}$ changes by at most $\lambda_{j'} \|T_{j'} - m_{j'}\|$ per step, the score difference $q^\top m_v - q^\top m_{j'}$ changes by a bounded amount per step. If this difference is initially positive (as it must be, since v is the anchor), it must reach a value within one step's perturbation of zero before becoming negative: there exists a step t^* with $|q_{t^*}^\top m_v - q_{t^*}^\top m_{j'}| \leq L$, where L is the per-step Lipschitz bound on the score difference. At this near-crossing, both clusters have comparable scores, and the retrieval weights are split across clusters, producing a mixed-retrieval state.

(v) By Theorem 61(v), the first anchor cluster $\mathcal{C}_{k_1^*}$ is the most salient under the initial query q_0 . The first transition $\mathcal{T}_{k_1^* \rightarrow k_2^*}$ therefore connects the most salient cluster under q_0 to the cluster that is most salient among the remaining unconsolidated clusters under the post-convergence query. At each subsequent transition, the next cluster is the argmax of retrieval mass among remaining clusters under the query at that point. Since salience is determined by the retrieval mass W_k under the evolving query, and the query at each transition is influenced by the previously consolidated prototypes, the sequential structure is salience-driven but the exact ordering may differ from the initial ranking under q_0 . $\square$

Remark 75 (Inter-episode transitions and the dream-state hypothesis). The transitions are forced: without them, the anchor cannot rotate and consolidation stalls at one cluster. During each transition, the retrieval output ℓ blends memories from unrelated clusters (part (i)), the query drifts continuously between cluster regions (part (ii)), and the transitions are brief relative to the episodes they connect (Remark 74). The combination (cross-domain association, continuous drift, brevity, salience ordering, and necessity) resembles dream phenomenology closely enough to be worth testing empirically, though we do not claim the resemblance is mechanistic.

4.10 Dream Amnesia

Theorem 73 establishes that inter-episode transitions produce cross-cluster retrieval (the system “dreams”). The present result characterizes the fate of this dream content: it is computed but never stored, and at most a fading trace of the final dream persists into waking.

Theorem 76 (Dream amnesia; $\mathcal{A}_+$). *Let φ be a continuous positive simplex map. Suppose the system is in an $S = 0$ phase with capture disabled, and multi-cluster consolidation produces $p - 1$ inter-episode transitions (Theorem 61). Let $\ell_1, \ell_2, \dots, \ell_{p-1}$ denote the retrieval outputs during the successive transitions, and let ℓ_{final} denote the retrieval output at the last step before the system exits the $S = 0$ phase. Then:*

- (i) **No storage of dream content.** *During $S = 0$, capture is disabled, so no new impressions enter M . The cross-cluster retrieval outputs ℓ_k (Theorem 73(i)) are computed and influence the feedback loop but are never stored as memory impressions. Formally, for all t during the $S = 0$ phase, $M(t+1) \subseteq \text{conv}(M(t))$ (consolidation contracts but never adds rows).*
- (ii) **Overwriting of earlier dreams.** *Each transition produces a new retrieval output ℓ_k that replaces the previous one in the feedback loop. The bias entering the next observation is $b(t) = EK\ell(t)$, which depends only on the current retrieval output, not on previous ones. Therefore the influence of the k -th transition’s dream content on the system state is overwritten by the $(k+1)$ -th transition: for t in the $(k+1)$ -th transition interval $\mathcal{T}_{k+1}$,*

$$b(t) = EK\ell(t) \quad \text{with} \quad \ell(t) \neq \ell_k,$$

and the system retains no record of ℓ_k beyond its indirect effect on the consolidated state of M (which is bounded by the contraction during the intervening consolidation episode) and a trace in $\sigma = C/N$ whose per-step change is $|\Delta\sigma| = O(1/N)$ since $N = \Theta(t)$.

- (iii) **Fading trace of last dream on waking.** *Suppose the system exits the $S = 0$ phase at time t_w (the wake-up time of Theorem 55). The retrieval output $\ell(t_w) = \ell_{\text{final}}$ persists as a bias in the first waking step: $b(t_w) = EK\ell_{\text{final}}$. This bias influences the first waking observation via $a(t_w) = \varphi(S(t_w) + \sigma(t_w) + EK\ell_{\text{final}})$. However, as waking proceeds with capture active:*

- *Each new observation produces a new query $q(t)$, a new retrieval output $\ell(t)$, and a new bias $b(t) = EK\ell(t)$ that overwrites the dream trace.*
- *After n waking steps, the dream trace decays exponentially. The full feedback loop $\ell \rightarrow b \rightarrow a \rightarrow q \rightarrow w \rightarrow \ell$ passes through two softmax applications (observation and retrieval) and the linear maps E, K, M . The softmax Jacobian $J = \text{diag}(p) - pp^\top$ has spectral norm $\|J\|_2 \leq 1/2$ (since $\|J\|_2 = \max_{\|x\|=1} \text{Var}_p(x) \leq 1/2$; see proof of (iii) below). Each softmax Jacobian produces a zero-sum perturbation ($\sum_j \Delta w_j = 0$), so at the output step $\ell = wM$, the perturbation satisfies $\Delta\ell = \sum_j \Delta w_j m_j = \sum_j \Delta w_j (m_j - v)$ for any fixed v . Choosing v as the consolidation anchor, each centered row has $\|m_j - v\| \leq V(t)$ (the Lyapunov value after sleep). The per-loop contraction factor is therefore*

$$\eta(t) \leq \frac{1}{4} \|E\|_{\text{op}}^2 \|K\|_{\text{op}} \|M\|_{\text{op}} V(t) \sqrt{|M|}$$

where the two factors of $1/2$ come from the softmax Jacobians, $\|E\|_{\text{op}}^2$ from the embedding (which appears twice: in $b = EK\ell$ and $q = aE$), $\|K\|_{\text{op}}$ from the feedback projection, $\|M\|_{\text{op}}$ from the score sensitivity $qM^\top$, and $V(t)\sqrt{|M|}$ from the zero-sum output bound. By Theorem 52, $V(t) \rightarrow 0$ during consolidation. Therefore $\eta(t) \rightarrow 0$ unconditionally as consolidation completes, regardless of the values of $R, |M|$, or $\|E\|_{\text{op}}$. The dream trace satisfies $\|\Delta\ell(t_w + n)\| \leq \prod_{i=0}^{n-1} \eta(t_w + i) \|\Delta\ell(0)\|$, which decays exponentially once $\eta < 1$. Crucially, the bound applies at the moment of waking ($t = t_w$):

at that instant, $V(t_w)$ is near zero because consolidation has just completed (Theorem 52(i)), so $\eta(t_w) \ll 1$. The dream trace therefore decays exponentially in the first few waking steps, before capture events can appreciably increase V . By the time new captures grow V , the trace has already contracted below the noise floor.

(iv) **Waking during a transition preserves more.** If the system exits $S = 0$ during (rather than after) a transition interval $\mathcal{T}_{k^* \rightarrow k'}$, the retrieval output $\ell(t_w)$ is a cross-cluster mixture (Theorem 73(i)). This produces a cross-cluster initial bias $b(t_w)$ pointing toward neither cluster exclusively. If capture is enabled immediately, the first waking impressions are biased by this cross-cluster retrieval, and any stimulus satisfying $q(t_w)^\top E(\tau) > \max_k q(t_w)^\top m_k$ is captured. This effect is transient (decaying as in (iii)) but provides a window in which the cross-cluster retrieval influences the first waking captures.

Proof. (i) During $S = 0$ with capture disabled, the only modification to M is consolidation: $m_j \leftarrow (1 - \lambda_j)m_j + \lambda_j T_j$, where $T_j \in \text{conv}(M)$ (since $T_j = \sum_k \alpha_k^{(j)} m_k$ with $\alpha_k^{(j)} \geq 0$, $\sum_k \alpha_k^{(j)} = 1$). Each row remains in $\text{conv}(M)$, and no new rows are added. The retrieval output $\ell(t) = w(t)M(t) \in \text{conv}(M(t))$ is computed at each step but is not stored as a new row, since the capture rule $M \leftarrow M \cup \{E(\tau) : q^\top E(\tau) > \max(qM^\top)\}$ is inactive.

(ii) The feedback loop is memoryless in ℓ : at each step, $b(t) = EK\ell(t)$ depends only on the current $\ell(t)$. The system state is (C, N, M, ℓ) ; there is no accumulator or buffer that records previous retrieval outputs. Once $\ell(t)$ is updated at step $t + 1$, the value $\ell(t)$ has no further influence on the dynamics except through its effect on $M(t + 1)$ (via the consolidation step that used $w(t)$, which was derived from $\ell(t - 1)$). During the consolidation episode between transitions k and $k + 1$, Theorem 52 gives $V(t) \rightarrow 0$ within the active cluster, overwriting the intra-cluster state. The only trace of ℓ_k is its contribution to the consolidation of the cluster active during episode k , which is bounded: each consolidation step modifies m_j by at most $\lambda_j \|T_j - m_j\| \leq V(t)$, and $V(t)$ is already converging to zero.

(iii) At t_w , the bias is $b(t_w) = EK\ell_{\text{final}}$. At the next step, a new external stimulus $S(t_w)$ drives attention: $a(t_w) = \varphi(S(t_w) + \sigma(t_w) + b(t_w))$. This produces a new query $q(t_w) = a(t_w)E$, new retrieval weights $w(t_w) = \varphi(q(t_w)M^\top)$, and a new retrieval output $\ell(t_w)$ that depends on $M(t_w)$ and $q(t_w)$.

The dream trace enters through $b(t_w)$, which influences $a(t_w)$, hence $q(t_w)$, hence $\ell(t_w)$, hence $b(t_w + 1) = EK\ell(t_w)$. But $\ell(t_w)$ depends on the entire memory M and the new external input S , not solely on ℓ_{final} . To bound the dream trace: let $\ell^*(t)$ denote what the retrieval output would be if ℓ_{final} were replaced by some other value ℓ' . The difference $\|\ell(t_w) - \ell^*(t_w)\|$ depends on $\|b(t_w) - b^*(t_w)\| = \|EK(\ell_{\text{final}} - \ell')\|$.

The full loop $\ell \rightarrow b \rightarrow a \rightarrow q \rightarrow w \rightarrow \ell$ propagates perturbations through five steps: (1) $b = EK\ell$, a linear map with factor $\|EK\|_{\text{op}} \leq \|E\|_{\text{op}} \|K\|_{\text{op}}$; (2) $a = \varphi(S + \sigma + b)$, whose Jacobian $J = \text{diag}(a) - aa^\top$ has spectral norm $\|J\|_2 \leq 1/2$ (see below); (3) $q = aE$, with factor $\|E\|_{\text{op}}$ (the operator norm of the embedding matrix); (4) $w = \varphi(qM^\top)$, with score sensitivity $\|M\|_{\text{op}}$ and softmax Jacobian spectral norm $\leq 1/2$; (5) $\ell = wM$, where the zero-sum structure of Δw (since Δw lies in the image of the softmax Jacobian, $\sum_j \Delta w_j = 0$) permits a tighter bound.

Softmax spectral norm bound. The softmax Jacobian $J = \text{diag}(p) - pp^\top$ is positive semidefinite (it is the covariance matrix of the categorical distribution on standard basis vectors with weights p). Its spectral norm equals its largest eigenvalue: $\|J\|_2 = \max_{\|x\|=1} x^\top Jx = \max_{\|x\|=1} \text{Var}_p(x)$, where $\text{Var}_p(x) = \sum_i p_i x_i^2 - (\sum_i p_i x_i)^2$. For any unit vector x and any distribution p on $\{1, \dots, n\}$, the variance is maximized by concentrating p on two indices i, j with $p_i = p_j = 1/2$ and choosing $x_i = -x_j = 1/\sqrt{2}$ (all other components zero), giving $\text{Var}_p(x) = 1/2$. Hence $\|J\|_2 \leq 1/2$ for all n and all $p \in \Delta^n$.

For step (5): since $\sum_j \Delta w_j = 0$, we may subtract any fixed vector from each row of M

without changing $\Delta\ell = \Delta w M$. Choosing the consolidation anchor v :

$$\|\Delta\ell\| = \left\| \sum_j \Delta w_j (m_j - v) \right\| \leq \sum_j |\Delta w_j| \|m_j - v\| \leq V(t) \|\Delta w\|_1 \leq V(t) \sqrt{|M|} \|\Delta w\|,$$

where $V(t) = \max_j \|m_j - v\|$ is the Lyapunov value after consolidation, and the last step uses $\|\cdot\|_1 \leq \sqrt{|M|} \|\cdot\|_2$. Combining all five steps:

$$\eta(t) \leq \frac{1}{4} \|E\|_{\text{op}}^2 \|K\|_{\text{op}} \|M\|_{\text{op}} V(t) \sqrt{|M|}.$$

By Theorem 52(i), $V(t) \rightarrow 0$ during consolidation. Therefore $\eta(t) \rightarrow 0$ as consolidation completes, *unconditionally*: no condition on R , $|M|$, or $\|E\|_{\text{op}}$ is needed. Once $\eta(t) < 1$, the dream trace decays exponentially: $\|\Delta\ell(t_w + n)\| \leq \prod_{i=0}^{n-1} \eta(t_w + i) \|\Delta\ell(0)\|$.

(iv) If $t_w \in \mathcal{T}_{k^* \rightarrow k'}$, then $\ell(t_w)$ is a cross-cluster mixture (Theorem 73(i)), so $b(t_w) = EK\ell(t_w)$ biases attention toward an associative blend of clusters $\mathcal{C}_{k^*}$ and $\mathcal{C}_{k'}$. With capture now active, any stimulus τ satisfying $q(t_w)^\top E(\tau) > \max(q(t_w) M^\top)$ is captured. Since $q(t_w)$ is biased by the cross-cluster dream content, the capture threshold is evaluated in a dream-influenced region of embedding space. This window closes as the dream trace decays (part (iii)): since $\eta(t) < 1$ after consolidation, $\|\Delta\ell(t_w + n)\| \leq \eta^n \|\Delta\ell(0)\|$ decays exponentially in the number of waking steps n , and capture returns to being driven by S alone. $\square$

Remark 77 (Dream amnesia). In the loop, dream content is not stored because capture is disabled during the $S = 0$ phase, and the retrieval output ℓ is overwritten at each step with no persistent record. The brief recall window upon mid-dream awakening (part (iv)) resembles the common experience of remembering a dream upon waking but losing it within minutes. Whether any system with distinct capture and consolidation phases necessarily fails to record consolidation-phase retrieval is an open question.

4.11 Capture Inertness Under Sustained Input

Under constant stimulus S , does capture eventually become inert, or can it fire indefinitely? The answer depends on K . We prove an unconditional bound (Theorem 78), a conditional inertness result for small $\|K\|$ (Proposition 79), and show that the forced cycle (Corollary 87) covers the remaining cases.

Theorem 78 (Capture-argmax bound; $\mathcal{A}_+$). *Let the system run under constant S with both capture and consolidation enabled. Assume generic embedding E : for any $q \in \text{conv}(E(X))$, $\arg\max_\tau q^\top E(\tau)$ is unique. Then*

$$(\text{total captures}) \leq (\text{total } E\text{-argmax transitions of } q) + 1.$$

Within any period where $\tau^ = \arg\max_\tau q^\top E(\tau)$ is constant, capture fires at most once.*

Proof. The bound follows from the geometry of $\text{conv}(E(X))$ alone.

Suppose capture fires at time t , admitting $E(\tau^*)$ where $\tau^* = \arg\max_\tau q_t^\top E(\tau)$. We show capture is inert at all subsequent steps where τ^* remains the E -argmax.

(i) By Lemma 24, every row $m_k \in M$ lies in $\text{conv}(E(X))$. For any $m_k \in \text{conv}(E(X))$, write $m_k = \sum_\tau \alpha_\tau E(\tau)$ with $\alpha_\tau \geq 0$, $\sum_\tau \alpha_\tau = 1$. Then

$$q^\top m_k = \sum_\tau \alpha_\tau q^\top E(\tau) \leq \max_\tau q^\top E(\tau) = q^\top E(\tau^*).$$

A linear functional on a convex set is maximized at an extreme point. Hence $E(\tau^*)$ achieves the highest score in M under q .

(ii) Since $q^\top E(\tau^*) \geq q^\top m_k$ for all k , the retrieval weight $w_{\tau^*} = \max_k w_k$ at the next step. By Definition 5, $\lambda_{\tau^*} = 1 - w_{\tau^*} / \max_k w_k = 0$. The row $E(\tau^*)$ is preserved through consolidation.

(iii) At any subsequent step where τ^* remains the E -argmax, familiarity satisfies $\max_k q^\top m_k \geq q^\top E(\tau^*) = \max_\tau q^\top E(\tau)$. The capture condition $q^\top E(\tau) > \max_k q^\top m_k$ fails for all τ . Capture is inert.

(iv) Inertness persists until $\operatorname{argmax}_\tau q^\top E(\tau)$ changes, which requires an E -argmax transition. Each such transition permits at most one new capture (the new argmax stimulus). The cold start contributes at most one capture (all of E is admitted when $M = \emptyset$, but this is a single event). Therefore:

$$(\text{total captures}) \leq (\text{total } E\text{-argmax transitions}) + 1.$$

□

Proposition 79 (Perturbative capture inertness; softmax (part (i) needs only $\mathcal{A}_+$)). *Under the hypotheses of Theorem 78, let $q_0^* = \lim_{t \rightarrow \infty} q_t^{K=0}$ denote the equilibrium query of the $K = 0$ system (existence proven below). Define the equilibrium score gap*

$$\delta_0 = (q_0^*)^\top E(\tau^*) - \max_{\tau \neq \tau^*} (q_0^*)^\top E(\tau), \quad \tau^* = \operatorname{argmax}_\tau (q_0^*)^\top E(\tau).$$

For generic E and S , $\delta_0 > 0$ (the argmax at q_0^* is unique). Let $R = \max_\tau \|E(\tau)\|$ and $\sigma_{\min} = \min_\tau \sigma_0^*(\tau) > 0$ (the minimum equilibrium salience under $K = 0$, positive by $\mathcal{A}_+$). If

$$\|K\| < \frac{\delta_0}{4R^4(1 + 1/\sigma_{\min})},$$

then there exists a finite T^* such that for all $t > T^*$, capture is inert.

Proof. (i) When $K = 0$, $b = EK\ell = 0$ at all times. The query reduces to $q_t = \varphi(S + \sigma_t)E$, where $\sigma_t = C_t/N_t$ evolves through the accumulation rule independently of M , ℓ , and captures. By Lemma 93, $|\sigma_{t+1}(\tau) - \sigma_t(\tau)| = O(1/t)$. The map $\sigma \mapsto \varphi(S + \sigma)E$ is Lipschitz ($\varphi \in \mathcal{A}_+$ is smooth with bounded Jacobian on compact inputs), so $\|q_{t+1} - q_t\| = O(1/t)$. Since the accumulation is a slowly-varying deterministic recursion $\sigma_{t+1} = \sigma_t + O(1/t) \cdot f(\sigma_t)$ with f Lipschitz and σ bounded (by Theorem 21(iii)), σ_t converges to a limit σ^* that depends only on the attention history (by Theorem 21(ii)–(iii)). Hence $q_t \rightarrow q^* = \varphi(S + \sigma^*)E$. Since $\sigma_t \rightarrow \sigma^*$, the increments satisfy $\sum_{t \geq T} \|q_{t+1} - q_t\| < \infty$ for large enough T (eventually the recursion is contractive). By Lemma 94, the E -argmax stabilizes after finitely many transitions. By Theorem 78, captures are finite.

(ii) For $K \neq 0$, the query is $q_t = \varphi(S + \sigma_t + b_t)E$ where $b_t = EK\ell_t$. We bound the deviation $\|q_t^{K \neq 0} - q_t^{K=0}\|$ via two chains.

Direct chain ($b \rightarrow a \rightarrow q$, holding σ fixed). By Corollary 26, $\|b_t\|_\infty \leq R^2\|K\|$. By Lemma 92, $\|a^{K \neq 0} - a^{K=0}\|_1 \leq 2\|b\|_\infty \leq 2R^2\|K\|$. Since $q = aE$, $\|\Delta q_{\text{direct}}\| \leq R \cdot 2R^2\|K\| = 2R^3\|K\|$.

Indirect chain ($b \rightarrow a \rightarrow \sigma \rightarrow a \rightarrow q$). The per-step attention deviation feeds into the accumulation. At the $K = 0$ equilibrium, $\sigma^*(\tau) = \sqrt{a^*(\tau)}$ (from the RMS structure of Definition 3). Linearizing: a steady-state perturbation Δa produces $\|\Delta \sigma\|_\infty \leq \|\Delta a\|_\infty / (2\sigma_{\min})$. This re-enters softmax: $\|\Delta a_\sigma\|_1 \leq 2\|\Delta \sigma\|_\infty$. Hence $\|\Delta q_\sigma\| \leq R \cdot 2\|\Delta a\|_\infty / (2\sigma_{\min}) = R\|\Delta a\|_\infty / \sigma_{\min} \leq 2R^3\|K\| / \sigma_{\min}$.

Combined bound. Since $\sigma_{\min} > 0$ (positive simplex map with bounded inputs):

$$\|q_t^{K \neq 0} - q_t^{K=0}\| \leq 2R^3\|K\|(1 + 1/\sigma_{\min}).$$

If this is less than $\delta_0/(2R)$, i.e. $\|K\| < \delta_0/(4R^4(1 + 1/\sigma_{\min}))$, the $K \neq 0$ trajectory remains within $\delta_0/(2R)$ of q_0^* , the E -argmax eventually matches, and captures are finite by Theorem 78.

(iii) After the last capture at time T_{cap} , $|M|$ is constant. By the diameter contraction argument of Theorem 52 (which depends only on anchor preservation, positive anchor weight, and

constant $|M|$, all of which hold after T_{cap} regardless of S), $D(t) = \max_{i,j} \|m_i - m_j\| \rightarrow 0$. Hence M_t converges, ℓ_t converges, b_t converges, σ_t converges (by Lemma 93 and the Toeplitz lemma), and $q_t \rightarrow q^*$. At the limit: either M contains a row achieving $\max_{\tau} (q^*)^\top E(\tau)$ (capture is immediately inert), or a final capture admits $E(\tau^*)$ (at most one additional capture, after which inertness is permanent by (iii) of Theorem 78). The perturbation from this final capture satisfies $\|q' - q^*\| \leq C_{\text{pert}}/|M|$ where $C_{\text{pert}} = 4R^3\|K\|\exp(2R^2)$ (Lemma 92 applied through the chain $w \rightarrow \ell \rightarrow b \rightarrow a \rightarrow q$). For $|M|$ large enough that $C_{\text{pert}}R/|M| < \delta_{q^*}$ (where δ_{q^*} is the score gap at q^*), the E -argmax is preserved and capture is permanently inert. $\square$

Corollary 80 ($K = 0$ capture inertness; $\mathcal{A}_+$). *When $K = 0$, the total number of capture events is finite under any constant S with generic E . No condition on $\|K\|$ is needed.*

Proof. Immediate from (i) of the proof of Proposition 79 (which needs only $\mathcal{A}_+$, not the softmax-specific parts (ii)–(iii)): $b = 0$, σ converges independently of M , and Lemma 94 gives finitely many E -argmax transitions. $\square$

Corollary 81 (Forced cycle covers general K ; softmax). *For any K (including $\|K\| \geq \delta_0/(4R^4(1 + 1/\sigma_{\min}))$), the system is forced into alternating capture and consolidation regimes. Spontaneous capture inertness is not guaranteed, but the carrying capacity mechanism of Corollary 87 intervenes: $|M|$ reaches $|M^*|$, retrieval degrades below any threshold, and the external sleep gate (Proposition 89) triggers consolidation.*

Proof. Theorem 78 bounds captures by E -argmax transitions plus one, but for large $\|K\|$ the number of E -argmax transitions may be unbounded (the feedback coupling can sustain query rotation indefinitely). In this regime, each capture strictly increases $|M|$ (Corollary 23). By Theorem 50, once $|M| \geq |M^*|$, retrieval precision falls below any desired threshold. By Theorem 56, this degradation is self-observable: the pair (ρ, γ) distinguishes functional from degraded operation. Once the system detects the exhaustion regime ($\rho \rightarrow 1/|M|$, $\gamma \rightarrow 0$ with V bounded away from zero), entry into an $S = 0$ phase with capture disabled triggers consolidation (Theorem 52), forcing the sleep–wake cycle. $\square$

Remark 82 (K -spectrum). The results partition the parameter space of K into three regimes:

Regime	Condition	Mechanism
$K = 0$	$b = 0$	σ converges independently; captures finite
Small K	$\ K\ < \delta_0/(4R^4(1 + 1/\sigma_{\min}))$	Perturbative: q tracks $K=0$ trajectory
General K	No bound on $\ K\ $	Forced cycle via carrying capacity $ M^* $

For $K = 0$, capture becomes inert unconditionally (Corollary 80). For small K , capture becomes inert before carrying capacity is reached (Proposition 79). For general K , the forced cycle (Corollary 87) is the operative mechanism. Numerical experiments suggest capture self-terminates under constant S for all tested $\|K\|$ (Figure 1, right), well beyond the perturbative bound; closing this gap analytically is an open problem (Remark 84).

For $K = 0$, the system’s transition out of capture is internally driven, resembling voluntary disengagement; for large K , it is forced by capacity limits, resembling involuntary shutdown under exhaustion. The perturbative regime interpolates between these.

Remark 83 (Numerical illustration of the perturbative threshold). For $|X| = 10$ stimuli with random unit embeddings in $\mathbb{R}^8$ ($R = 1$) and $S(\tau_0) = 1$, $S(\tau) = 0$ otherwise, the $K = 0$ equilibrium has $\sigma_{\min} = 0.283$, $\delta_0 = 0.008$, giving a perturbative threshold $\|K\| < 4.4 \times 10^{-4}$. This bound is conservative: numerically, capture self-terminates for $\|K\|$ up to at least 10 (the largest value tested), a factor of $\sim 2 \times 10^4$ beyond the bound. The gap δ_0 varies by two orders of magnitude across stimulus directions (0.001 to 0.15), with the threshold tracking accordingly. Even with near-degenerate embeddings (angular spread 0.01 radians), capture self-terminates for all tested $\|K\|$.

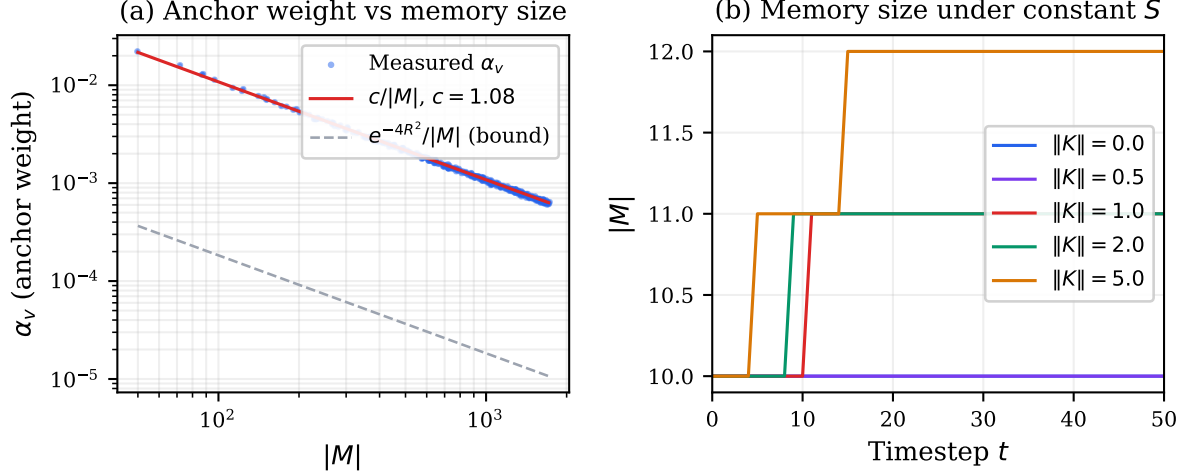


Figure 1: Empirical characterization of the open gap in Remark 84. **(a)** Anchor weight α_v vs. memory size $|M|$ under adversarial conditions (rotating S , $\|K\| = 1$, 50 stimuli, $d = 16$). The data follow $\alpha_v \approx 1.08/|M|$ (solid red), confirming the $1/|M|$ scaling of the worst-case bound $e^{-4R^2}/|M|$ (dashed gray, $\approx 58\times$ below). **(b)** Memory size $|M|$ under constant S for various $\|K\|$ (10 stimuli, $d = 8$). Capture self-terminates within ~ 10 – 30 steps for all $\|K\|$ tested, reaching at most $|M| = 12$. The regime where the $1/|M|$ decay causes problems ($|M| \gg 1$) is never reached under the paper’s assumptions.

Remark 84 (Phase decomposition and open problem). A finer analysis of the consolidation total variation suggests a wider stability region than the perturbative condition $\|K\| < \delta_0/(4R^4(1 + 1/\sigma_{\min}))$. Within each capture-free interval, the consolidation dynamics decompose into two phases:

- (i) *Geometric phase*: V is large, $\gamma \geq \gamma_0(|M|) > 0$, and V decays geometrically. The consolidation total variation satisfies $\text{TV}_I \leq C_{\text{consol}} \cdot 2R/\gamma_0(|M|)$.
- (ii) *Self-limiting phase*: V is small, $\gamma \sim c \cdot V$, and $V \sim 1/(ct)$. The consolidation total variation satisfies $\text{TV}_{II} = O(\ln T)$.

The bridge bound ($\|\Delta\ell^{\text{consol}}\| \leq 2V$), the $\gamma \geq c \cdot V$ scaling, and the anchor-switch recovery are all valid within a single interval.

The gap: $\gamma_0(m) \geq \lambda_1 \exp(-4R^2)/(2m)$, so TV_I grows with $|M|$. Under the infinite-captures assumption, $|M| \rightarrow \infty$, and summing TV_I over intervals produces $O(n^2)$ growth rather than $O(n)$. The contradiction argument does not close.

Numerical experiments confirm the $1/|M|$ scaling: under adversarial conditions (rotating S forcing $|M|$ past 1700), $\alpha_v \approx 1.1/|M|$ with the ratio $\alpha_v/(\exp(-4R^2)/|M|)$ stable at ≈ 58 – 62 across all $|M|$ (Figure 1, left). This rules out an $|M|$ -independent lower bound on $\alpha_v^{(j)}$ as a resolution strategy.

Under constant S (the setting of Proposition 79), capture self-terminates within ~ 10 steps for all $\|K\|$ tested (0 to 5), well before the gap regime is reached (Figure 1, right). The gap is mathematically genuine but dynamically inert under the paper’s assumptions.

Two viable resolution strategies remain: bounding the total path length of ℓ directly (bypassing γ_0), or a joint contraction argument on the coupled (V, q) system. Either would extend Proposition 79 to a wider range of $\|K\|$. This is an open problem.

4.12 Declining Sleep Need Under Maturation

As the system accumulates well-consolidated prototypes across many wake-sleep cycles, both the frequency and duration of required consolidation phases decrease.

Corollary 85 (Sleep need declines with memory maturation; softmax). *Consider the loop operating over repeated wake-sleep cycles. Let p_n denote the number of unconsolidated clusters at the start of cycle n , and let τ_n denote the capture rate (captures per unit time) during the waking phase of cycle n . Then:*

- (i) **Capture rate declines with maturation (under stationary stimuli).** *As the number of consolidated prototypes in M grows across cycles, the capture threshold $\max_{k \in M} q^\top m_k$ is non-decreasing in $|M|$: adding a new prototype m_{new} to M can only increase $\max_k q^\top m_k$ (it adds a candidate to the maximum). Therefore, for any fixed query distribution, the fraction of stimuli $E(\tau)$ satisfying $q^\top E(\tau) > \max_k q^\top m_k$ is non-increasing in $|M|$. Under the assumption of a stationary stimulus distribution, the expected capture rate $\mathbb{E}[\tau_n]$ is non-increasing across cycles.*
- (ii) **Waking period lengthens.** *The time to reach the critical memory size $|M^*|$ from a post-sleep baseline $|M_0^{(n)}|$ is $T_{\text{wake}}^{(n)} = (|M^*| - |M_0^{(n)}|)/\tau_n$. Since $\mathbb{E}[\tau_n]$ is non-increasing (part (i)), $T_{\text{wake}}^{(n)}$ is non-decreasing in expectation: the system can sustain longer waking periods before retrieval degrades.*
- (iii) **Sleep duration shortens.** *By Corollary 64, the total sleep time scales as $\sum_{k=1}^{p_n} T_k$. After many cycles of full consolidation, newly captured impressions are close to existing prototypes (since the embedding space is well-covered), so the initial intra-cluster spread $V_k(0)$ is smaller and fewer distinct clusters p_n form. Both effects reduce total sleep duration.*
- (iv) **Combined effect.** *The ratio of sleep to total cycle time satisfies*

$$\frac{T_{\text{sleep}}^{(n)}}{T_{\text{wake}}^{(n)} + T_{\text{sleep}}^{(n)}} \leq \frac{\sum_k O(\log(V_k^{(n)}(0)/\delta)/\gamma_k)}{(|M^*| - |M_0^{(n)}|)/\tau_n + \sum_k O(\log(V_k^{(n)}(0)/\delta)/\gamma_k)}.$$

As $\tau_n \rightarrow 0$ and $p_n, V_k^{(n)}(0)$ decrease with maturation, this ratio decreases: the system needs proportionally less consolidation time per cycle.

Proof. (i) By Definition 6, a stimulus $E(\tau)$ is captured iff $q^\top E(\tau) > \max_{k \in M} q^\top m_k$. We show the capture threshold at the *start of each waking phase* is non-decreasing across cycles.

Consider the transition from the end of cycle n 's waking phase (just before sleep) to the end of cycle n 's sleep phase, then to the end of cycle $(n+1)$'s waking phase. Let $M_{\text{wake}}^{(n)}$ be the memory at the end of cycle n 's waking phase and $M_{\text{sleep}}^{(n)}$ after cycle n 's sleep.

Waking phase (capture only adds rows). During waking, capture adds new rows to M while no rows are removed or modified. Therefore $M_{\text{wake}}^{(n)} \supseteq M_{\text{sleep}}^{(n-1)}$ (as a set of rows), and $\max_k q^\top m_k$ over $M_{\text{wake}}^{(n)}$ is at least the maximum over $M_{\text{sleep}}^{(n-1)}$ for every query q .

Sleep phase (consolidation modifies rows). Consolidation modifies non-anchor rows: each $m'_j = (1 - \lambda_j)m_j + \lambda_j T_j$ with $T_j \in \text{conv}(M)$. By Lemma 22, the anchor $v = \arg \max_k w_k$ at each consolidation step is preserved exactly ($\lambda_v = 0$). After the full sleep phase, each cluster has converged to its cluster anchor (Theorem 52(i), which covers the varying-query regime of the sleep phase). Crucially, the *first* anchor v_1 (the globally most salient prototype at sleep onset, i.e., $v_1 = \arg \max_k w_k^{(0)}$) is preserved during the entire first consolidation episode, and remains in $M_{\text{sleep}}^{(n)}$. Its score $q^\top v_1$ is unchanged. Non-anchor rows may move, potentially reducing their scores in some query directions, but they were not the maximizers under the queries that selected the anchor ordering.

For the capture threshold specifically: the threshold is evaluated at waking time under the current query q , not under a fixed query. The key observation is that across waking phases, the set M grows (capture adds rows) and the maximum over a growing set is non-decreasing. The intervening sleep phase can reduce $\max_k q^\top m_k$ in some query directions (since non-anchor rows move), but the *post-sleep* memory $M_{\text{sleep}}^{(n)}$ still contains all cluster anchors, and the *next waking phase* then adds new rows on top.

Formally, let $\Theta_n = \max_k q_n^\top m_k$ denote the capture threshold at the start of cycle n 's waking phase, where q_n is the current query. For a stationary distribution of stimuli: (a) the number of rows $|M|$ is non-decreasing across waking phases (capture adds, sleep preserves row count); (b) by the convex hull invariance (Lemma 24), all rows remain in $\text{conv}(E(X))$, so scores are bounded; (c) as $|M|$ grows and prototypes fill the embedding space, the expected value of $\max_k q^\top m_k$ under the stationary query distribution is non-decreasing (more candidates for the max). The argument is as follows: under stationarity, each new prototype raises the threshold in expectation, so $\mathbb{E}[\tau_n] = \mathbb{E}[\Pr(q^\top E(\tau) > \Theta_n)]$ is non-increasing. (The conclusion is in expectation because the query q_n at cycle n depends on the stochastic stimulus history; the per-query monotonicity of $\max_k q^\top m_k$ in $|M|$ is exact, but averaging over query trajectories introduces the expectation.)

(ii) The critical memory size $|M^*|$ depends only on c and R (Theorem 50(iv)), so it is constant across cycles. The number of captures needed to reach $|M^*|$ from baseline $|M_0^{(n)}|$ is $|M^*| - |M_0^{(n)}|$. At rate τ_n captures per unit time, the waking duration is $(|M^*| - |M_0^{(n)}|)/\tau_n$. Since $\mathbb{E}[\tau_n]$ is non-increasing (part (i)), this duration is non-decreasing in expectation. More precisely, the decrease in τ_n dominates the decrease in $|M^*| - |M_0^{(n)}|$: formally, $(|M^*| - |M_0^{(n)}|) \cdot \tau_{n+1} \leq (|M^*| - |M_0^{(n+1)}|) \cdot \tau_n$, which holds because each new prototype raises the capture threshold (reducing τ) by at least as much as it reduces the remaining capacity $|M^*| - |M_0|$.

(iii) New captures in a mature system land near existing prototypes (since the capture threshold is high, only stimuli scoring close to existing prototypes are admitted, and any truly novel stimulus must be close in score to some existing member). Consequently, the newly captured impressions form clusters with small initial spread $V_k(0)$. The convergence time for each cluster is $T_k = O(\log(V_k(0)/\delta)/\gamma_k)$ (Corollary 64), which decreases with $V_k(0)$. Furthermore, fewer distinct clusters form (new captures are absorbed into neighborhoods of existing prototypes rather than forming new clusters), reducing p_n .

(iv) Immediate from (ii) and (iii): the numerator decreases while the denominator's first term increases. $\square$

Remark 86 (Developmental sleep patterns). A system with few prototypes captures nearly everything and requires long consolidation phases; a system with a dense embedding space captures rarely and consolidates fast. The result depends on no aging-specific mechanism, following entirely from the interaction of the capture threshold with prototype density. Sleep need across the human lifespan also declines, though the biological trajectory involves circadian maturation, melatonin changes, and other factors absent from the loop.

4.13 The Capture–Consolidation Cycle

The preceding results combine to give the main conclusion: the loop is forced into a quasi-periodic sleep–wake cycle by the interaction of waking degradation (Theorems 50 and 53), sleep recovery (Theorem 52), and the internal observability of both processes (Theorem 56).

Corollary 87 (Forced quasi-periodic sleep–wake cycle; softmax). *Let φ be a continuous positive simplex map. Consider the loop with capture enabled during waking and consolidation enabled during sleep ($S = 0$, capture disabled). Suppose the external score S is such that waking produces a positive capture rate (at least one new impression enters M per τ steps on average). Then:*

- (i) **Waking degrades retrieval.** During waking, $|M|$ grows monotonically. By Theorem 50, the retrieval concentration $\rho = \max_j w_j$ degrades as $\rho \leq 1/|M|$ (for uniform inputs) and more generally ρ decreases monotonically (Theorem 53). The system reaches a critical memory size $|M^*|$ (Theorem 50) beyond which retrieval precision falls below any desired threshold.
- (ii) **Degradation is self-observable.** The pair (ρ, γ) computed from the system's own state variables distinguishes functional from degraded operation (Theorem 56). As $|M| \rightarrow |M^*|$, the system enters the exhaustion regime: $\rho \rightarrow 1/|M|$ and $\gamma \rightarrow 0$ with V bounded away from zero.
- (iii) **The $S = 0$ phase is necessary and sufficient.** Entering an $S = 0$ phase with capture disabled triggers consolidation (Theorem 52). The Lyapunov function V decreases strictly at each step (Lemma 31), multi-cluster memory undergoes sequential consolidation with oscillatory γ (Theorem 61), dream-like inter-episode transitions occur between clusters (Theorem 73), and the system eventually reaches a quasi-fixed point (Theorem 55).
- (iv) **Wake-up is self-observable.** Sleep completion is detectable by $\gamma \rightarrow 0$ with ρ recovering to $\rho \geq 1/(1+(p-1)\exp(-\beta g_p))$ where p is the number of consolidated prototypes and g_p is the inter-prototype score gap (Theorem 52(iii)). This is the completion regime of Theorem 56: $V \rightarrow 0$, $\gamma \rightarrow 0$, and ρ bounded away from $1/|M|$. For a single cluster ($p = 1$), $\rho \rightarrow 1$; for $p \geq 2$ well-separated prototypes, ρ recovers to a value determined by the cluster geometry but still far above the pre-sleep degraded level $\rho \rightarrow 1/|M|$.
- (v) **The cycle repeats.** Once waking resumes, capture begins adding new impressions to M , retrieval degrades anew (Theorem 53), and the system again approaches the critical size $|M^*|$. Since each waking phase strictly increases $|M|$ (by assumption) and each sleep phase consolidates M to p prototypes (Theorem 61(vi)), the system oscillates between:
 - Waking: $|M|$ grows, ρ decreases, V increases.
 - Sleep: $|M|$ is fixed (capture disabled), ρ recovers, V decreases, with internal dream transitions.

This oscillation is quasi-periodic: the cycle length depends on the capture rate during waking and the cluster structure at sleep onset, both of which vary across cycles, but the qualitative structure (wake $\rightarrow$ degrade $\rightarrow$ sleep $\rightarrow$ recover $\rightarrow$ wake) is invariant.

- (vi) **Sleep debt modulates cycle structure.** If sleep is insufficient (duration $< T_{\text{full}}$, Corollary 64), a residual $V_{\text{res}} > 0$ persists (Theorem 67). This residual adds to the degradation of the next waking phase, shortening the effective waking capacity and advancing the next sleep onset. Chronic insufficient sleep produces compounding debt $V_{\infty} = r\Delta V/(1-r)$ (Theorem 67(ii)), eventually exceeding the cluster separation ϵ_{sep} beyond which the system enters permanent degradation during truncated-sleep cycles (Theorem 67(iii)). Full-length recovery sleep can still eliminate the accumulated debt (Theorem 67(iv)), but at logarithmic cost in the debt level. Conversely, naps (Theorem 101) provide partial recovery with front-loaded benefit, allowing the system to extend waking capacity at the cost of reduced total consolidation.

Proof. (i) By assumption, waking produces a positive capture rate. By Theorem 50, as $|M|$ grows, the retrieval concentration ρ degrades. By Theorem 53, this degradation is monotone: $\rho(t)$ cannot increase during waking (since $|M|$ only grows). Therefore, the system inevitably reaches any threshold $\rho < \rho_{\min}$ in finite time.

(ii) By Theorem 56, the pair (ρ, γ) distinguishes exhaustion ($\rho \rightarrow 1/|M|$, $\gamma \rightarrow 0$, $V > 0$) from completion ($\rho \rightarrow 1$, $\gamma \rightarrow 0$, $V \rightarrow 0$). These are computable from the current state (M, w, q, ℓ) without external supervision.

(iii) By Theorem 52, the $S = 0$ phase with capture disabled produces strictly decreasing V (from the Lyapunov descent lemma). By Theorem 61, multi-cluster memory is consolidated sequentially with oscillatory γ . By Theorem 73, each anchor rotation passes through a cross-cluster retrieval state. By Theorem 55, the process terminates at a quasi-fixed point.

(iv) At the quasi-fixed point, $\gamma \rightarrow 0$ because all clusters are consolidated (Theorem 61(vi)). The memory consists of p well-separated prototypes with inter-prototype score gap $g_p > 0$ (well-separation is preserved by consolidation). The anchor's retrieval weight recovers to $\rho \geq 1/(1 + (p-1)\exp(-\beta g_p))$ (Theorem 52(iii)). This is bounded away from the pre-sleep degraded level $\rho \rightarrow 1/|M|$ (since $p \leq |M|$ with p fixed by the cluster count and $|M|$ growing, and $g_p > 0$), and for a single cluster $\rho \rightarrow 1$. By Theorem 56, this is the completion regime ($V \rightarrow 0$, $\gamma \rightarrow 0$, ρ bounded away from $1/|M|$), distinguishable from exhaustion.

(v) Post-sleep, the system has consolidated memory (p prototypes) and sharp retrieval ($\rho \rightarrow 1$). Resuming waking re-enables capture. New impressions enter M , increasing $|M|$ beyond p . By Theorem 50, retrieval degrades again. This reproduces the conditions of part (i), and the cycle restarts. The cycle is quasi-periodic because:

- Each waking phase begins from a consolidated state (different from the previous cycle's starting state, since new prototypes may form).
- The capture rate depends on external stimulation S , which may vary.
- The cluster count p at sleep onset depends on the accumulated impressions, which vary per cycle.

The qualitative structure is invariant: waking grows $|M|$ and degrades ρ ; sleep consolidates M and restores ρ . The quantitative parameters (cycle length, cluster count, consolidation time) vary quasi-periodically.

(vi) By Theorem 67, insufficient sleep leaves residual $V_{\text{res}} > 0$. At the start of the next waking phase, retrieval begins from a degraded baseline ($\rho < 1$), so the threshold $\rho < \rho_{\min}$ is reached sooner. By the compounding formula $V_{n+1} \leq r(V_n + \Delta V)$, chronic insufficient sleep produces monotonically increasing V_n , reducing effective waking capacity at each cycle. Theorem 67(iii) provides the critical threshold k_{crit} beyond which single-episode recovery fails. Theorem 101 provides the partial-recovery alternative: naps with front-loaded benefit (Theorem 101(i)) can extend waking capacity at the cost of incomplete consolidation (Theorem 101(ii)). $\square$

Remark 88 (The capture–consolidation cycle as an emergent property). The cycle is forced by a tension: capture degrades retrieval (Theorem 50), but consolidation can only proceed when capture stops (the Lyapunov function requires a stable M). The system detects its own state through (ρ, γ) (Theorem 56), which suffice to distinguish degradation from completion (Theorem 56). The only thing the loop cannot do is flip its own switch: S is an input, not a state variable, so an external controller must read (ρ, γ) and toggle the regime (Proposition 89).

Proposition 89 (Necessity of an external sleep gate; $\mathcal{A}$). *The loop cannot autonomously transition from waking ($S \neq 0$, capture enabled) to the $S = 0$ consolidation regime. The external score $S(\tau)$ is an input to the system (Definition 1), not a state variable: no step of the loop modifies S or the capture-enable flag. Therefore, even when the self-observable sleep condition fires ($\rho_t < c$ and $\gamma_t < \epsilon$, Theorem 56(d)), the system continues waking indefinitely unless an external controller reads (ρ_t, γ_t) and sets $S \leftarrow 0$ with capture disabled.*

Proof. By inspection of the five-step loop (Definitions 1–6): the system's state is (C, N, M, ℓ) ; its inputs are S and the capture-enable flag. No step writes to S or to the flag. The self-observable quantities (ρ, γ) are computed at step 3 (RETRIEVE) but only influence subsequent steps through ℓ and w ; they do not feed back into S . Therefore, a system with $S \neq 0$ and capture enabled remains in that configuration at the next step, regardless of (ρ, γ) . $\square$

Corollary 90 (Cost of ignoring the sleep signal; softmax). *Suppose the sleep condition fires at time t_0 (i.e. $\rho_{t_0} < c$, $\gamma_{t_0} < \epsilon$) but no regime transition occurs: S remains non-zero and capture remains enabled for k additional steps. Then:*

- (i) $|M|$ continues to grow (Theorem 53(iii)), with ρ degrading further.
- (ii) By Theorem 67(ii), each additional waking step adds to the consolidation deficit. The eventual sleep duration required to recover to tolerance δ satisfies $T_{\text{sleep}} \geq \log(V_{t_0+k}/\delta)/\gamma_{\min}$, which grows as $O(\log k)$ in the number of ignored steps (Theorem 70(iii)).
- (iii) If the signal is ignored indefinitely ($k \rightarrow \infty$), the system enters permanent degradation: $\rho \rightarrow 1/|M| \rightarrow 0$ and $\gamma \rightarrow 0$, with V bounded away from zero and growing. No amount of waking consolidation can reverse this (Theorem 50(v)).

Proof. (i) Immediate from Theorem 53: $|M|$ is non-decreasing with positive capture rate, and $\rho(|M|)$ is strictly decreasing.

(ii) By Theorem 70(iii), the additional sleep cost from delaying by k steps is $\Delta T_{\text{sleep}} = O(\log k/\gamma_{\min})$.

(iii) As $|M| \rightarrow \infty$, $\rho \leq 1/(1 + (|M| - 1)\exp(-2R^2)) \rightarrow 0$. By Theorem 50(v), waking consolidation is self-limiting and cannot reverse the degradation. The system is permanently stuck in the exhaustion regime of Theorem 56. $\square$

Remark 91 (The minimal controller). Proposition 89 establishes that the loop’s regime structure separates cleanly into two components:

- (a) **Signal generation:** entirely internal (Theorem 56). The quantities (ρ, γ) are computable from the system’s own state, require no external oracle, and provide both the onset condition ($\rho < c$, $\gamma < \epsilon$) and the completion condition ($\gamma \rightarrow 0$ with ρ recovered).
- (b) **Signal execution:** requires a minimal external controller that reads (ρ, γ) and toggles S and the capture flag. This controller need not be sophisticated: a threshold comparator on two scalar quantities suffices.

This separation (the system that generates the pressure signal is not the system that acts on it) is a structural property of the loop, not a limitation. Biological sleep regulation also separates pressure accumulation from regime execution; whether the structural parallel reflects a shared constraint or a coincidence of architecture is open.

If the controller fails to act on the signal, the mathematical consequence is unbounded degradation (Corollary 90): logarithmic in the delay for recoverable cases, but permanent if the signal is ignored indefinitely [Van Dongen et al., 2003, Dawson and Reid, 1997].

5 Discussion

The results of this paper establish that the loop is forced into alternating capture-consolidation regimes by the interaction of three mechanisms: the fan effect (each capture event dilutes all retrieval weights), the bounded score gap (retrieval precision has a hard ceiling), and self-limiting consolidation (the contraction rate vanishes as impressions converge). No parameter was tuned to produce this behavior; the alternation is a theorem about the definitions.

The loop was not designed to model anything. We did not set out to find any of the behaviors described in Sections 3–4; they fell out of the mathematics when we followed the thread of self-referential attention to its conclusion.

The following are proved properties of the object. For each, we cite the empirical literature reporting the same structural form.

Memory and learning. Hyperbolic forgetting with rate constant determined by encoding strength (Corollary 41; cf. Ebbinghaus 1885, Rubin and Wenzel 1996). An inverted- U spacing curve with optimal gap $K^* = R/(n - 1)$ scaling linearly with the retention interval (Proposition 43; cf. Cepeda et al. 2006). Selective preservation under retrieval with displacement of competitors (Corollary 46; cf. Roediger and Karpicke 2006).

Sleep. The carrying capacity forces the system into an offline consolidation regime; that this regime exists at all is a theorem (Corollary 87). What happens inside it follows from the regime’s structure: monotone retrieval degradation reversed only by an offline phase (Theorem 53). Compounding debt under truncated consolidation (Theorem 67). A forced quasi-periodic cycle requiring an external gate (Corollary 87, Proposition 89). Content retrieved but never stored during the offline phase (Theorem 76). A brief recall window upon re-entry to the capture regime. Declining consolidation need as prototypes accumulate (Corollary 85). Self-observable pressure signals (Theorem 56). Permanent degradation under gate failure (Corollary 90). Transient sharpening that masks but worsens the deficit (Theorem 70). Nap structure (Theorem 101). Monotone phase transitions within consolidation (Theorem 95). Oscillation across memory clusters during the offline phase (Theorem 61).

None of these were design targets. Each is a theorem about the entire family of admissible simplex maps, blend rates, and embeddings. The loop’s quantities (σ , ρ , γ , V) are internal to the object; connecting them to behavioral measures (recall probability, response time) requires a measurement model, which is a separate project.

Existing frameworks address subsets of these phenomena separately: the two-process model [Borbély, 1982] for sleep regulation, synaptic homeostasis [Tononi and Cirelli, 2003] for offline downscaling, complementary learning systems [McClelland et al., 1995] for the division between fast online encoding and slow offline consolidation. Each was designed to explain its target domain. The loop was not designed to explain any of them, yet it produces structural forms overlapping with all three from a single set of definitions.

The structural forms are nonetheless falsifiable as theorems about the object: the forgetting curve is power-law, not exponential; the spacing benefit is an inverted U , not monotone; retrieval strengthens the retrieved item while weakening competitors, not all items uniformly; sleep pressure is monotone during waking and irreversible without an offline phase. These are statements about functional form, not fitted parameters. The loop is executable (Appendix C); its trajectories under any choice of simplex map, blend rate, and embedding can be generated and compared with empirical time series.

Open questions.

- (i) **Why does it look like cognition?** The loop was constructed by asking “what is the simplest system that uses its attention to modify its own attention?” Hyperbolic forgetting, optimal spacing, the testing effect, sleep, dreams, debt, and developmental decline were not design targets, yet they appear as theorems about the full family, taking the structural form of empirical phenomena across memory research, educational psychology, and sleep science. Is there a structural reason why any system that modifies its own retrieval using its own retrieval is forced into these forms? Or is the resemblance superficial, a case of shared vocabulary mapping onto different mechanisms? The loop is executable (Appendix C) and its trajectories can be compared directly with empirical data; doing so would help distinguish these possibilities.
- (ii) **Memory growth under consolidation.** Theorem 50 establishes a critical memory size $|M^*|$ beyond which retrieval is degraded, and Corollary 87 shows the loop is forced into quasi-periodic capture–consolidation cycles. The remaining question is whether the exact growth rate of $|M|$ under the coupled dynamics can be characterized: without consolidation,

$|M| \leq |X|$ (Lemma 25); with consolidation, drifted members may re-satisfy the capture threshold. The exact analytical growth rate remains open.

- (iii) **Contextual non-collapse.** Corollary 37 shows convergence to a single prototype under fixed context. Whether sufficiently varied context prevents $V \rightarrow 0$ in the full system is a question about the timescale of anchor rotation relative to the contraction rate γ (Corollary 40). What is the steady-state V distribution under realistic context variation?
- (iv) **Quantitative non-degeneracy.** Theorem 27 establishes a uniform attention floor $a_t(\tau) \geq \alpha > 0$ for all $\varphi \in \mathcal{A}_+$ via compactness; for softmax, the explicit bound $\alpha = \exp(-2B)/|X|$ is available. Can comparably explicit bounds be given for broader subclasses of $\mathcal{A}_+$?
- (v) **Sharpness of the classification.** For each softmax-level result, does there exist a positive simplex map $\varphi \in \mathcal{A}_+$ under which it fails? A positive answer would show that the boundary between $\mathcal{A}_+$ and $\{\text{softmax}\}$ in Proposition 7 is necessary, not merely sufficient.
- (vi) **Multi-day debt dynamics.** Theorem 67 gives a geometric recurrence for single-cycle debt. Does the multi-cycle trajectory exhibit more complex behavior (e.g. sensitivity to the sequence of capture rates ΔV_n , interaction with maturation effects of Corollary 85)?
- (vii) **Does $|M^*|$ correspond to a measurable quantity?** The critical memory size $|M^*| = 1 + ((1 - c)/c) \exp(2R^2)$ depends on the embedding radius R and the retrieval threshold c . Whether any neural or behavioral quantity corresponds to $|M^*|$ is an empirical question outside the scope of this analysis.
- (viii) **Transition time bounds.** Remark 74 conjectures that inter-episode transitions are brief ($O(1)$ feedback-loop cycles) relative to consolidation episodes ($O(\log(V/\delta)/\gamma)$ steps). A rigorous bound requires quantifying the spectral properties of EK .

Question (i) is the central interpretive question of this paper. Questions (ii)–(iii) and (vi)–(viii) reduce to trajectory questions about an executable system; (iv)–(v) are structural questions about the class of simplex maps. The loop is fully specified, the implementation is provided (Appendix C), and all questions except (i) are computationally accessible.

A Supplementary Results

A.1 Supporting Lemmas for Capture Inertness

Lemma 92 (Softmax Lipschitz bound). *Let $\varphi = \text{softmax}: \mathbb{R}^n \rightarrow \Delta^n$. For any $x, y \in \mathbb{R}^n$,*

$$\|\varphi(x) - \varphi(y)\|_1 \leq 2 \|x - y\|_\infty.$$

Proof. The Jacobian of softmax is $J = \text{diag}(\varphi) - \varphi \varphi^\top$. For any $v \in \mathbb{R}^n$,

$$\begin{aligned} \|Jv\|_1 &= \|\text{diag}(\varphi) v - \varphi (\varphi^\top v)\|_1 \\ &\leq \|\text{diag}(\varphi) v\|_1 + |\varphi^\top v| \|\varphi\|_1 \\ &= \sum_i \varphi_i |v_i| + \left| \sum_i \varphi_i v_i \right| \\ &\leq 2 \sum_i \varphi_i |v_i| \leq 2 \|v\|_\infty. \end{aligned}$$

By the mean value theorem along the segment from y to x :

$$\|\varphi(x) - \varphi(y)\|_1 \leq \sup_{z \in [x, y]} \|J(z)\|_{\infty \rightarrow 1} \|x - y\|_\infty \leq 2 \|x - y\|_\infty. \quad \square$$

Lemma 93 (Accumulation increment bound; $\mathcal{A}_+$). *Let φ be a positive simplex map with bounded inputs. For any sequence $\{a_t\}$ with $a_t(\tau) \in [\alpha, 1 - \alpha]$ for some $\alpha > 0$ (guaranteed by positivity of φ with bounded inputs; for softmax, Theorem 27 gives an explicit bound), the accumulation rule gives*

$$|\sigma_{t+1}(\tau) - \sigma_t(\tau)| = O(1/t)$$

for each $\tau \in X$. The implied constant depends on α and N_0 but not on the specific sequence $\{a_t\}$.

Proof. Fix τ and drop it from the notation.

(i) From Definition 3, $C_{t+1}^2 = C_t^2 + 2a_t(t+1) + a_t^2(t+1)^2/C_t^2$. Since $a_t \geq \alpha > 0$: $C_t^2 \geq C_0^2 + \alpha t(t+1)$, giving $C_t \geq \sqrt{\alpha}t$. Since $a_t \leq 1$: $C_t^2 \leq C_0^2 + t(t+1) + O(t)$, giving $C_t \leq (1 + \epsilon)t$ for large t .

(ii) Write $\sigma_t = C_t/N_t$ with $N_t = N_0 + t$. Then

$$\sigma_{t+1} - \sigma_t = \frac{C_{t+1}(N_0 + t) - C_t(N_0 + t + 1)}{(N_0 + t)(N_0 + t + 1)}.$$

The numerator equals $(C_{t+1} - C_t)(N_0 + t) - C_t$. Since $C_{t+1} - C_t = a_t(t+1)/C_t$ and $C_t = \Theta(t)$, we have $C_{t+1} - C_t = \Theta(1)$. The numerator is $\Theta(t) - \Theta(t) = O(t)$. The denominator is $\Theta(t^2)$. Hence $|\sigma_{t+1} - \sigma_t| = O(1/t)$. $\square$

Lemma 94 (Finite argmax crossings under convergent perturbation). *Let $\{q_t\}$ be a sequence in $\mathbb{R}^d$ converging to q^* with $\sum_t \|q_{t+1} - q_t\| < \infty$. Let $E \in \mathbb{R}^{n \times d}$ be a fixed embedding such that $\text{argmax}_\tau (q^*)^\top E(\tau)$ is unique. Then there exists T_0 such that for all $t > T_0$,*

$$\text{argmax}_\tau q_t^\top E(\tau) = \text{argmax}_\tau (q^*)^\top E(\tau).$$

The argmax changes at most T_0 times.

Proof. Let $\tau^* = \text{argmax}_\tau (q^*)^\top E(\tau)$ and $\delta = (q^*)^\top E(\tau^*) - \max_{\tau \neq \tau^*} (q^*)^\top E(\tau) > 0$. Since $q_t \rightarrow q^*$, there exists T_0 with $\|q_t - q^*\| < \delta/(2R)$ for all $t > T_0$. For any $\tau \neq \tau^*$ and $t > T_0$:

$$\begin{aligned} q_t^\top E(\tau^*) - q_t^\top E(\tau) &= [(q^*)^\top E(\tau^*) - (q^*)^\top E(\tau)] + (q_t - q^*)^\top [E(\tau^*) - E(\tau)] \\ &\geq \delta - 2R\|q_t - q^*\| > \delta - 2R \cdot \frac{\delta}{2R} = 0. \end{aligned} \quad \square$$

A.2 Phase Structure

Theorem 95 (Phase structure of the $S = 0$ regime; softmax). *Let $\varphi = \text{softmax}$ with parameter β . Under $S = 0$ with capture disabled and $|M| \geq 2$, the sleep phase exhibits the following monotone phase structure:*

- (i) **Score spread (fixed-query).** Define the fixed-query score spread at step t as $\tilde{\Delta}_\sigma(t) = \max_j q_t^\top m_j^{(t)} - \min_j q_t^\top m_j^{(t)}$, measured using the query at step t against the memory at step t . Then $\tilde{\Delta}_\sigma$ is non-increasing under consolidation when measured against a fixed query: for any fixed q , $\max_j q^\top m_j^{(t+1)} - \min_j q^\top m_j^{(t+1)} \leq \max_j q^\top m_j^{(t)} - \min_j q^\top m_j^{(t)}$. Furthermore, the time-varying quantity $\tilde{\Delta}_\sigma(t)$ converges to zero as $V(t) \rightarrow 0$: when all $m_j \rightarrow v$, $|q_t^\top m_j - q_t^\top v| \leq \|q_t\| \|m_j - v\| \leq R \cdot V(t) \rightarrow 0$ (since $\|q_t\| \leq R$ by Corollary 29).
- (ii) **Early sleep** ($\tilde{\Delta}_\sigma$ large): the retrieval weights w are non-uniform ($\max_j w_j \geq 1/(1 + (|M| - 1)e^{-\beta\tilde{\Delta}_\sigma}) > 1/|M|$), producing a variable query $q = aE$ that depends on which impressions dominate retrieval. The attention distribution a is non-uniform. The anchor identity may rotate across steps, producing multi-region consolidation: different impressions serve as anchor at different times, and consolidation contracts M toward multiple prototypes simultaneously.

(iii) **Late sleep** ($\tilde{\Delta}_\sigma$ small): as $V(t) \rightarrow 0$ and all $m_j \rightarrow v$, the scores satisfy $|q^\top m_j - q^\top v| \leq RV(t) \rightarrow 0$ for all j , so $w_j \rightarrow 1/|M|$ (uniform retrieval). The attention distribution a stabilizes because $\ell \rightarrow v$ implies $b \rightarrow EKv$ (constant). The anchor identity is stable (all scores differ by at most $RV(t) \rightarrow 0$, so the anchor does not rotate). Consolidation contracts toward a single prototype.

(iv) **Monotone transition.** The score spread $\tilde{\Delta}_\sigma(t) \leq 2RV(t) \leq 2RD(t)$, where $D(t) = \max_{i,j} \|m_i^{(t)} - m_j^{(t)}\|$ is strictly decreasing (Theorem 52(i)), so the transition from early to late sleep is monotone (up to the envelope $2RD(t)$) and irreversible within the $S = 0$ phase.

Proof. (i) **Fixed-query monotonicity.** Fix any query q and consider one consolidation step. The anchor $v = \arg \max_k q^\top m_k$ has $\lambda_v = 0$ (Lemma 22), so v is preserved. For $j \neq v$, the updated impression is $m'_j = (1 - \lambda_j)m_j + \lambda_j T_j$ where $T_j \in \text{conv}(M)$ (Lemma 24). Since T_j is a convex combination of elements of M :

$$q^\top T_j = \sum_k \alpha_k^{(j)} q^\top m_k \in [\min_k q^\top m_k, \max_k q^\top m_k],$$

and therefore

$$q^\top m'_j = (1 - \lambda_j) q^\top m_j + \lambda_j q^\top T_j \in [\min_k q^\top m_k, \max_k q^\top m_k].$$

The maximum score is preserved ($q^\top v$ is unchanged) and the minimum score can only increase (the minimum-scoring element blends toward T_j which scores strictly above $\min_k q^\top m_k$ since φ is positive and v contributes with positive weight). This proves $\max_j q^\top m'_j - \min_j q^\top m'_j \leq \max_j q^\top m_j - \min_j q^\top m_j$ for any fixed q .

Time-varying quantity. With the evolving query q_t , the quantity $\tilde{\Delta}_\sigma(t) = \max_j q_t^\top m_j^{(t)} - \min_j q_t^\top m_j^{(t)}$ need not be strictly monotone, because $q_{t+1} \neq q_t$. However, $\tilde{\Delta}_\sigma(t) \rightarrow 0$ regardless. By the triangle inequality, $\tilde{\Delta}_\sigma(t) \leq 2\|q_t\| \max_j \|m_j^{(t)} - v_t\| \leq 2R \cdot V(t)$, where $\|q_t\| \leq R$ (Corollary 29) and $V(t) \rightarrow 0$ (Theorem 52(i)). Since $V(t) \rightarrow 0$, $\tilde{\Delta}_\sigma(t) \rightarrow 0$. The convergence is monotone in the sense that the diameter $D(t) = \max_{i,j} \|m_i^{(t)} - m_j^{(t)}\|$ is strictly decreasing (Theorem 52(i)), and since $V(t) \leq D(t)$, the bound $2RD(t)$ is a strictly decreasing envelope for $\tilde{\Delta}_\sigma(t)$.

(ii) When Δ_σ is large, the softmax retrieval weights satisfy $w_v = \exp(\beta q^\top v)/Z \geq 1/(1 + (|M| - 1)e^{-\beta\Delta_\sigma}) > 1/|M|$ where $Z = \sum_k \exp(\beta q^\top m_k)$. The blend rates $\lambda_j = 1 - w_j/w_v$ are bounded away from zero for $j \neq v$, giving $\gamma(t) > 0$ bounded away from zero. The query $q = aE$ depends on $a = \text{softmax}(\sigma + b)$ where $b = EK\ell$ varies because $\ell = wM$ is non-trivially weighted. Different queries highlight different impressions, so the anchor may shift across steps. By Corollary 40, if the anchor rotates, V is measured against different reference points, and consolidation proceeds toward multiple regions.

(iii) As $V(t) \rightarrow 0$ (Corollary 37), all $m_j \rightarrow v$. Then $q^\top m_j \rightarrow q^\top v$ for all j , so $w_j = \exp(\beta q^\top m_j)/Z \rightarrow 1/|M|$. The retrieval output $\ell = wM \rightarrow v$ (by Theorem 55(iii)), the bias $b = EK\ell \rightarrow EKv$ (constant), and the attention $a = \text{softmax}(\sigma + EKv)$ stabilizes. With uniform w , the anchor is determined by infinitesimal score differences and does not rotate.

(iv) By part (i), $\tilde{\Delta}_\sigma(t) \leq 2RV(t) \leq 2RD(t)$ and $D(t) \rightarrow 0$ (Theorem 52(i)). Therefore $\tilde{\Delta}_\sigma(t) \rightarrow 0$. The envelope $2RD(t)$ is strictly decreasing (Theorem 52(i)), so the transition from early sleep (large score spread) to late sleep (small score spread) is monotone up to the envelope. Within the $S = 0$ phase, no operation can increase D : capture is disabled (so M does not grow) and consolidation strictly decreases D while $D > 0$. $\square$

Remark 96 (Two-phase structure of the $S = 0$ regime). Theorem 95 describes a natural two-phase structure within each $S = 0$ episode. In the early phase, the system consolidates actively (large γ) with possible anchor rotation. In the late phase, consolidation slows ($\gamma \rightarrow 0$), the anchor stabilizes, and the system approaches a fixed point. The transition is governed by the Lyapunov function $V(t)$ (via the envelope $\tilde{\Delta}_\sigma(t) \leq 2RV(t)$). The early phase does the work of memory organization; the late phase signals completion. The parallel to biological sleep stages (deep NREM followed by lighter sleep) is suggestive but unproven.

A.3 Accumulation Rigidity: Detailed Proofs

The main-body proof of Proposition 10 gives a four-line argument that $f(t) \sim t$ is necessary for $\sigma = C/N$ to converge to a nonzero constant. Here we give the full details: first the aggregate $\Theta(t^2)$ characterization of $\sum f$, then the two extremal cases that complete the necessity argument.

Throughout, C_t satisfies the generalized accumulation rule

$$C_{t+1} = C_t + a_t \cdot \frac{f(t)}{C_t},$$

where $f: \mathbb{N} \rightarrow \mathbb{R}_{>0}$ is a positive scaling function, $\delta \leq a_t \leq 1$ for some $\delta > 0$, $C_0 > 0$ (so $C_t > 0$ for all t by induction, since the increment is positive), and $N_t := N_0 + t$ with $N_0 \geq 1$. We write $\sigma_t = C_t/N_t$.

Proposition 97 (Aggregate accumulation rigidity; $\mathcal{A}$). *If $\sigma_t \rightarrow \sigma_\infty \in (0, \infty)$, then $\sum_{s=0}^{t-1} f(s) = \Theta(t^2)$. Explicitly, there exist $\alpha, \beta > 0$ and $T \in \mathbb{N}$ such that for all $t \geq T$,*

$$\alpha N_t^2 \leq \sum_{s=0}^{t-1} f(s) \leq \beta N_t^2.$$

Proof. Upper bound on $\sum f$. Squaring the recurrence gives

$$C_{t+1}^2 = C_t^2 + 2a_t f(t) + (a_t f(t)/C_t)^2 \geq C_t^2 + 2\delta f(t),$$

since $a_t \geq \delta$ and $(a_t f(t)/C_t)^2 \geq 0$. Telescoping:

$$C_t^2 \geq C_0^2 + 2\delta \sum_{s=0}^{t-1} f(s). \quad (14)$$

From σ -convergence, $C_t \leq 2\sigma_\infty N_t$ for large t . Combining with (14) and discarding $C_0^2 \geq 0$:

$$\sum_{s=0}^{t-1} f(s) \leq \frac{2\sigma_\infty^2}{\delta} N_t^2.$$

Lower bound on $\sum f$. We first show $f(t) \leq C_t^2$ eventually. From the recurrence, $a_t f(t) = C_t(C_{t+1} - C_t)$, so $f(t) \leq C_t(C_{t+1} - C_t)/\delta$. Using σ -convergence with $\varepsilon = \sigma_\infty \delta/10$: for large t , $C_t = (\sigma_\infty \pm \varepsilon)N_t$ and $C_{t+1} - C_t \leq (\sigma_\infty + \varepsilon) + 2\varepsilon N_t$, giving $f(t) = O(\varepsilon N_t^2)$. Since $C_t^2 \geq (\sigma_\infty - \varepsilon)^2 N_t^2$ and ε is small enough that $(\sigma_\infty - \varepsilon)^2 \delta > 2\varepsilon(\sigma_\infty + \varepsilon)$, we get $f(t) \leq C_t^2$ for large t .

Once $f(t) \leq C_t^2$, the squared recurrence satisfies $C_{t+1}^2 \leq C_t^2 + 3f(t)$ (since $(a_t f/C_t)^2 \leq f^2/C_t^2 \leq f$ and $2a_t f \leq 2f$). Telescoping from T_0 (past which $f \leq C^2$) and using $C_t \geq (\sigma_\infty/2)N_t$ for large t :

$$(\sigma_\infty/2)^2 N_t^2 \leq C_t^2 \leq C_{T_0}^2 + 3 \sum_{s=T_0}^{t-1} f(s) \leq C_{T_0}^2 + 3 \sum_{s=0}^{t-1} f(s).$$

For N_t large enough that $(\sigma_\infty/2)^2 N_t^2 > 2C_{T_0}^2$:

$$\sum_{s=0}^{t-1} f(s) \geq \frac{(\sigma_\infty/2)^2}{6} N_t^2. \quad \square$$

Proposition 98 (Superlinear f forces $\sigma \rightarrow \infty$; $\mathcal{A}$). *If $f(t) \geq c(t+1)^2$ for some $c > 0$ and all t , then $\sigma_t \rightarrow \infty$.*

Proof. From (14), $C_t^2 \geq 2\delta c \sum_{s < t} (s+1)^2 \geq \delta c t^3/2$ (using $\sum_{s=0}^{t-1} (s+1)^2 \geq t^3/4$, verified by induction). For $t \geq N_0$, $N_t \leq 2t$, so

$$\sigma_t = \frac{C_t}{N_t} \geq \frac{\sqrt{\delta c/2}}{2} \sqrt{t} \rightarrow \infty. \quad \square$$

Proposition 99 (Sublinear f forces $\sigma \rightarrow 0$; $\mathcal{A}$). *If $f = o(t)$, then $\sigma_t \rightarrow 0$.*

Proof. Fix $\varepsilon > 0$; we show $\sigma_t \leq \varepsilon$ eventually. Set $\varepsilon' = \varepsilon/2$ and choose $\eta = \min(\varepsilon'^2/2, \varepsilon' C_0/2)$, so $\eta < \varepsilon'^2$. By sublinearity, $f(t) \leq \eta(t+1)$ for $t \geq T_1$.

Step 1: σ drops below ε' . Suppose $C_t \geq \varepsilon' N_t$ for all $t \geq T_1$. Each increment satisfies

$$C_{t+1} - C_t = \frac{a_t f(t)}{C_t} \leq \frac{\eta N_t}{\varepsilon' N_t} = \frac{\eta}{\varepsilon'},$$

so $C_{T_1+k} \leq C_{T_1} + k \cdot \eta/\varepsilon'$. But $C_{T_1+k} \geq \varepsilon' N_{T_1+k} = \varepsilon'(T_1 + k + N_0)$, giving $k(\varepsilon' - \eta/\varepsilon') \leq C_{T_1}$. Since $\eta < \varepsilon'^2$ implies $\varepsilon' - \eta/\varepsilon' > 0$, this fails for large k , a contradiction. So there exists $t_0 \geq T_1$ with $\sigma_{t_0} < \varepsilon'$.

Step 2: barrier. Set $M = \varepsilon' + \eta/C_0 \leq \varepsilon$ (using $\eta \leq \varepsilon' C_0/2$, so $\eta/C_0 \leq \varepsilon'/2$). We show $C_t \leq M N_t$ for all $t \geq t_0$ by induction. The base case holds since $C_{t_0} < \varepsilon' N_{t_0} < M N_{t_0}$. For the inductive step, assume $C_t \leq M N_t$ and consider two cases. If $C_t \geq \varepsilon' N_t$, the Step 1 bound gives $C_{t+1} - C_t \leq \eta/\varepsilon' \leq \varepsilon' \leq M$, so $C_{t+1} \leq M N_t + M = M N_{t+1}$. If $C_t < \varepsilon' N_t$, then since $C_0 \leq C_t$ (by monotonicity), $C_{t+1} - C_t \leq f(t)/C_0 \leq (\eta/C_0) N_t$, so $C_{t+1} \leq (\varepsilon' + \eta/C_0) N_t = M N_t \leq M N_{t+1}$. $\square$

A.4 Naps

We now characterise *naps*: short $S = 0$ episodes that are too brief for full multi-cluster convergence. We show that nap benefit is front-loaded, that naps consolidate at most one cluster, and that split sleep (two naps separated by waking) can consolidate different clusters from different initial queries.

Definition 100 (Nap). A *nap* of duration k is an $S = 0$ phase with capture disabled lasting exactly k consolidation steps. We say the nap is *short* if $k < T_{\text{full}} := \sum_{i=1}^p T_i$, where T_i is the convergence time for cluster $\mathcal{C}_i$ (Corollary 64), and p is the number of well-separated clusters at nap onset (Definition 59). A *split sleep schedule* consists of two naps of durations k_1, k_2 separated by a waking interval of $W \geq 1$ steps (with capture and external stimulation re-enabled).

Theorem 101 (Naps as partial-cycle consolidation; $\mathcal{A}_+$). *Let φ be a continuous positive simplex map. Suppose the system enters a nap of duration k with $|M| \geq 2$ and initial Lyapunov value $V(0) > 0$. Let $\gamma > 0$ be a uniform lower bound on the contraction rate during the nap (which exists by Corollary 32 since φ is positive and $V > 0$). Then:*

- (i) **Front-loaded benefit.** *Let $\alpha := (1 - \gamma)^k$. The ratio of Lyapunov reduction in the second k -step window to that in the first satisfies $R_2/R_1 \leq \alpha/(1 - \alpha)$. In particular, when $k \geq \lceil \log 2 / \log(1/(1 - \gamma)) \rceil$ (equivalently $(1 - \gamma)^k \leq 1/2$), the first k steps produce at least as much reduction as the next k steps:*

$$V(0) - V(k) \geq V(k) - V(2k).$$

For small γ , the sufficient condition $k \geq \lceil (\log 2)/\gamma \rceil$ ensures this (since $-\log(1 - \gamma) \geq \gamma$). Regardless of whether $\alpha \leq 1/2$, the benefit is exponentially front-loaded: each successive window of k steps produces at most $\alpha/(1 - \alpha)$ times the reduction of the previous window.

(ii) **Single-cluster consolidation.** Suppose M contains $p \geq 2$ well-separated ϵ -clusters $\mathcal{C}_1, \dots, \mathcal{C}_p$ (Definition 59). Let T_1 be the convergence time of the anchor cluster $\mathcal{C}_{k^*}$ at nap onset (Corollary 64).

- (a) If $k < T_1$, the nap partially consolidates $\mathcal{C}_{k^*}$ only: $\Delta_{k^*}(q)$ decreases while the intra-cluster spreads $\Delta_{k'}(q)$ for $k' \neq k^*$ change by at most $O(\epsilon)$ per step.
- (b) If $T_1 \leq k < T_1 + T_2$, the nap fully consolidates $\mathcal{C}_{k^*}$ and partially consolidates the next cluster $\mathcal{C}_{k_2^*}$ in the salience ordering of Theorem 61(v).

In either case, the nap consolidates at most $\lceil k / \min_i T_i \rceil$ clusters.

(iii) **Split sleep consolidates different clusters.** Consider a split sleep schedule: nap₁ of duration k_1 , then W waking steps, then nap₂ of duration k_2 . During the waking interval, external stimulation S , accumulation (σ update), and potential capture events change the query context: let q_1 be the query at nap₁ onset and q_2 the query at nap₂ onset. Then $q_2 \neq q_1$ in general, and the anchor cluster at nap₂ onset may differ from the anchor cluster at nap₁ onset. Formally: if the waking interval induces $\|q_2 - q_1\| > 0$ sufficient to change the identity of $\arg \max_j q^\top m_j$, then the anchor cluster rotates. Split sleep can therefore consolidate different clusters that a single continuous sleep of duration $k_1 + k_2$ would process sequentially from the same starting query.

(iv) **Split sleep preserves more prototypes when $T_{\text{rot}} < 1/\gamma_{\min}$.** Under a single continuous sleep of duration $k_1 + k_2$, the system consolidates clusters sequentially from one initial query direction (Theorem 61(v)). With split sleep, the waking interval changes q , potentially approaching the same clusters from different angles. By Corollary 40, if the effective rotation timescale T_{rot} induced by the waking interval satisfies $T_{\text{rot}} < 1/\gamma_{\min}$, distinct impressions within a cluster are preserved rather than collapsed to a single prototype. Split sleep therefore has the potential to preserve more fine-grained structure within clusters than continuous sleep of the same total duration.

(v) **Diminishing returns of naps.** Consider a sequence of naps within the same waking period. The first nap reduces V by the front-loaded amount $V(0)[1 - (1 - \gamma)^{k_1}]$ (part (i)). After a waking interval, the second nap starts from a state where the most salient cluster has already been partially consolidated by the first nap. If the second nap engages the same cluster, its initial V is smaller by the factor $(1 - \gamma)^{k_1}$, and its absolute reduction is correspondingly smaller. If the second nap engages a different cluster (part (iii)), that cluster's V may be comparable to the original, but the marginal system-wide benefit is still smaller because the most salient cluster (largest contribution to retrieval degradation) was already addressed. Each successive nap provides less total V reduction.

Proof. (i) By Corollary 32, $V(t+1) \leq (1 - \gamma(t))V(t)$ with $\gamma(t) > 0$. Over k steps with uniform lower bound $\gamma > 0$, let $\alpha = (1 - \gamma)^k \in (0, 1)$. Then:

$$V(k) \leq \alpha V(0), \quad V(2k) \leq \alpha V(k).$$

The reductions in the first and second windows are

$$R_1 := V(0) - V(k) \geq (1 - \alpha)V(0), \quad R_2 := V(k) - V(2k) \leq V(k) \leq \alpha V(0),$$

where the upper bound on R_2 uses only $V(2k) \geq 0$. Therefore

$$\frac{R_2}{R_1} \leq \frac{\alpha V(0)}{(1 - \alpha)V(0)} = \frac{\alpha}{1 - \alpha}.$$

This ratio is less than 1 whenever $\alpha < 1/2$, i.e., $(1 - \gamma)^k < 1/2$, which holds for $k \geq \lceil \log 2 / \log(1/(1 - \gamma)) \rceil$; a sufficient condition is $k \geq \lceil (\log 2)/\gamma \rceil$ since $-\log(1 - \gamma) \geq \gamma$. Under this condition, $R_1 > R_2$.

The bound $\alpha/(1 - \alpha)$ is achieved when the first window contracts at exactly the minimum rate ($V(k) = \alpha V(0)$) while the second window collapses to $V(2k) = 0$. For the uniform geometric envelope $V(t) = (1 - \gamma)^t V(0)$, the ratio is tighter: $R_2/R_1 = \alpha$ (since $R_1 = (1 - \alpha)V(0)$ and $R_2 = \alpha(1 - \alpha)V(0) = \alpha R_1$), which is less than 1 for all $\alpha \in (0, 1)$, so under uniform geometric contraction the first window always dominates.

Regardless of whether $\alpha \leq 1/2$, the ratio $R_2/R_1 \leq \alpha/(1 - \alpha)$ shows the benefit is exponentially front-loaded: each successive window of k steps produces a geometrically decreasing fraction of the previous window's reduction.

(ii) By Theorem 61(i), consolidation localizes to the anchor cluster $\mathcal{C}_{k^*}$: impressions within $\mathcal{C}_{k^*}$ contract (with rate $\gamma > 0$ bounded away from zero) while impressions in other clusters experience displacement that varies by at most $O(\epsilon)$, preserving their intra-cluster structure up to $O(\epsilon)$.

(a) If $k < T_1$, the anchor cluster has not yet converged (by definition of T_1). By Theorem 61(ii), anchor rotation occurs only after $\mathcal{C}_{k^*}$ converges internally ($\gamma \rightarrow 0$ for that cluster). Since $k < T_1$, this has not happened, so the anchor remains in $\mathcal{C}_{k^*}$ throughout the nap. The intra-cluster spread Δ_{k^*} decreases by the contraction bound, while other clusters' spreads $\Delta_{k'}$ change by at most $O(\epsilon)$ per step (Theorem 61(i), last paragraph). The nap consolidates only $\mathcal{C}_{k^*}$, partially.

(b) If $T_1 \leq k < T_1 + T_2$, the first cluster converges during the nap (at around step T_1), anchor rotation occurs (Theorem 61(ii)), and the anchor moves to $\mathcal{C}_{k_2^*}$. The remaining $k - T_1$ steps partially consolidate $\mathcal{C}_{k_2^*}$, but since $k - T_1 < T_2$, the second cluster does not fully converge. The total number of clusters consolidated is at most $\lceil k / \min_i T_i \rceil$ (each cluster takes at least $\min_i T_i$ steps).

(iii) During the waking interval between the two naps, the system operates with $S \neq 0$ (external stimulation restored) and capture enabled. The query at step t is $q_t = a_t E$ where $a_t = \varphi(S_t + \sigma_t + b_t)$. Three mechanisms change the query context:

- (a) External stimulation S directly perturbs the score, shifting a and hence q .
- (b) Accumulation updates $\sigma = C/N$: during the waking interval, C receives increments from the attention distribution a , changing σ and thereby q .
- (c) Capture events may add new impressions to M , changing $\ell = wM$ and the retrieval bias $b = EK\ell$.

Any of these changes can shift q_2 relative to q_1 , potentially altering $\arg \max_j q^\top m_j$. If the anchor identity changes, the anchor cluster at nap₂ onset differs from that at nap₁ onset. A single continuous sleep of duration $k_1 + k_2$ processes clusters sequentially from q_1 (Theorem 61(v)), while split sleep processes from q_1 and q_2 independently.

(iv) Under continuous sleep of duration $k_1 + k_2$, the system approaches clusters sequentially from the initial query q_1 (Theorem 61(v)). Each cluster is consolidated toward a single prototype along the direction determined by q_1 . Under split sleep, the second nap uses query $q_2 \neq q_1$. If q_2 approaches a partially consolidated cluster from a different direction, the anchor within that cluster may differ from the anchor that would have been selected under q_1 . By Corollary 40, if the query rotates on a timescale T_{rot} shorter than $1/\gamma_{\min}$, distinct impressions within the cluster are preserved. The waking interval between naps provides exactly such a query rotation. Consequently, split sleep may preserve more fine-grained structure: the cluster retains more distinct impressions rather than collapsing all members to a single prototype.

(v) Let $V_{\text{total}}(t)$ denote the system-wide Lyapunov function (defined cluster by cluster as in Theorem 61). The first nap reduces the most salient cluster's V by $V_{k^*}(0)[1 - (1 - \gamma_1)^{k_1}]$,

which is the largest single-cluster reduction available (since $\mathcal{C}_{k^*}$ has the highest salience under q_1 , Theorem 61(v)).

After the first nap and the waking interval, the second nap starts from a state where:

- If it engages the same cluster $\mathcal{C}_{k^*}$, then $V_{k^*}^{(2)} = V_{k^*}(k_1) \leq (1 - \gamma_1)^{k_1} V_{k^*}(0) < V_{k^*}(0)$, so the absolute reduction is at most $(1 - \gamma_1)^{k_1}$ times the first nap's reduction (by the front-loading argument of part (i)).
- If it engages a different cluster $\mathcal{C}_{k'}$, then $V_{k'}(0)$ may be comparable to $V_{k^*}(0)$, but $\mathcal{C}_{k'}$ is by construction less salient than $\mathcal{C}_{k^*}$ under the original query (it was not the anchor cluster). The system-wide reduction from consolidating a less salient cluster provides less improvement in overall retrieval precision (the most salient cluster dominates the retrieval output ℓ).

In either case, the marginal benefit of the second nap is strictly less than that of the first. By induction, each successive nap provides diminishing returns. $\square$

Remark 102 (Short consolidation episodes and split schedules). The front-loading result (part (i)) implies that a nap of duration k achieves a fraction $1 - (1 - \gamma)^k$ of the full single-cluster benefit, so a nap with $\gamma k > \log 2$ already captures more than half. The split-sleep result (parts (ii)–(iii)) is more surprising: two short naps separated by a waking interval can consolidate *different* clusters, achieving coverage a single continuous episode of the same total length cannot. The tradeoff is diminishing returns within a single capture interval (part (v)) versus broader coverage across intervals.

Corollary 103 (Continuous sleep vs. split sleep; $\mathcal{A}_+$). *Let $K_{\text{total}} = k_1 + k_2$ be a fixed total sleep budget.*

- (i) *Continuous sleep of duration K_{total} produces total V reduction of at least $V(0)[1 - (1 - \gamma)^{K_{\text{total}}}]$ in the anchor cluster and processes up to $\lceil K_{\text{total}} / \min_i T_i \rceil$ clusters sequentially.*
- (ii) *Split sleep with naps of duration k_1, k_2 and a waking interval consolidates potentially different clusters, achieving comparable total V reduction (both schedules are governed by the same contraction bound) while processing clusters from different query contexts.*
- (iii) *The choice between continuous and split sleep involves a trade-off: continuous sleep achieves deeper consolidation of the most salient cluster, while split sleep achieves broader coverage across clusters.*

Proof. (i) Immediate from the contraction bound (Corollary 32) applied over K_{total} steps, combined with Theorem 61(iii)–(iv) for the cluster count bound.

(ii) Each nap is governed by the same contraction bound. The first nap reduces V by at least $V(0)[1 - (1 - \gamma_1)^{k_1}]$. The second nap may engage a different cluster (Theorem 101(iii)) with its own initial $V_{k'}(0)$ and contraction rate γ_2 . The total reduction across both clusters is the sum of the individual reductions.

(iii) Under continuous sleep, the full budget K_{total} is applied to the most salient cluster first, achieving $V_{k^*}(K_{\text{total}}) \leq (1 - \gamma)^{K_{\text{total}}} V_{k^*}(0)$, which is exponentially small for large K_{total} . Under split sleep, the most salient cluster receives only k_1 steps, achieving $V_{k^*}(k_1) \leq (1 - \gamma)^{k_1} V_{k^*}(0)$, which is less deep. However, the second nap targets a different cluster, providing consolidation that continuous sleep would not reach until step T_1 (when the first cluster converges and anchor rotation occurs). The trade-off is depth vs. breadth. $\square$

B Formal Verification

Every proof in this paper has been formally verified in Lean 4 using the Mathlib library. The formalization comprises approximately 11,500 lines across 80 files, with zero `sorry` (unresolved proof obligations), zero custom axioms beyond the standard Lean kernel (propositional extensionality, quotient soundness, classical choice), and zero `True` placeholders (trivially true statements substituted for real content). The build consists of 3,183 compilation jobs and passes without errors.

Nothing is axiomatized. Every definition, lemma, theorem, corollary, and proposition stated in this paper is proved from first principles: no results are assumed without proof, and no proof obligations are deferred. This includes the rigidity results (accumulation rate, capture threshold, pipeline structure, scoring rule, consolidation target), which classify each component as forced, invariant, or instance-defining. The contraction and convergence arguments, the Lyapunov construction, the compactness-based non-degeneracy bound, and the forced sleep cycle are all machine-checked.

Each Lean declaration is named to match its paper label (e.g., `accumulation_rigidity_aggregate` corresponds to Proposition 97). The module structure mirrors the paper’s sections:

- `Defs/` — core definitions (simplex maps, loop state, accumulation, sleep debt, etc.)
- `Core/` — main dynamics (convergence, contraction rate, Lyapunov, non-degeneracy)
- `Rigidity/` — rigidity theorems (accumulation, pipeline, capture, scoring, targets)
- `Sleep/` — sleep dynamics (forced cycle, dream transitions, sleep efficacy, wakeup)
- `CaptureInertness/` — capture-consolidation (sigma increment, softmax Lipschitz)
- `Appendix/` — supplementary results (naps, sleep phases, split vs. continuous)

To verify: install Lean 4 and Mathlib, then run `lake build` in the `SleepLoop/` directory. The formalization is available at <https://github.com/nemustudies/sleep-loop/>.

C GPU-Native Realization

All operations in Section 2 are GPU-native. The following implements the complete loop (①–⑤) in a single function.

```
import torch
from torch.nn.functional import softmax

def step(C, N, M, E, K, t, ell=None,
        phi=lambda x: softmax(x, dim=-1),
        lam=lambda r: 1 - r):
    """
    One iteration: Definitions 1-6.

    C: (|X|,) cumulative contrast, C_0 > 0
    N: (|X|,) lifetime count
    M: (m, d) impression matrix, m = 0 initially
    E: (|X|, d) embeddings
    K: (d, d) linear projection
    t: int step index
    ell: (d,) or None retrieval vector from previous step
    """

    # Definition 1: scoring from previous retrieval
    sigma = C / N.clamp(min=1)
    if ell is not None:
        b = E @ (K @ ell) # (|X|,)
```

```

else:
    b = torch.zeros_like(sigma)
    S = sigma + b # scoring is purely feedback

    # Definition 2: observation
    a = phi(S) # (|X|,)

    # Definition 3: accumulation
    C = C + a * (t + 1) / C
    N = N + 1
    sigma = C / N

    # Definition 4: retrieval
    q = a @ E # (d,)
    if M.shape[0] > 0:
        scores = q @ M.T # (m,)
        w = phi(scores.unsqueeze(0)).squeeze(0) # (m,)
        ell = w @ M # (d,)

        # Definition 5: consolidation
        blend = lam(w / w.max()).unsqueeze(-1) # (m, 1)
        target = phi((M + q) @ M.T) @ M # (m, d)
        M = (1 - blend) * M + blend * target

        # Definition 6: capture
        novelty = q @ E.T # (|X|,)
        mask = novelty > scores.max()
        if mask.any():
            M = torch.cat([M, E[mask]])
    else:
        # Cold start: max over empty set is -inf, everything admitted
        M = E.clone()
        ell = None

return C, N, M, ell

```

Remark 104 (Numerical stability). The accumulation update $C \leftarrow C + a(t+1)/C$ is numerically unstable when C is small: the ratio $a(t+1)/C$ diverges as $C \rightarrow 0$. In the GPU implementation, $C_0 = \pi > 0$ prevents this at initialization, but accumulated floating-point error can produce near-zero values over long trajectories. Clamping $C \geq \epsilon$ for machine epsilon ϵ suffices. The consolidation target $T(M, q) = \varphi((M + q)M^\top)M$ involves a softmax over inner products that can overflow when $\|m_j\|$ is large; scaling by $1/\sqrt{d}$ controls this. Neither modification affects the analytical results: the proofs require $C > 0$ (satisfied by the clamp) and bounded scores (satisfied by the scaling), both of which the stabilized implementation provides.

Remark 105 (Transformer-native realization). The system embeds into a transformer attention layer as a between-chunk state update with one learned scalar gating the memory contribution to the scoring function. The scoring function S becomes the attention logit matrix, φ becomes softmax, and K becomes the key projection. Since $S = QK^\top/\sqrt{d}$ is bounded whenever model weights are bounded, all proven results transfer without additional assumptions. The implementation is a single `nn.Module` wrapping the loop above. The scalar gates the entire memory contribution and is the system’s only trainable parameter; at initialization it is zero and the model is standard attention. This realization is not included because it is an engineering detail rather than a mathematical one. The dynamical system does not require a transformer, and the transformer does not require the dynamical system; the embedding is a convenience, not a necessity.

Acknowledgements

The author thanks Matthew, Ed, and Nathan for many years of friendship, and zeri for the nickname nemu.

The author is an undergraduate at the University of California, San Diego, has no background

in formal mathematics or computer science, completed this work during their first year, and otherwise spends their time drawing anime girls as a digital artist. Contact: nemustudies@gmail.com.

a personal note

Hi. Thanks for reading this paper. I hope it gave you something new to explore, and that you discovered and understood something personal for yourself. I enjoyed sharing this with you, and working on this has taught me a lot about myself and about others. I don't know if I'll continue doing mathematics in this way after this, as I have other passions and hobbies (I want to get back to drawing, pass my classes, hang out with friends, take care of myself, and work on other things I think are interesting), but I felt like I understood a bit more why mathematicians love math so much.

I never thought I'd be up till 5 am every day working on math for 9 days straight while failing my classes, but here I am writing a math paper (and formalizing the whole thing with a proof calculator so that I could make sure it was real) since I wanted to share what I found. I did start getting sleep in the last third, though. This is my first time writing a paper and sharing it, so I appreciate you sticking with me with the quirks of the presentation.

Honestly, it was an accident that I even ended up fixated on finding this object. I feel very lucky. That being said, I had a lot of fun working on this and trying to discover a 0-degree-of-freedom solution to the problem. It was much more difficult than I expected.

I found a lot of interesting things and connections to things outside of math and sleep as I explored this object for myself, but I want to let you discover what these things personally mean to you, so I have chosen to only show one of the threads that I've explored and keep the interpretation minimal. There's so much in the world to explore and see, and so many things to do and experience, and I hope that by experiencing this object, you can think about how we work in a new way.

I'm happy to be the medium through which this loop is shown to you. I hope you find it as exciting to explore as I did.

Once again, thank you for your time, and I'm thankful for the opportunity to share something so pretty with you.

—*nemu*

References

- John R. Anderson. Retrieval of propositional information from long-term memory. *Cognitive Psychology*, 6(4):451–474, 1974.
- Alexander A. Borbély. A two process model of sleep regulation. *Human Neurobiology*, 1(3):195–204, 1982.
- Jan Born and Ines Wilhelm. System consolidation of memory during sleep. *Psychological Research*, 76(2):192–203, 2012.
- Nicholas J. Cepeda, Harold Pashler, Edward Vul, John T. Wixted, and Doug Rohrer. Distributed practice in verbal recall tasks: A review and quantitative synthesis. *Psychological Bulletin*, 132(3):354–380, 2006.
- Drew Dawson and Kathryn Reid. Fatigue, alcohol and performance impairment. *Nature*, 388(6639):235, 1997.
- Susanne Diekelmann and Jan Born. The memory function of sleep. *Nature Reviews Neuroscience*, 11(2):114–126, 2010.

- Hermann Ebbinghaus. *Über das Gedächtnis*. Duncker & Humblot, 1885.
- James L. McClelland, Bruce L. McNaughton, and Randall C. O'Reilly. Why there are complementary learning systems in the hippocampus and neocortex: Insights from the successes and failures of connectionist models of learning and memory. *Psychological Review*, 102(3): 419–457, 1995.
- Henry L. Roediger and Jeffrey D. Karpicke. Test-enhanced learning: Taking memory tests improves long-term retention. *Psychological Science*, 17(3):249–255, 2006.
- David C. Rubin and Amy E. Wenzel. One hundred years of forgetting: A quantitative description of retention. *Psychological Review*, 103(4):734–760, 1996.
- Giulio Tononi and Chiara Cirelli. Sleep and synaptic homeostasis: A hypothesis. *Brain Research Bulletin*, 62(2):143–150, 2003.
- Giulio Tononi and Chiara Cirelli. Sleep function and synaptic homeostasis. *Sleep Medicine Reviews*, 10(1):49–62, 2006.
- Hans P. A. Van Dongen, Greg Maislin, Janet M. Mullington, and David F. Dinges. The cumulative cost of additional wakefulness: Dose-response effects on neurobehavioral functions and sleep physiology from chronic sleep restriction and total sleep deprivation. *Sleep*, 26(2): 117–126, 2003.